

SingleRAN

Common Transmission Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised the MML-based configurations for IP-based multimode co-transmission on the base station side. For details, see , [5.4.3.12.2 Using MML Commands](#), [5.4.3.13.2 Using MML Commands](#), and [5.4.3.14.2 Using MML Commands](#).

Revised other descriptions in this document.

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for co-transmission by the UMPThb. For details, see: • 5.1.2.3 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT MBTS Side • 5.1.2.5 Co-transmission on the Co-MPT MBTS Side • 5.4.3.1 Main-Control-Board-based Co-transmission Through Panel Interconnection on the Separate-MPT NL/NT/NM MBTS Side in IPv6 over FE/GE Mode • 5.4.3.2 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT NL/NT/NM MBTS Side in IPv6 over FE/GE Mode • 5.4.3.3 Main-Control-Board-based Co-transmission Through Panel Interconnection on the Separate-MPT N +GL/GT/GM MBTS Side in IP over FE/GE Mode • 5.4.3.4.1 Data Preparation • 5.4.3.5.1 Data Preparation • 5.4.3.6.1 Data Preparation • 5.4.3.7.1 Data Preparation • 5.4.3.8.1 Data Preparation • 5.4.3.9.1 Data Preparation • 5.4.3.10.1 Data Preparation • 5.4.3.11.1 Data Preparation • 5.4.3.12.1 Data Preparation • 5.4.3.13.1 Data Preparation • 5.4.3.14.1 Data Preparation	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
5.4.3.16 Co-transmission on the Co-MPT MBTS Side in IPv4/IPv6 Dual-Stack over FE/GE/10GE/25GE/50GE Mode 5.4.3.17.1 Data Preparation 5.4.3.18.1 Data Preparation		

Editorial Changes

Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and New Radio (NR).

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

Unless otherwise specified, in this document, LTE always includes FDD, TDD, and NB-IoT. In scenarios where they need to be distinguished, LTE FDD, LTE TDD, and LTE NB-IoT are used. The "L", "T", and "M" in RAT acronyms refer to LTE FDD, LTE TDD, and LTE NB-IoT, respectively. Unless otherwise specified, in this document, NR always includes FDD and TDD.

2.3 Features in This Document

RAT	Feature ID	Feature Name	Chapter/Section
GSM/ UMTS	MRFD-1211 01	IP-Based BSC and RNC Co-Transmission on MBSC Side	4 IP-based BSC and RNC Co-transmission on the MBSC Side (GSM/UMTS)
GSM	MRFD-1211 14	IP-Based Multi-mode Co-Transmission on BS side(GBTS)	5 IP-Based Multimode Co- transmission on the Base Station Side
UMTS	MRFD-1211 24	IP-Based Multi-mode Co-Transmission on BS side(NodeB)	
FDD	MRFD-1211 34	IP-Based Multi-mode Co-Transmission on BS side(eNodeB)	
TDD	MRFD-1211 44	IP-Based Multi-mode Co-Transmission on BS side(LTE TDD)	
NB-IoT	MRFD-1211 54	IP-Based Multi-mode Co-Transmission on BS side(NB-IoT)	
NR	MRFD-1511 67	IP-Based Multi-mode Co-Transmission on BS side(NR)	

3 Overview

3.1 Introduction

In a co-transmission network, transmission ports and transport networks are shared, achieving transmission resource sharing.

This document describes the co-transmission networking on the multimode base station controller (MBSC) side and on the multimode base station (MBTS) side.

NOTE

- An MBSC mentioned in this document can be a BSC, a radio network controller (RNC), or a GU MBSC. In addition, the GSM side of an MBSC is referred to as BSC, and the UMTS side of an MBSC is referred to as RNC in this document.
- In this document, the GSM side of an MBTS is referred to as GBTS or eGBTS, the UMTS side of an MBTS is referred to as NodeB, the LTE side of an MBTS is referred to as eNodeB, and the NR side of an MBTS is referred to as gNodeB. The operation and maintenance (O&M) of a GBTS is implemented on a BSC whereas the O&M of an eGBTS, which has an independent O&M channel (OMCH), is not dependent on the BSC.
- In this document, the first of letters describing the RATs of a separate-MPT MBTS indicates the RAT that provides a co-transmission port. For example, in a UG MBTS, UMTS provides a co-transmission port. In a UG+L MBTS, UMTS provides a co-transmission port, UMTS and GSM share one BBU, and LTE uses an independent BBU.

3.2 Application Scenarios

3.2.1 Co-transmission on the MBSC Side (GSM/UMTS)

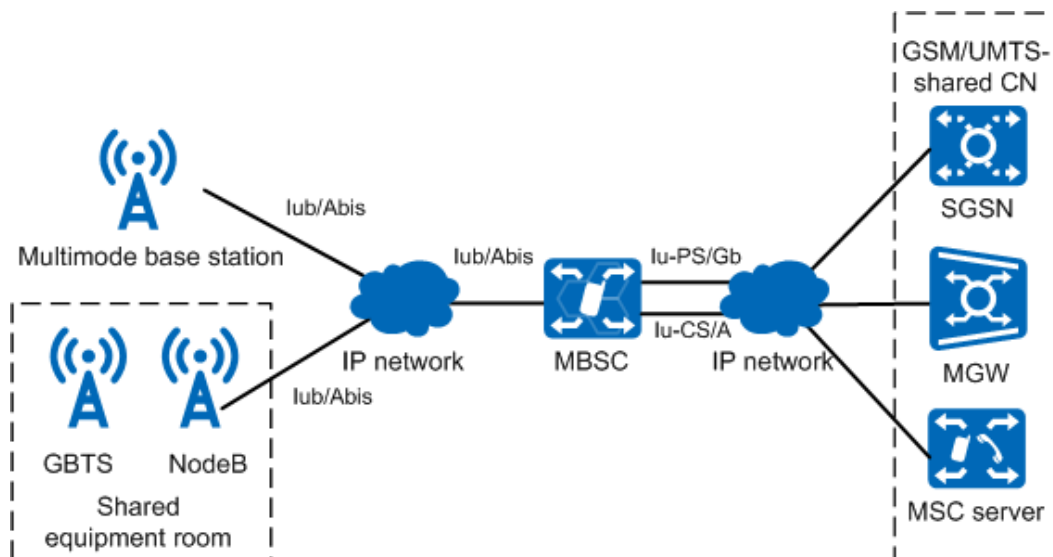
This document describes IPv4-based co-transmission on the MBSC side. IPv6-based co-transmission is not supported.

Figure 3-1 shows a network topology of co-transmission on the MBSC side. GSM and UMTS are deployed on an MBSC, with co-transmission implemented in the following ways:

- On the core network side, the Iu-CS and A interfaces as well as the Iu-PS and Gb interfaces of the MBSC share an IP transport network when GSM and UMTS share the core network (including the MSC server, MGW, and SGSN).

- On the radio access network (RAN) side, the lub and Abis interfaces of the MBSC share an IP transport network.

Figure 3-1 Co-transmission on the MBSC side



For details about co-transmission on the MBSC side, see [4 IP-based BSC and RNC Co-transmission on the MBSC Side \(GSM/UMTS\)](#).

3.2.2 Co-transmission on the MBTS Side

MBTSs are classified into separate-MPT and co-MPT MBTSs based on the configuration of main processing and transmission (MPT) boards.

- A separate-MPT MBTS is a physical base station in which different RATs use different main control boards. Such an MBTS consists of multiple logical base stations, each of which has an independent OMCH.
- A co-MPT MBTS is a physical base station in which different RATs share one main control board. Such an MBTS has only one logical base station and one OMCH.

IP-based co-transmission is supported by separate-MPT and co-MPT MBTSs. MBTSs support both IPv4- and IPv6-based co-transmission. The restrictions on IPv6-based co-transmission are as follows:

- Only LTE and NR support IPv6 transmission.
- IPv6 transmission supports only IPv6 over Ethernet but not IPv6 over E1/T1.
- A UMPT or UMDU must be configured to support IPv6 transmission.
- For IPv6 co-transmission through panel or backplane interconnection, the IP addresses of the lower-level and upper-level base stations cannot be the same when multiple VRFs are configured.

IP-based Co-transmission of a Separate-MPT MBTS

Table 3-1 describes different types of co-transmission for a separate-MPT MBTS based on the interconnection modes between RATs.

Table 3-1 Different types of co-transmission for a separate-MPT MBTS

Co-transmission Type		Co-transmission Description
Co-transmission through panel interconnection	Main-control-board-based co-transmission through panel interconnection	The main control board of one RAT provides an FE/GE port or an E1/T1 port as the co-transmission port to connect to the transport network. The main control boards of all RATs are connected through panel interconnection.
Co-transmission through panel interconnection	UTRP-based co-transmission through panel interconnection	The UTRP of one RAT provides an FE/GE port or an E1/T1 port as the co-transmission port to connect to the transport network. The main control boards of all RATs are connected through panel interconnection or the main control boards of other RATs are connected to the UTRP through panel interconnection.
Co-transmission through backplane interconnection	Main-control-board-based co-transmission through backplane interconnection	The main control board of one RAT provides an FE/GE port as the co-transmission port to connect to the transport network. The main control boards of all RATs are connected through backplane interconnection.
Co-transmission through backplane interconnection	UTRPc-based co-transmission through backplane interconnection	The UTRPc of one RAT provides an FE/GE port as the co-transmission port to connect to the transport network. The main control boards of all RATs are connected to the UTRPc through backplane interconnection.

IP-based Co-transmission of a Co-MPT MBTS

NOTE

The BBU3910A/BBU3910C/BookBBU5901/BookBBU5901a is an outdoor BBU with a built-in UMDU/UMDC board, which integrates the main control, transmission, and baseband processing functions. It can be used only in co-MPT base stations.

Different RATs of a co-MPT MBTS share one main control board and one OMCH. An eGBTS must be used in a co-MPT MBTS for GSM.

The main control board or UTRP of a co-MPT MBTS provides an FE/GE port or an E1/T1 port as the co-transmission port to connect to the transport network.

 NOTE

The co-transmission principle for macro base stations is the same as that for distributed base stations (DBSs). This document uses a DBS as an example to describe co-transmission on the base station side.

If the link between the co-transmission port and the transport network is disconnected, services of all RATs are interrupted.

3.3 Transmission Configuration Model

As network traffic volume increases, network transmission scenarios become increasingly complicated. For example, transmission ports or boards need to be adjusted to support inter-board functions. The traditional transport network configuration is complicated and struggles to meet the requirement for fast deployment on the live network. Therefore, a new transmission configuration model is introduced to decouple transmission configurations from physical devices, thereby simplifying operations.

A transmission configuration model can either be new or old, which is controlled by the **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** parameter.

- When this parameter is set to **OLD**, the old transmission configuration model (hereinafter referred to as old model) is used. In the old model, location information such as cabinet, subrack, and slot numbers are configured, the transmission configuration is bound to physical devices, and IPv4 and IPv6 configuration objects above the IP layer are independent of each other.
- When this parameter is set to **NEW**, the new transmission configuration model (hereinafter referred to as new model) is used. In the new model, the transmission model is decoupled from the device model. That is, the transmission configuration does not contain cabinet, subrack, and slot numbers, and IPv4 and IPv6 configuration objects above the IP layer are combined. This model facilitates the expansion of new transmission functions and requires fewer transmission configuration parameters.

Compared with the old model, the new model incorporates an **INTERFACE MO** to isolate the upper layers from the physical layer. In this way, transmission configuration objects are decoupled from physical devices.

In addition, with the new model, when a transmission link is configured or a transmission board or port is modified, the information such as the cabinet, subrack, and slot numbers as well as the subboard type only needs to be configured for the physical layer and data link layer.

The new model is applicable to base stations excluding GBTs and pico base stations.

4 IP-based BSC and RNC Co-transmission on the MBSC Side (GSM/UMTS)

4.1 Principles

4.1.1 Co-transmission for the lub and Abis Interfaces

4.1.1.1 Network Topologies

When the MBSC works in GU mode, the lub and Abis interfaces share an IP transport network and transmission ports on the interface boards of the MBSC. [Figure 4-1](#) shows the network topology of co-transmission in FE/GE/10GE mode. [Figure 4-2](#) shows the network topology of co-transmission in E1/T1 over OC-3 mode. Co-transmission in FE/GE/10GE mode is recommended because an FE/GE/10GE port has a wider bandwidth than an E1/T1 port.

The MBSC Abis/lub interface board identifies GSM/UMTS data, and sends the data to the corresponding GSM/UMTS service or signaling processing board.

Figure 4-1 Co-transmission in FE/GE/10GE mode for the lub and Abis interfaces on the MBSC side

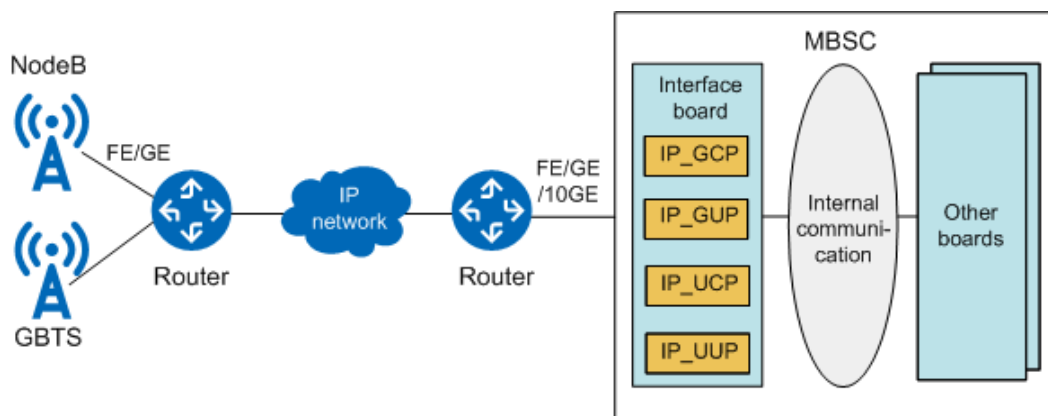
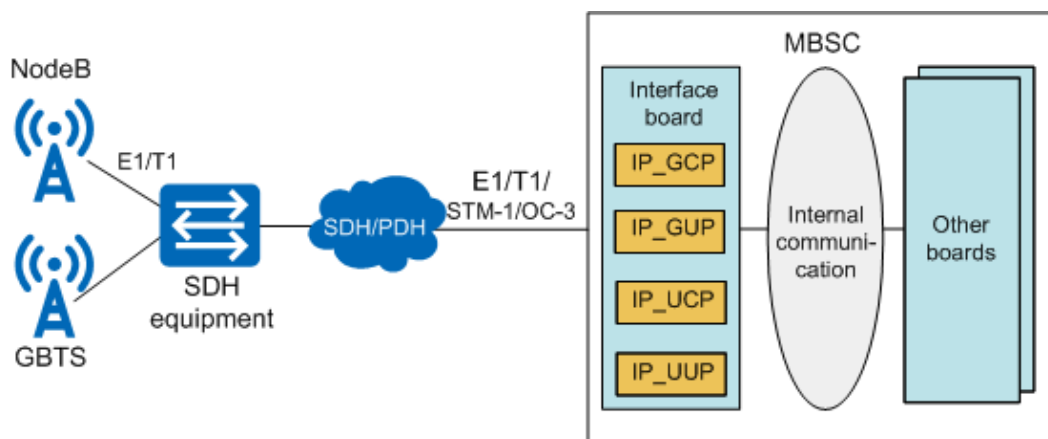


Figure 4-2 Co-transmission in E1/T1 over OC-3 mode for the lub and Abis interfaces on the MBSC side



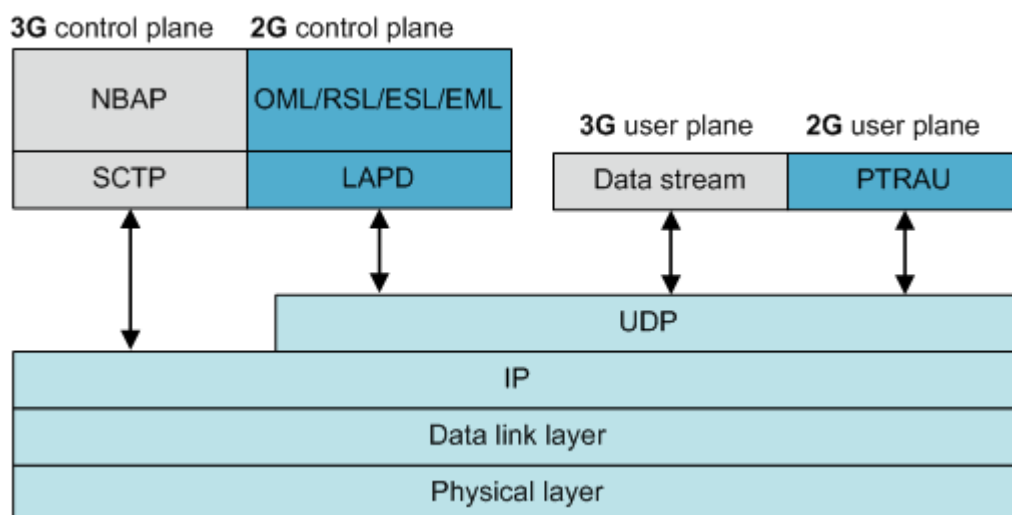
NOTE

- IP_GCP: GSM control-plane IP address
- IP_GUP: GSM user-plane IP address
- IP_UCP: UMTS control-plane IP address
- IP_UUP: UMTS user-plane IP address

4.1.1.2 lub/Abis Co-transmission Protocol Stack

If the GSM base station is a GBTS, the lub and Abis interfaces use the IP-based co-transmission protocol stack shown in [Figure 4-3](#).

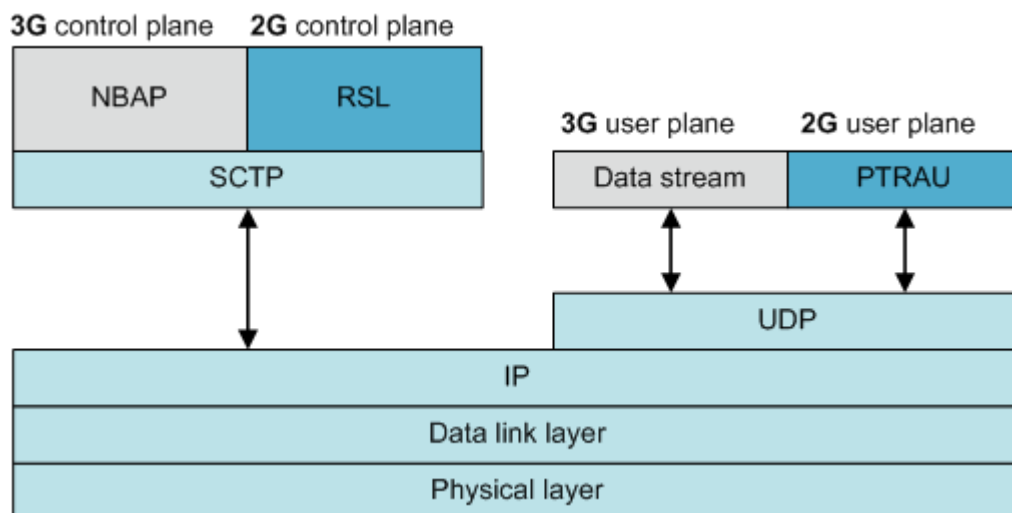
Figure 4-3 IP-based co-transmission protocol stack of the lub and Abis interfaces (1)



As shown in [Figure 4-3](#), the protocols at the physical layer, data link layer, network layer (IP), and transport layer (UDP) are processed by the interface board shared by the lub and Abis interfaces. The protocols at other layers are processed by the control plane processing boards and user plane processing boards of UMTS and GSM separately.

If the GSM base station is an eGBTS, the lub and Abis interfaces use the IP-based co-transmission protocol stack shown in [Figure 4-4](#).

Figure 4-4 IP-based co-transmission protocol stack of the lub and Abis interfaces (2)



For the O&M-plane co-transmission protocol stack of the lub and Abis interfaces, see the descriptions of interface protocol stacks in *IPv4 Transmission*.

4.1.1.3 Protocol Stack Application

Physical Layer

The following types of ports on MBSC interface boards support IP-based co-transmission for the lub and Abis interfaces:

- GE optical ports on a GOUa/GOUc/GOUe/GOUf
- FE/GE electrical ports on an FG2a/FG2c/FG2e
- 10GE optical ports on an EXOUa/EXOUb
- Channelized OC-3 optical ports on a POUa/POUc

Data Link Layer

When co-transmission in FE/GE mode is used for the lub and Abis interfaces, the Ethernet protocol is used at the data link layer.

When co-transmission in E1/T1 over OC-3 mode is used for the lub and Abis interfaces, Point-to-Point Protocol (PPP) or Multi-Link Point-to-Point Protocol (MLPPP) is used at the data link layer. A PPP link can be bound with one to thirty-one 64 kbit/s timeslots, and timeslots of two E1/T1 links cannot be bound. In an MLPPP group, each PPP link is bound with the same number of timeslots (at least eight timeslots). GBTS/eGBTS data and NodeB data can be transmitted over different PPP links or MLPPP groups. Alternatively, GSM data and UMTS data can be transmitted over the same PPP link or MLPPP group and be distinguished by UDP port numbers.

Network Layer

Based on communication types, IP addresses can be classified into port IP addresses and device IP addresses. A port IP address refers to a physical IP address of a port on an interface board. A device IP address refers to a logical IP address of an interface board. Port IP addresses must be configured and the device IP address configuration is optional. The lub and Abis interfaces can share one port IP address or use different port IP addresses. Either port IP addresses or device IP addresses can be used as the control-plane and user-plane IP addresses on the lub and Abis interfaces or OMCH IP addresses. The IP address planning principles are as follows:

- Control-plane and user-plane IP addresses of the lub interface
The control-plane and user-plane IP addresses of the lub interface can be the same or different.
- Control-plane and user-plane IP addresses of the Abis interface
The control-plane and user-plane IP addresses of the Abis interface can be the same or different.
The lub and Abis interfaces can use the same or different control-plane and user-plane IP addresses.
- OMCH IP address
On the lub interface:
 - If the OMCH between the NodeB and the MAE passes through the MBSC, the OMCH IP address must be configured on the NodeB and the MAE, and OM packet forwarding routes must be configured on the MBSC. In this scenario, the MBSC forwards packets between the MAE and the NodeB.
 - If the OMCH between the NodeB and the MAE does not pass through the MBSC, the OMCH IP address must be configured on the NodeB and the MAE.On the Abis interface:
The GBTS IP addresses must be configured on the MBSC. If the BSC is not enabled with the GBFD-118613 BTS Multi-IP Address feature, the OMCH, control-plane, and user-plane IP addresses of the GBTS must be the same. If the BSC is enabled with the GBFD-118613 BTS Multi-IP Address feature, the OMCH IP address of the GBTS can be different from the control-plane and user-plane IP addresses of the GBTS.

4.1.2 Co-transmission for the lu-CS and A Interfaces

4.1.2.1 Network Topologies

When the MBSC works in GU mode, the lu-CS and A interfaces share an IP transport network and transmission ports on the interface boards of the MBSC. **Figure 4-5** shows the network topology of co-transmission in FE/GE mode. **Figure 4-6** shows the network topology of co-transmission in E1/T1 over OC-3 mode. Co-transmission in FE/GE mode is recommended because an FE/GE port has a wider bandwidth than an E1/T1 port.

The MBSC lu-CS/A interface board identifies GSM/UMTS data and sends the data to the corresponding GSM/UMTS service or signaling processing board.

Figure 4-5 Co-transmission in FE/GE mode for the lu-CS and A interfaces

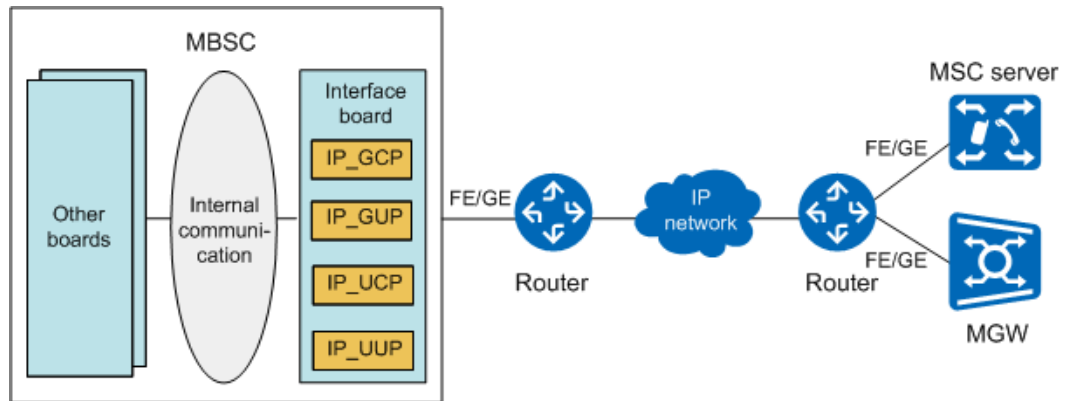
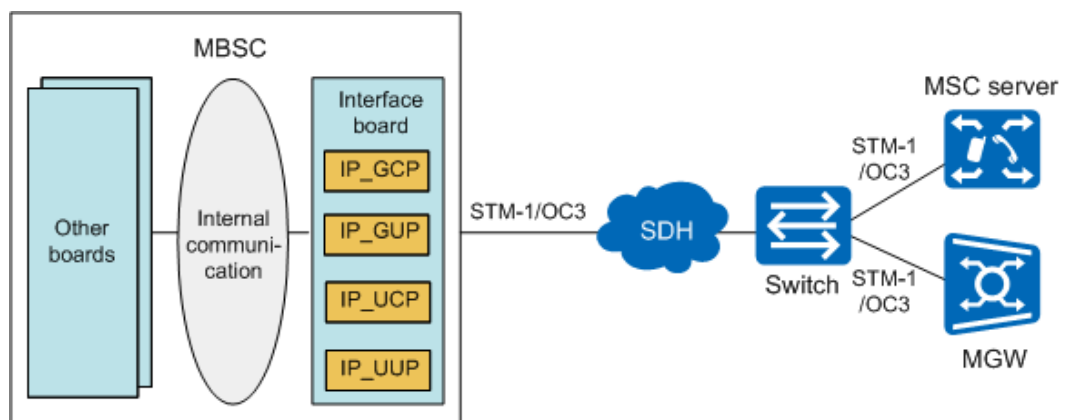


Figure 4-6 Co-transmission in E1/T1 over OC-3 mode for the lu-CS and A interfaces



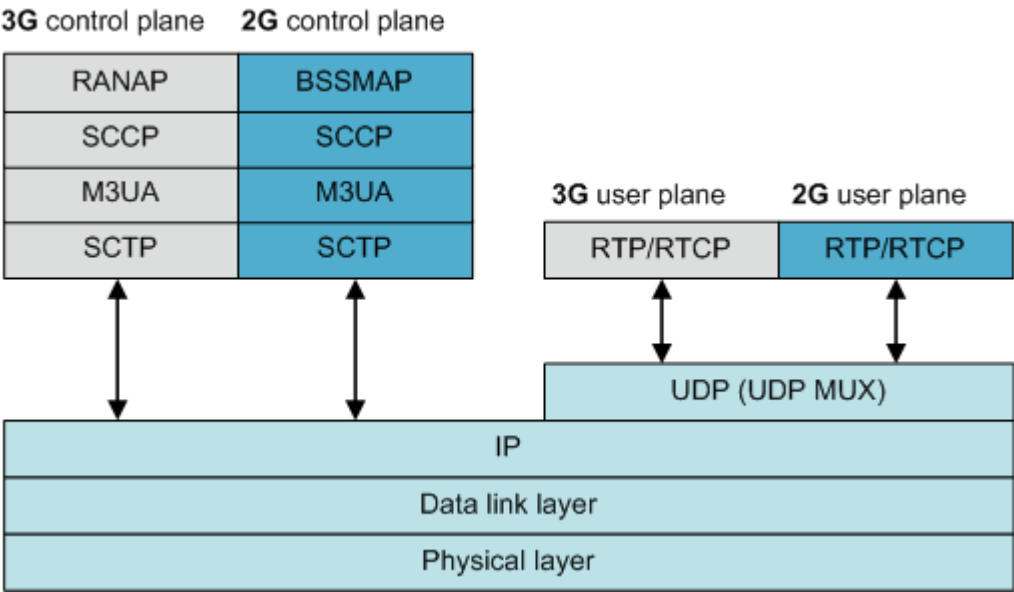
NOTE

- IP_GCP: GSM control-plane IP address
- IP_GUP: GSM user-plane IP address
- IP_UCP: UMTS control-plane IP address
- IP_UUP: UMTS user-plane IP address

4.1.2.2 lu-CS/A Co-transmission Protocol Stack

Figure 4-7 shows the IP-based co-transmission protocol stack of the lu-CS and A interfaces.

Figure 4-7 IP-based co-transmission protocol stack of the lu-CS and A interfaces



As shown in Figure 4-7, the protocols at the physical layer, data link layer, network layer (IP), and transport layer (UDP and UDP MUX) are processed by the interface board shared by the lu-CS and A interfaces. The protocols at other layers are processed by the control plane processing boards and user plane processing boards of UMTS and GSM separately.

4.1.2.3 Protocol Stack Application

Physical Layer

The following types of ports on MBSC interface boards support IP-based co-transmission for the lu-CS and A interfaces:

- GE optical ports on a GOUa/GOUc/GOUe/GOUf
- FE/GE electrical ports on an FG2a/FG2c/FG2e
- 10GE optical ports on an EXOUa/EXOUb
- Channelized OC-3 optical ports on a POUa/POUc

Data Link Layer

When co-transmission in FE/GE mode is used for the lu-CS and A interfaces, the Ethernet protocol is used at the data link layer.

When co-transmission in E1/T1 over OC-3 mode is used for the lu-CS and A interfaces, PPP or MLPPP is used at the data link layer. A PPP link can be bound with one to thirty-one 64 kbit/s timeslots, and timeslots of two E1/T1 links cannot be bound. In an MLPPP group, each PPP link is bound with the same number of timeslots (at least eight timeslots).

Network Layer

Control-plane and user-plane IP addresses of the lu-CS and A interfaces can be either port IP addresses or device IP addresses. The lu-CS and A interfaces can use

the same or different port IP addresses. It is recommended that the lu-CS and A interfaces use different port IP addresses.

The IP address planning principles are as follows:

- Control-plane IP addresses
The lu-CS and A interfaces can use the same control-plane IP address.
Generally, SCTP links are dual-homed. One interface board is configured with two control-plane IP addresses, one IP address used as the primary IP address and the other used as the secondary IP address. The SCTP links over the lu-CS and A interfaces on the MBSC side use the two IP addresses. lu-CS interface signaling and A interface signaling are distinguished by the SCTP port numbers in them. For details about SCTP links on the lu-CS and A interfaces, see *IPv4 Transmission*.
- User-plane IP addresses
The lu-CS and A interfaces use different user-plane IP addresses.
The UDP MUX technique uses different Real-time Transport Protocol (RTP) compression algorithms on the lu-CS and A interfaces. Therefore, different user-plane IP addresses must be planned for the lu-CS and A interfaces to distinguish lu-CS interface data from A interface data.

On the lu-CS interface or A interface, the control-plane IP address can be the same as the user-plane IP address.

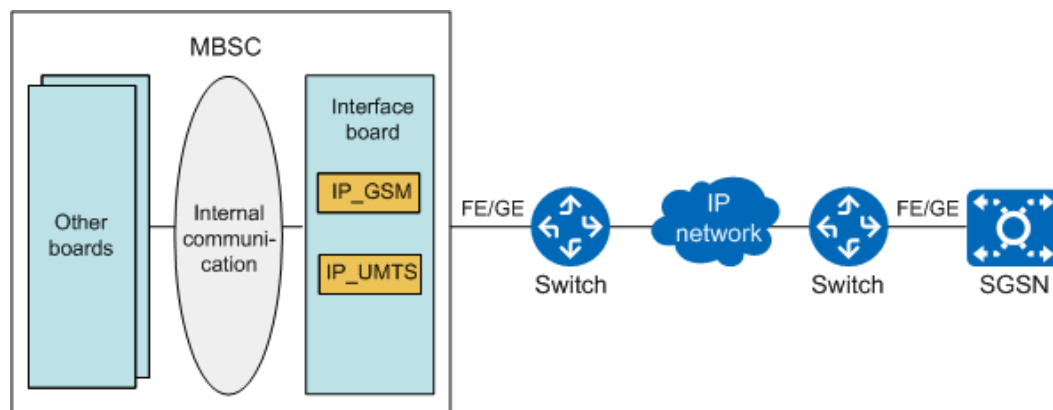
4.1.3 Co-transmission for the lu-PS and Gb Interfaces

4.1.3.1 Network Topologies

When the MBSC works in GU mode, the lu-PS and Gb interfaces share an IP transport network and transmission ports on the interface boards of the MBSC. The Gb interface does not support IP over E1. Therefore, the lu-PS and Gb interfaces can only use co-transmission in FE/GE mode. [Figure 4-8](#) shows the network topology of co-transmission in FE/GE mode for the lu-PS and Gb interfaces.

The MBSC Gb/lu-PS interface board identifies GSM/UMTS data and sends the data to the corresponding GSM/UMTS service or signaling processing board.

Figure 4-8 Co-transmission in FE/GE mode for the lu-PS and Gb interfaces

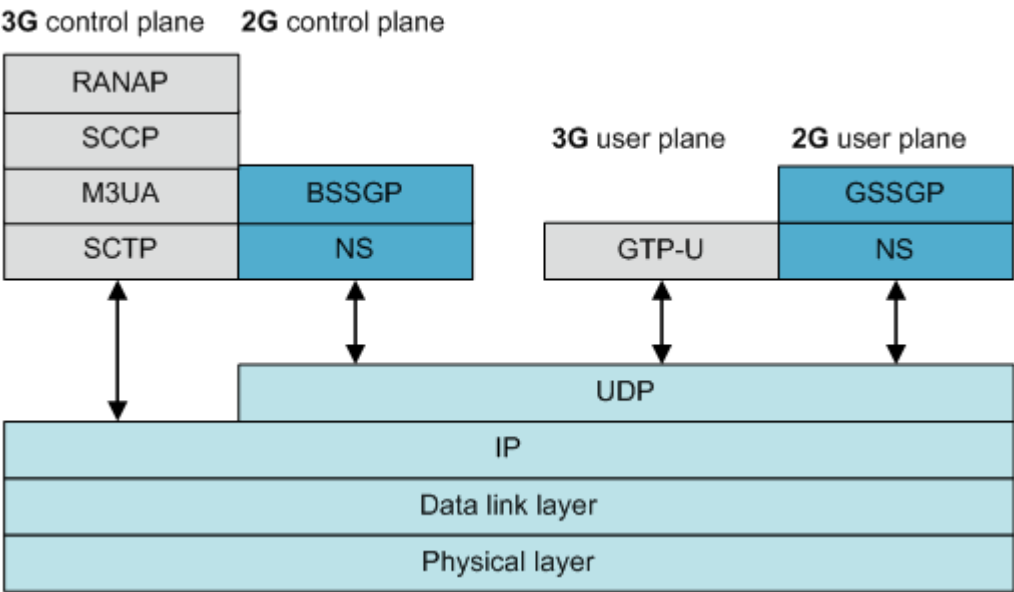


- NOTE
- IP_GSM: GSM IP address
 - IP_UMTS: UMTS IP address

4.1.3.2 lu-PS/Gb Co-transmission Protocol Stack

Figure 4-9 shows the IP-based co-transmission protocol stack of the lu-PS and Gb interfaces.

Figure 4-9 IP-based co-transmission protocol stack of the lu-PS and Gb interfaces



As shown in Figure 4-9, the protocols at the physical layer, data link layer, network layer (IP), and transport layer (UDP) are processed by the interface board shared by the lu-PS and Gb interfaces. The protocols at other layers are processed by the control plane processing boards and user plane processing boards of UMTS and GSM separately.

4.1.3.3 Protocol Stack Application

Physical Layer

The following types of ports on MBSC interface boards support IP-based co-transmission for the lu-PS and Gb interfaces:

- GE optical ports on a GOUa/GOUc/GOUe/GOUf
- FE/GE electrical ports on an FG2a/FG2c/FG2e
- 10GE optical ports on an EXOUa/EXOUb

Data Link Layer

When co-transmission is implemented on the lu-PS and Gb interfaces, the Ethernet protocol is used at the data link layer.

Network Layer

Control-plane and user-plane IP addresses of the lu-PS and Gb interfaces can be either port IP addresses or device IP addresses. The lu-PS and Gb interfaces can use the same or different port IP addresses. It is recommended that the lu-PS and Gb interfaces use different port IP addresses.

- Control-plane IP addresses
Generally, SCTP links on the lu-PS interface are dual-homed. One interface board is configured with two control-plane IP addresses, one IP address used as the primary IP address and the other used as the secondary IP address. For details about SCTP links on the lu-PS interface, see *IPv4 Transmission*.
No control-plane IP address is assigned to the Gb interface.
- User-plane IP addresses
The lu-PS and Gb interfaces can use the same or different IP addresses.
On the lu-PS interface, the tunnel endpoint identifier (TEID) at the GPRS Tunneling Protocol-User Plane (GTP-U) layer is used to identify users. GTP-U only uses UDP port 2152.

On the lu-PS interface, the control-plane IP address can be the same as the user-plane IP address.

4.2 Network Analysis

4.2.1 Benefits

For operators, co-transmission reduces the capital expenditure (CAPEX) and operational expenditure (OPEX) and simplifies operation and maintenance in the following ways:

- Transmission port sharing reduces the number of transmission links.
- Transport network sharing simplifies transmission configuration and maintenance.
- Smooth evolution from GSM to UMTS/LTE/NR can be achieved, reducing transport network adjustments.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
GSM	Gb over IP	None	<i>IPv4 Transmission</i>	This function is required when co-transmission is applied on the Gb interface of GSM.
GSM	Abis over IP	None	<i>IPv4 Transmission</i>	This function is required when co-transmission in FE/GE mode is applied on the Abis interface of GSM.
GSM	Abis IP over E1/T1	None	<i>IPv4 Transmission</i>	This function is required when co-transmission in IP over E1/T1 mode is applied on the Abis interface of GSM.
GSM	A over IP	None	<i>IPv4 Transmission</i>	This function is required when co-transmission in FE/GE mode is applied on the A interface of GSM.

RAT	Function Name	Function Switch	Reference	Description
GSM	A IP over E1/T1	None	<i>IPv4 Transmission</i>	This function is required when co-transmission in IP over E1/T1 mode is applied on the A interface of GSM.
UMTS	IP Transmission Introduction on lub Interface	None	<i>IPv4 Transmission</i>	This function is required when co-transmission is applied on the lub interface of UMTS.
UMTS	IP Transmission Introduction on lu Interface	None	<i>IPv4 Transmission</i>	This function is required when co-transmission is applied on the lu interface of UMTS.

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

The MBSC supports lub/Abis co-transmission on the following interface boards: GOUa/GOUc/GOUe/GOUf/FG2a/FG2c/FG2e/EXOUa/EXOUb/POUa/POUc.

The MBSC supports lu-CS/A co-transmission on the following interface boards: GOUa/GOUc/GOUe/GOUf/FG2a/FG2c/FG2e/EXOUa/EXOUb/POUa/POUc.

The MBSC supports Iu-PS/Gb co-transmission on the following interface boards:
GOUa/GOUc/GOUe/GOUf/FG2a/FG2c/FG2e/EXOUa/EXOUb.

RF Modules

This function does not depend on RF modules.

NEs

The BSC6960 cannot be used.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 When to Use

It is recommended that co-transmission for the Iub and Abis interfaces in IP over FE/GE mode be enabled on the MBSC side if a BSC is reconstructed from single-mode to multimode. This section describes the engineering guidelines for IP-based co-transmission for Iub and Abis interfaces on the MBSC side.

4.4.2 Precautions

No additional hardware is required. You only need to connect the FE/GE or E1/T1 port on the Abis/Iub interface board to the intermediate transmission equipment. When co-transmission in IP over FE/GE mode is used, connect the FE/GE port on the Abis/Iub interface board to the next-hop router. When co-transmission in IP over E1/T1 mode is used, connect the Abis/Iub interface board to the intermediate synchronous digital hierarchy (SDH)/plesiochronous digital hierarchy (PDH) transmission equipment.

4.4.3 Data Configuration

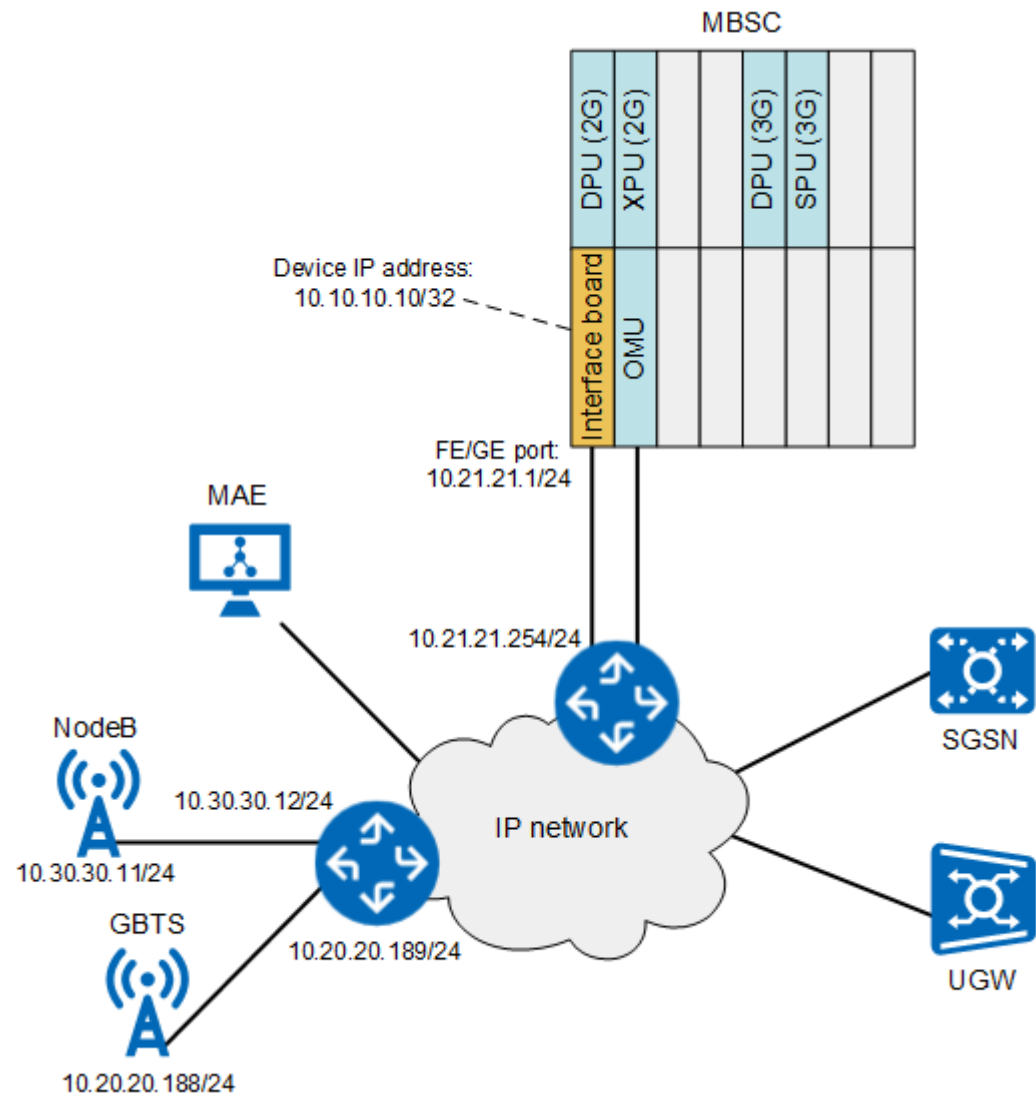
4.4.3.1 Data Preparation

This section describes the data to be prepared for co-transmission for the Iub and Abis interfaces on the MBSC side in IP over FE/GE and IP over E1/T1 modes.

Co-transmission for the Iub and Abis Interfaces on the MBSC Side in IP over FE/GE Mode

Figure 4-10 shows an example of network topology, and **Table 4-1** lists the MBSC, GBTS/eGBTS, and NodeB IP address plan.

Figure 4-10 Example of network topology for co-transmission for the lub and Abis interfaces on the MBSC side in IP over FE/GE mode



NOTE

Data configurations at the physical layer, such as the duplex mode and rate in FE/GE mode, must be consistent between the MBSC and the next-hop port.

Table 4-1 Data plan

Item	Instance	Remarks
Device IP address of the MBSC	10.10.10.10/32	Logical IP addresses of the lub and Abis interface boards on the MBSC side. The logical IP address of the lub interface board on the RNC side and the logical IP address of the Abis interface board on the BSC side must be configured on the same board. This table assumes that the logical IP address of the lub interface board on the RNC side is the same as that of the Abis interface board on the BSC side.
Port IP address of the MBSC	10.21.21.1/24	Physical IP addresses of the lub and Abis interface boards on the MBSC side. The port IP address of the Abis/lub co-transmission interface board on the MBSC side must be on the same network segment as the IP address of the port on the next-hop router in IP over FE/GE mode. If the MBSC is directly connected to a base station, the port IP address of the co-transmission interface board must be on the same network segment as the port IP address of the base station. The physical IP address of the lub interface board on the RNC side and the physical IP address of the Abis interface board on the BSC side must be configured on the same port of the same board. This table assumes that the physical IP address of the lub interface board on the RNC side is the same as that of the Abis interface board on the BSC side.
IP address of the port on the router that is directly connected to the MBSC	10.21.21.254/24	None
Port IP address of the NodeB	10.30.30.11/24	None
Port IP address of the GBTS/eGBTS	10.20.20.188/24	None

Item	Instance	Remarks
IP address of the port on the router that is directly connected to the NodeB	10.30.30.12/24	None
IP address of the port on the router that is directly connected to the GBTS/eGBTS	10.20.20.189/24	None

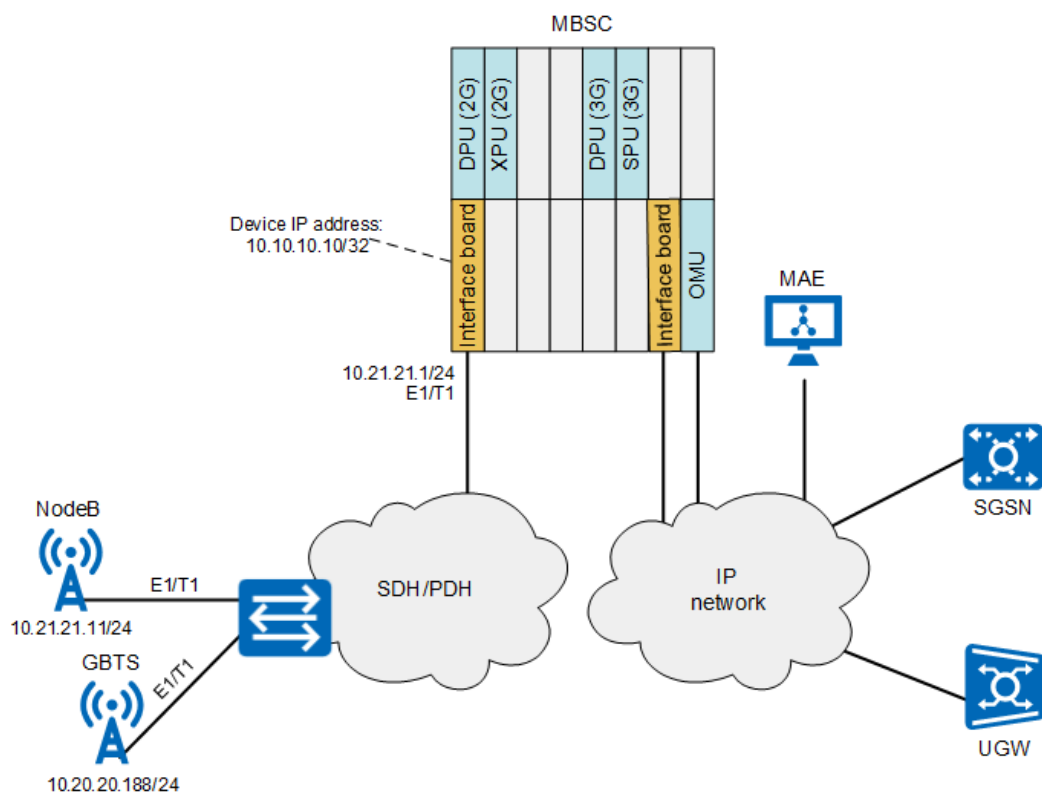


Table 4-2 Data plan

Item	Instance	Remarks
Device IP address of the MBSC	10.10.10.10/32	Logical IP addresses of the lub and Abis interface boards on the MBSC side
Port IP address of the MBSC	10.21.21.1/24	Physical IP addresses of the lub and Abis interface boards on the MBSC side
Port IP address of the NodeB	10.21.21.11/24	The MBSC terminates messages that are compliant with the Point-to-Point Protocol (PPP)/Multi-Link Point-to-Point Protocol (MLPPP) from the NodeB or GBTS/eGBTS. Therefore, the port IP addresses of the MBSC, NodeB, and GBTS/eGBTS must be on the same network segment.
Port IP address of the GBTS/eGBTS	10.21.21.188/24	

4.4.3.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

For details about data configurations at the data link layer and network layer on the BSC side, see "Configuring a GBTS and Its Cells" > "Configuring Transmission Data over the Abis Interface" in *BSC6900 GU Initial Configuration Guide* of *BSC6900 GU Product Documentation* or *BSC6910 GU Initial Configuration Guide* of *BSC6910 GU Product Documentation*.

For details about data configurations at the data link layer and network layer on the RNC side, see "Configuring the UMTS Interfaces" > "Configuring the lub Interface (over IP)" in *BSC6900 GU Initial Configuration Guide* of *BSC6900 GU Product Documentation*. Compared with independent transmission, co-transmission has no configuration change. Only the following information needs to be noted: The physical IP addresses of the BSC and the RNC must be configured on the same port of the same transmission board. The logical IP addresses of the Abis and lub interfaces for the BSC and RNC must be configured on the same transmission board. According to the data plan, port IP addresses or device IP addresses of the BSC and RNC can be the same or different.

4.4.3.3 Using the MAE-Deployment

On the MAE-Deployment, set the parameters based on the data plan listed in [4.4.3.1 Data Preparation](#).

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.4 Activation Verification

Co-transmission in IP over FE/GE Mode

Run the MBSC MML command **PING IP** to ping the port IP address of the GBTS/eGBTS/NodeB. If the port IP address can be pinged, the transmission link between the MBSC and the GBTS/eGBTS/NodeB is normal.

```
PING IP: SIPADDR="10.10.10.10", DESTIP="10.30.30.11", CONTPING=NO;  
PING IP: SIPADDR="10.10.10.10", DESTIP="10.20.20.188", CONTPING=NO;
```

Co-transmission in IP over E1/T1 Mode

Run the MBSC MML command **PING IP** to ping the port IP address of the GBTS/eGBTS/NodeB. If the port IP address can be pinged, the transmission link between the MBSC and the GBTS/eGBTS/NodeB is normal.

```
PING IP: SIPADDR="10.21.21.1", DESTIP="10.21.21.188", CONTPING=NO;  
PING IP: SIPADDR="10.21.21.1", DESTIP="10.21.21.11", CONTPING=NO;
```

4.4.5 Network Monitoring

None

5 IP-Based Multimode Co-transmission on the Base Station Side

5.1 Principles

5.1.1 Overview

Introduction to Main Control Boards and UTRPs

Co-transmission ports can be provided by main control boards or UTRPs of an MBTS. For the descriptions of transmission ports on the main control boards and UTRPs, see the hardware description of BBUs in *3900 & 5900 Series Base Station Product Documentation*.

Co-transmission Networking Scenarios

For co-transmission on the separate-MPT MBTS side, it is recommended that a board having higher processing capabilities provide the co-transmission port. If the boards of different RATs have the same processing capabilities, it is recommended that the inventory board provide the co-transmission port. The co-transmission solution is the same in NSA and SA networking scenarios.

NOTE

In the scenarios of separate-MPT MBTS co-transmission through backplane interconnection, the base station of a RAT that does not provide the co-transmission port cannot be connected to the transport network through a port on a board panel.

Due to the capability limitation of the GTMUc/UMPT board, the maximum MTU supported by the tunnel of the GTMUc board is 1476 when the GTMUc/UMPT board implements co-transmission through backplane interconnection.

5.1.2 IP-based Co-transmission

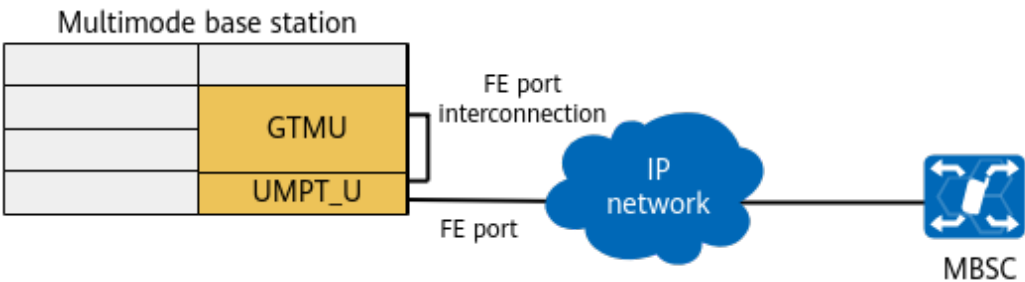
5.1.2.1 Main-Control-Board-based Co-transmission Through Panel Interconnection on the Separate-MPT MBTS Side

For co-transmission of a separate-MPT MBTS, the co-transmission port is provided by the NR, LTE, UMTS, or GSM in descending order of priority. This section uses a separate-MPT GU/GL MBTS as an example to describe main-control-board-based IP co-transmission through panel interconnection.

Co-transmission in IP over FE/GE Mode

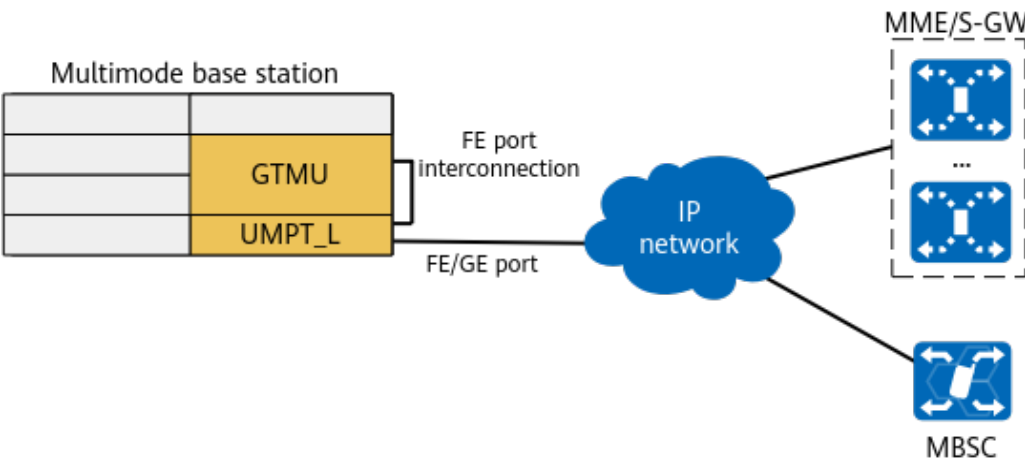
In IP over FE/GE mode, the main control boards of different RATs are interconnected through FE/GE ports, and the main control board of a RAT provides an FE/GE port as the co-transmission port to connect to the IP transport network. [Figure 5-1](#) and [Figure 5-2](#) show the main-control-board-based co-transmission through panel interconnection on the separate-MPT GU/GL MBTS sides in IP over FE/GE mode.

Figure 5-1 Main-control-board-based co-transmission through panel interconnection on the separate-MPT GU MBTS side in IP over FE mode



As shown in [Figure 5-1](#), the UMPT_U of the NodeB provides an FE port as the co-transmission port, and the GTMU is connected to the UMPT_U through panel interconnection. In this case, uplink data of the GBTS/eGBTS is transmitted from the GTMU to the UMPT_U. The downlink data to the GBTS/eGBTS is transmitted from the UMPT_U to the GTMU.

Figure 5-2 Main-control-board-based co-transmission through panel interconnection on the separate-MPT GL MBTS side in IP over FE/GE mode

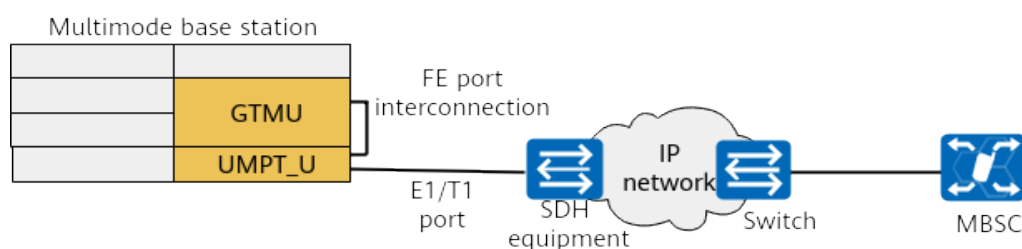


As shown in [Figure 5-2](#), the UMPT_L of the eNodeB provides an FE/GE port as the co-transmission port, and the GTMU is connected to the UMPT_L through panel interconnection. In this case, uplink data of the GBTS/eGBTS is transmitted from the GTMU to the UMPT_L. The downlink data to the GBTS/eGBTS is transmitted from the UMPT_L to the GTMU.

Co-transmission in IP over E1/T1 Mode

[Figure 5-3](#) shows main-control-board-based co-transmission through panel interconnection on the separate-MPT GU MBTS side in IP over E1/T1 mode. NR does not support co-transmission in IP over E1/T1 mode.

Figure 5-3 Main-control-board-based co-transmission through panel interconnection on the separate-MPT GU MBTS side in IP over E1/T1 mode



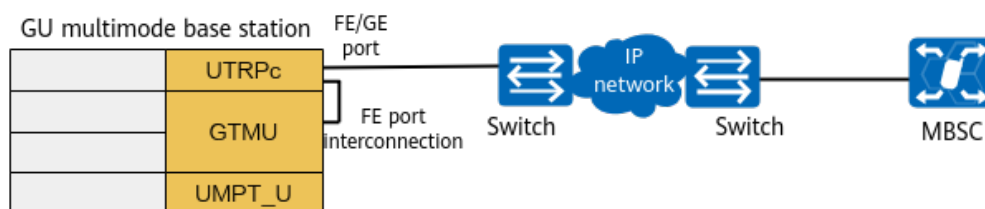
As shown in [Figure 5-3](#), the UMPT_U provides an E1/T1 port as the co-transmission port, and the GTMU is connected to the UMPT_U through panel interconnection. The UMPT_U uses MLPPP to bind E1/T1 links. In this case, uplink data of the GBTS/eGBTS is transmitted from the GTMU to the UMPT_U. The downlink data to the GBTS/eGBTS is transmitted from the UMPT_U to the GTMU.

5.1.2.2 UTRP-based Co-transmission Through Panel Interconnection on the Separate-MPT MBTS Side

UTRP-based co-transmission through panel interconnection on the separate-MPT MBTS is supported only in IP over FE/GE mode. UTRP-based co-transmission through panel interconnection is not supported in NR.

This section uses a separate-MPT GU MBTS as an example to describe UTRPc-based co-transmission through panel interconnection in IP over FE/GE mode. [Figure 5-4](#) shows UTRPc-based co-transmission through panel interconnection on the separate-MPT GU MBTS side in IP over FE/GE mode.

Figure 5-4 UTRPc-based co-transmission through panel interconnection on the GU MBTS side in IP over FE/GE mode



As shown in [Figure 5-4](#), the UTRPc managed by the NodeB provides an FE/GE port as the co-transmission port, and the GTMU is connected to the UTRPc

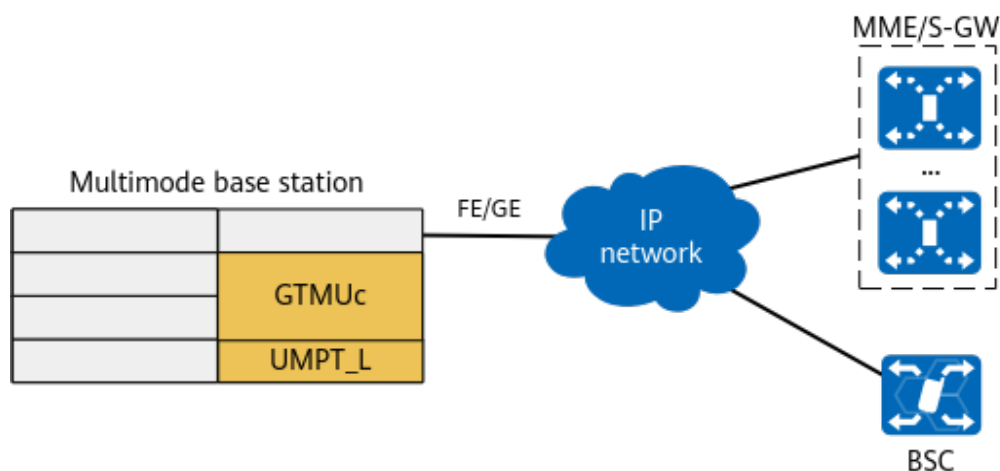
through panel interconnection. In this case, uplink data of the GBTS/eGBTS is transmitted from the GTMU to the UTRPc. The downlink data to the GBTS/eGBTS is transmitted from the UTRPc to the GTMU.

5.1.2.3 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT MBTS Side

Main-control-board-based co-transmission through backplane interconnection is supported by separate-MPT MBTSs. The main control boards provide a co-transmission port in ascending order of priority as follows: GTMU < UMPT. If LTE and NR coexist and the UMPT_L/UMPT_T/UMPT_M is a UMPTe/UMPTg/UMPTga/UMPTTha/UMPTThb board, it is recommended that a high-performance UMPT be preferentially used to provide a co-transmission port. Otherwise, it is recommended that the UMPT_N provide a co-transmission port.

Figure 5-5 uses the separate-MPT GL MBTS as an example to show the main-control-board-based co-transmission through backplane interconnection between the GTMUc and the UMPT_L.

Figure 5-5 Main-control-board-based co-transmission through backplane interconnection on the separate-MPT GL MBTS side



As shown in **Figure 5-5**, the GTMUc and the UMPT_L communicate with each other through the backplane. In this case, uplink data of the GBTS/eGBTS is transmitted from the GTMUc to the UMPT_L, and downlink data to the GBTS/eGBTS is transmitted from the UMPT_L to the GTMUc.

When the main control boards of different RATs communicate through the backplane, each main control board must be configured with a tunnel to the peer main control board. The related MML commands are as follows:

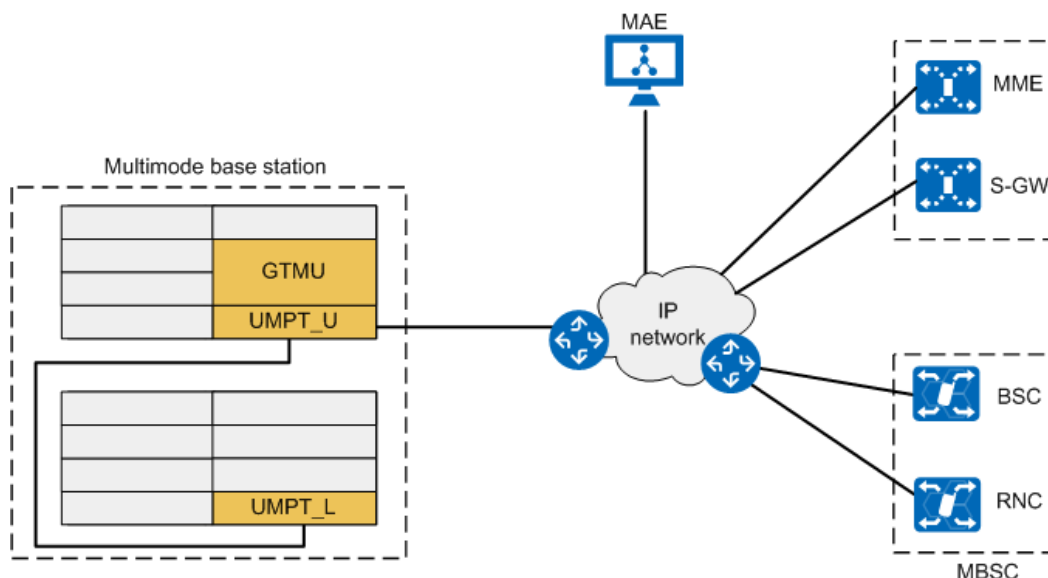
- GBTS: **ADD BTSTUNNEL**
- When the old transmission configuration model is used:
eGBTS/NodeB/eNodeB/gNodeB: **ADD TUNNEL**
- When the new transmission configuration model is used:
eGBTS/NodeB/eNodeB/gNodeB: **ADD BACKTUNNEL**

Each tunnel is managed only by the RAT of the main control board on which the tunnel is configured. Tunnels of the same RAT must each have a unique ID, and tunnels of different RATs can have the same ID. The corresponding parameters are as follows:

- GBTS: **BTSTUNNEL.7N**
- When the old transmission configuration model is used:
eGBTS/NodeB/eNodeB/gNodeB: **TUNNEL.TUNNELID (5G gNodeB, LTE eNodeB)**
- When the new transmission configuration model is used:
eGBTS/NodeB/eNodeB/gNodeB: **BACKTUNNEL.PORTID (5G gNodeB, LTE eNodeB)**

When a separate-MPT GUL/GUT/GLN/GTN/ULN/UTN/GULN/GUTN MBTS uses main-control-board-based IP co-transmission through backplane interconnection, two BBUs must be configured to implement inter-BBU interconnection. For details about inter-BBU interconnection, see *BBU Interconnection* in *3900 & 5900 Series Base Station Product Documentation*. **Figure 5-6** uses a separate-MPT GUL MBTS (with two BBUs interconnected in UMPT+UMPT interconnection mode) as an example to show the main-control-board-based IP co-transmission through backplane interconnection.

Figure 5-6 Main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT UG+L MBTS side



 NOTE

- When a RAT is served by two interconnected BBUs, the primary BBU must provide a co-transmission port.
- In scenarios of main-control-board-based co-transmission through backplane interconnection on the separate-MPT LN MBTS,
 - When the **gNodeBParam.X2uTransmissionType** parameter is set to **ROUTING_TRANS**, **Tunnel Type** (which is specified by the **TUNNEL.TUNNELTYPE** (old model)/**BACKTUNNEL.TUNNELTYPE** (new model) parameter) can be set to **NON_SHAPING** and **SHAPING** for both RATs.
 - When the **gNodeBParam.X2uTransmissionType** parameter is set to **INTRA_TRANS**, it is recommended that **Tunnel Type** be set to **NON_SHAPING** for the RAT that provides the co-transmission port. **Tunnel Type** can be set to **NON_SHAPING** or **SHAPING** for the RAT that does not provide the co-transmission port. It is not recommended that the corresponding **USERPLANEPEER** and **UPPEER2EPGRP** MOs be set manually. Otherwise, the base station reports ALM-26245 Configuration Data Inconsistency.
 - When VRF is deployed for the RAT that does not provide the co-transmission port, the value of the **BACKTUNNEL.TUNNELNO (5G gNodeB, LTE eNodeB)** parameter configured for the RAT that does not provide the co-transmission port must be the same as the value of the **BACKTUNNEL.TUNNELNO (LTE eNodeB, 5G gNodeB)** parameter configured for the RAT that provides the co-transmission port. In this scenario, only the new model and UMPT series main control boards are supported.
- In scenarios of main-control-board-based co-transmission through backplane interconnection on the non-LN separate-MPT MBTS,
 - **Tunnel Type** can be set to **NON_SHAPING** or **SHAPING** for the RAT that does not provide the co-transmission port. It is recommended that **Tunnel Type** be set to **NON_SHAPING** for the RAT that provides the co-transmission port.
 - When the Address Resolution Protocol (ARP) proxy or Neighbor Discovery (ND) proxy function is enabled on the transmission port of the RAT that provides the co-transmission port, the route to the service address of the RAT that does not provide the co-transmission port must be configured so that the transmission device can resolve the service address of the RAT that does not provide the co-transmission port using the ARP proxy or ND proxy function.

In scenarios of IPv6 co-transmission on the separate-MPT MBTS, the backplane tunnel of the RAT that does not provide the co-transmission port and the co-transmission port support ACL packet filtering to prevent attacks from invalid packets. For details, see *Equipment Security*.

Run the **ADD PACKETFILTERING** command with **PACKETFILTERING.ACLTYPE (LTE eNodeB, 5G gNodeB)** set to **IPV6**, **PACKETFILTERING.PT (LTE eNodeB, 5G gNodeB)** set to **INTERFACE**, and **PACKETFILTERING.PORTID (LTE eNodeB, 5G gNodeB)** set to the port ID of the backplane tunnel of the RAT that does not provide the co-transmission port to bind a packet filtering policy to the backplane tunnel.

In separate-MPT scenarios, IPsec cannot be configured for the backplane tunnel of the RAT that does not provide the co-transmission port and IPsec is recommended for the co-transmission port.

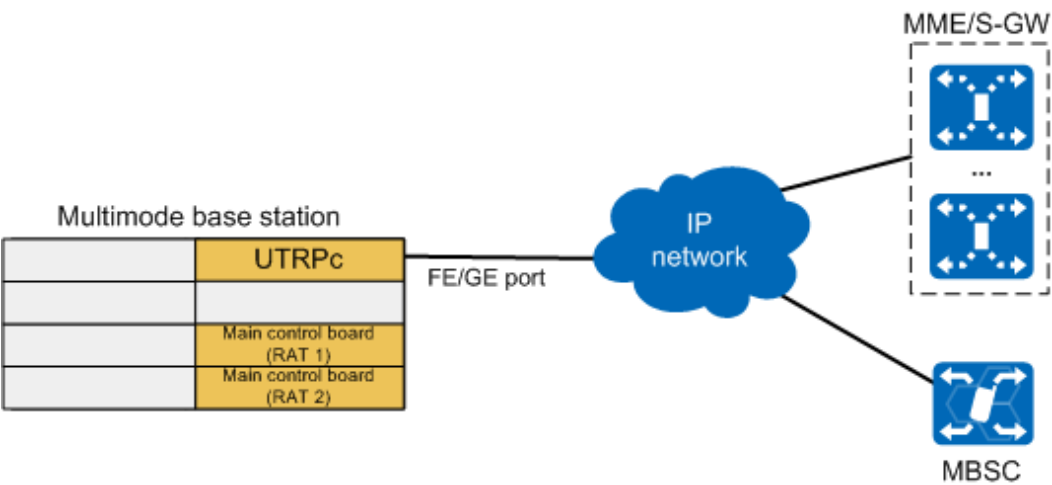
5.1.2.4 UTRPc-based Co-transmission Through Backplane Interconnection on the Separate-MPT MBTS Side

When a UTRPc is used for co-transmission, the UTRPc forwards data for multiple RATs but is managed by only one RAT. The management includes software

management, hardware management, and configuration management. The RAT that manages the UTRPc is called the managing RAT, and other RATs are called non-managing RATs. If an MBTS has more than one BBU, the UTRPc must be managed by the main control board that is located in the same BBU with the UTRPc. In a newly deployed site, it is recommended that UMTS be preferentially used as the managing RAT, then LTE, and the last GSM. In the scenarios of network reconstruction for co-transmission, the managing RAT that has minimum impact on the services in the live network is selected. NR does not support UTRPc-based co-transmission through backplane interconnection.

Figure 5-7 shows the UTRPc-based co-transmission through backplane interconnection on the separate-MPT GU/UL MBTS side.

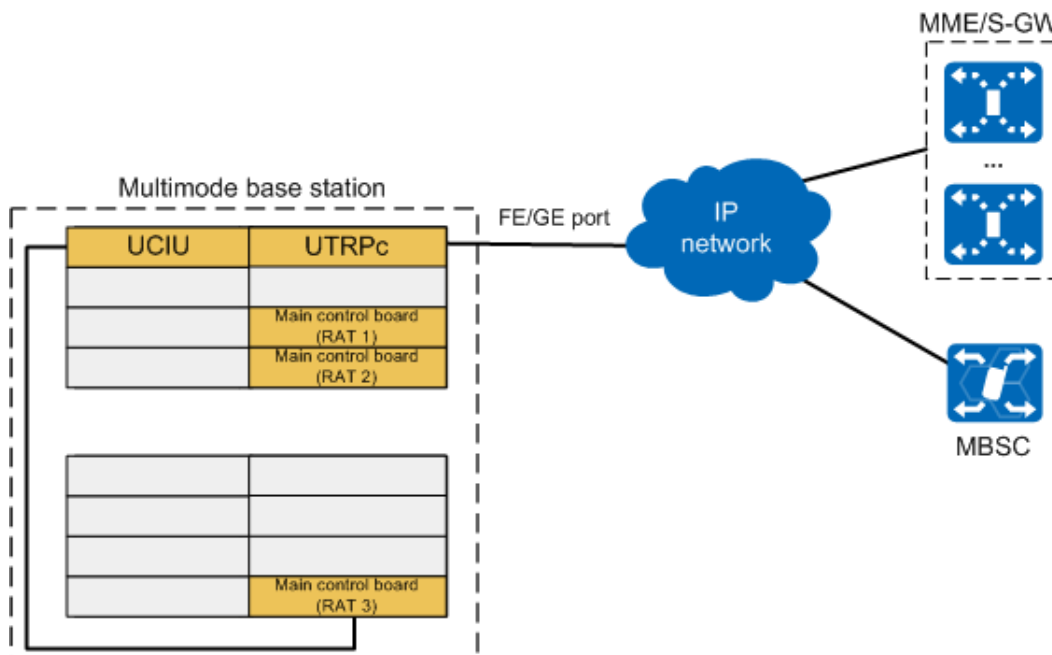
Figure 5-7 UTRPc-based co-transmission through backplane interconnection on the separate-MPT GU/UL MBTS side



As shown in Figure 5-7, the UTRPc provides an FE/GE port as the co-transmission port. The main control board of each RAT communicates with the UTRPc through the backplane. The data of the managing RAT is transmitted directly between the baseband processing unit and the UTRPc. Both the main control board of the non-managing RAT and the UTRPc must be configured with a tunnel to each other. The data of the non-managing RAT is transmitted between the baseband processing unit and the UTRPc through the main control board of the non-managing RAT.

In the UTRPc-based IP co-transmission through backplane interconnection on the separate-MPT GUL/GUT/GUM MBTS side, two BBUs are required and they must be connected through a Universal Cascading Interface Unit (UCIU). For details about inter-BBU interconnection, see *BBU Interconnection* in *3900 & 5900 Series Base Station Product Documentation*. Figure 5-8 shows UTRPc-based IP co-transmission through backplane interconnection on the separate-MPT GUL MBTS side as an example.

Figure 5-8 UTRPc-based IP co-transmission through backplane interconnection on the separate-MPT GUL MBTS side



As shown in [Figure 5-8](#), the UTRPc provides the co-transmission port and the main control board of RAT 3 (UMPT) is connected to the UCIU through the CI port. The UCIU is managed by RAT 1 or RAT 2 that is in the same BBU of the UCIU. It is recommended that GSM be preferentially used as the managing RAT of the UCIU, then UMTS, and the last LTE. Service data of RAT 1 and RAT 2 is directly transmitted between the UTRPc and the board of each RAT. Service data of RAT 3 is transmitted between the UTRPc and the UMPT through the UCIU.

NOTE

- During the software upgrade, cold patch installation, or main control board replacement of the managing RAT, the UTRPc is reset and ongoing services of all RATs are interrupted. During hot patch installation on the main control board of the managing RAT, services on other RATs are not affected.
- If the main control boards of the managing and non-managing RATs are installed in the same BBU, resetting the main control board of the managing RAT does not affect ongoing services of non-managing RATs. If the main control boards of the managing and non-managing RATs are installed in different BBUs, resetting the main control board of the managing RAT affects ongoing services of non-managing RATs.
- The UTRPc is invisible to the non-managing RATs. The software upgrade, cold patch installation, or main control board replacement of the non-managing RAT does not affect the UTRPc. Therefore, ongoing services on other RATs are not affected.

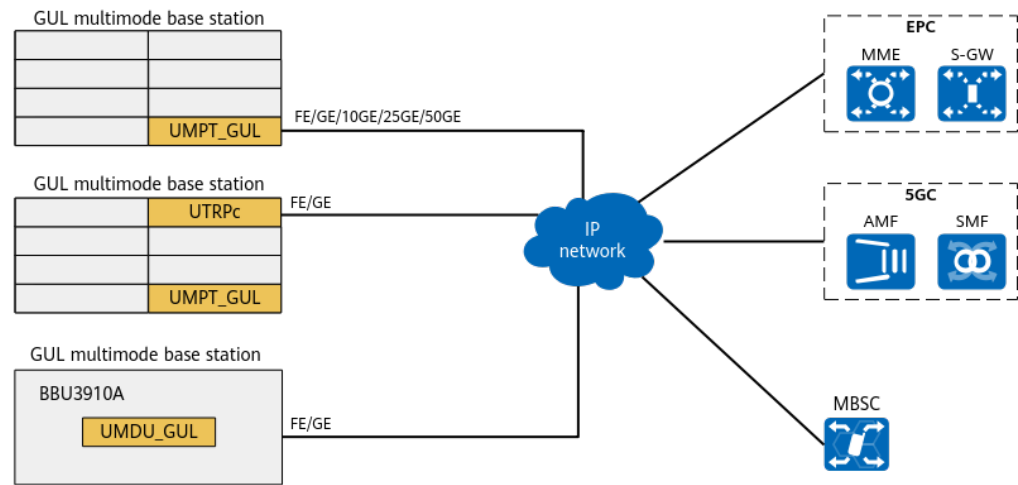
5.1.2.5 Co-transmission on the Co-MPT MBTS Side

Co-transmission in IP over FE/GE/10GE/25GE/50GE Mode

In IP over FE/GE/10GE/25GE/50GE mode, the UMPT board of a co-MPT MBTS provides an FE/GE/10GE/25GE/50GE port as the co-transmission port, or the UMDU/UMDC board of a co-MPT MBTS provides an FE/GE port as the co-transmission port. The UTRPc board can also provide an FE/GE port as the co-

transmission port. The implementation of co-MPT MBTSs is the same. **Figure 5-9** uses a co-MPT GUL MBTS as an example to show the co-transmission on the co-MPT MBTS side in IP over FE/GE mode.

Figure 5-9 Co-transmission on the co-MPT GUL MBTS side in IP over FE/GE/10GE/25GE/50GE mode



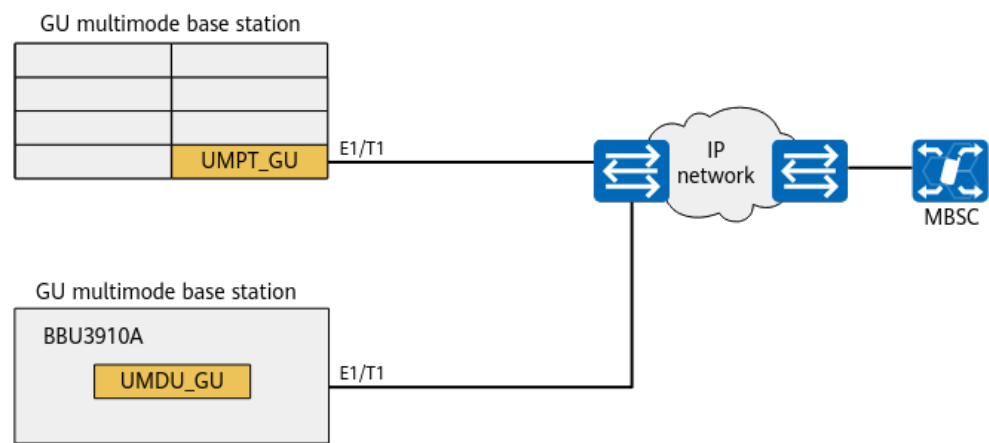
NOTE

If the co-transmission on the co-MPT MBTS side is in IP over FE/GE/10GE/25GE/50GE mode, multiple RATs use the same OM IP address but different service IP addresses.

Co-transmission in IP over E1/T1 Mode

Only a co-MPT GU MBTS supports co-transmission in IP over E1/T1 mode. In IP over E1/T1 mode, the UMPG_GU or UMDU_GU/UMDC_GU board of a co-MPT GU MBTS provides an E1/T1 port as the co-transmission port to connect to the transport network. **Figure 5-10** shows co-transmission on the co-MPT GU MBTS side in IP over E1/T1 mode.

Figure 5-10 Co-transmission on the co-MPT GU MBTS side in IP over E1/T1 mode



5.2 Network Analysis

5.2.1 Benefits

For operators, co-transmission reduces the CAPEX and OPEX and simplifies operation and maintenance in the following ways:

- Transmission port sharing reduces the number of transmission links.
- Transport network sharing simplifies transmission configuration and maintenance.
- Smooth evolution from GSM to UMTS/LTE/NR can be achieved, reducing transport network adjustments.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

This function has no impact on the network.

5.3 Requirements

5.3.1 Licenses

None

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

- MRFD-121114 IP-Based Multi-mode Co-Transmission on BS side(GBTS)

RAT	Function Name	Function Switch	Reference
GSM	Abis over IP	None	<i>IPv4 Transmission</i>

- MRFD-121124 IP-Based Multi-mode Co-Transmission on BS side(NodeB)

RAT	Function Name	Function Switch	Reference	Description
UMTS	IP Transmission Introduction on lub Interface	None	<i>IPv4 Transmission</i>	This function is required when co-transmission in FE/GE mode is applied on the lub interface.
UMTS	Fractional ATM Function on lub Interface	None	<i>ATM Transport</i> in RAN feature documentation	This function is required when co-transmission in ATM mode is applied on the lub interface.
UMTS	Fractional IP Function on lub Interface	None	<i>IPv4 Transmission</i>	This function is required when co-transmission in IP over E1/T1 mode is applied on the lub interface.

- MRFD-121134 IP-Based Multi-mode Co-Transmission on BS side(eNodeB)
None
- MRFD-121144 IP-Based Multi-mode Co-Transmission on BS side(LTE TDD)
None
- MRFD-121154 IP-Based Multi-mode Co-Transmission on BS side(NB-IoT)
None
- MRFD-151167 IP-Based Multi-mode Co-Transmission on BS side(NR)
None

5.3.2.2 Mutually Exclusive Functions

- MRFD-121144 IP-Based Multi-mode Co-Transmission on BS side(LTE TDD)

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE TDD	Virtual Routing and Forwarding	None	<i>VRF</i>

- Others: none

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Source-based IPv6 routing	None	<i>IPv6 Transmission</i>	None
NR FDD Low-frequency NR TDD	Source-based IPv6 routing	None	<i>IPv6 Transmission</i>	
LTE TDD	Virtual Routing and Forwarding	None	<i>VRF</i>	None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations
UMTS	3900 and 5900 series base stations - DBS3900 LampSite and DBS5900 LampSite
LTE	3900 and 5900 series base stations. 5900 series base stations must be configured with the BBU5910, BBU5900, BBU5900A, BookBBU5901, or BookBBU5901a. - DBS3900 LampSite and DBS5900 LampSite
NR	3900 and 5900 series base stations. 5900 series base stations must be configured with the BBU5910, BBU5900, BBU5900A, BookBBU5901, or BookBBU5901a. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

When a separate-MPT LN MBTS uses main-control-board-based IPv4/IPv6 co-transmission through backplane interconnection, VRF deployment is supported only when UMPT series main control boards are used and both LTE and NR versions are upgraded to SRAN18.1 and later.

Other functions are supported by all boards.

RF Modules

This function does not depend on RF modules.

5.3.4 Networking

For details about how to configure inter-BBU interconnection, see *BBU Interconnection* in *3900 & 5900 Series Base Station Product Documentation*.

5.3.5 Others

None

5.4 Operation and Maintenance

This section uses LTE FDD as an example to describe the engineering guidelines. The engineering guidelines for LTE TDD and LTE NB-IoT are the same as those for LTE FDD.

The scenarios in which co-MPT base stations use the UMDU/UMDC board are similar to those in which co-MPT base stations use the UMPT board. Unless otherwise specified, the scenarios in which UMPT board is used are described as examples, and those in which the UMDU/UMDC board is used are not described again.

Only the MML-based configurations for an MBTS in co-transmission scenarios that are different from those for a single-mode base station are described in this section. For details about data configurations at the data link layer and transport layer on a single-mode base station, see *3900 & 5900 Series Base Station Initial Configuration Guide* in *3900 & 5900 Series Base Station Product Documentation*.

For details about the slots available for new boards, see the hardware description of BBUs in *3900 & 5900 Series Base Station Product Documentation*.

If VLANs are configured and differentiated services code point (DSCP) values are used to distinguish VLANs, different DSCP values need to be configured for data packets of different RATs. For details about the settings of DSCP values, see *Bandwidth Sharing of Multimode Base Station Co-Transmission* for SingleRAN.

5.4.1 When to Use

It is recommended that co-transmission in IP over FE/GE mode be enabled if an IP-based base station is reconstructed from single-mode to multimode or from dual-mode to triple-mode and the number of transmission ports on the base station side is limited.

5.4.2 Precautions

- When main-control-board-based co-transmission is implemented through panel interconnection, only the panel interconnection cable is required and no additional board is required.
- A UTRPc is required when UTRPc-based co-transmission is implemented through backplane interconnection.

- For co-transmission on a triple-mode or quadruple-mode base station, UCIU+UMPT interconnection or UMPT+UMPT interconnection can be used to connect two BBUs.
- For a separate-MPT MBTS that uses main-control-board-based co-transmission through backplane interconnection, if different RATs are deployed in the same BBU and the base station and external network devices are on the same network segment, run the **SET ETHPORT** command in IPv4 scenarios to enable the ARP proxy function, and run the **MOD INTERFACEPARA6** command in IPv6 scenarios to enable the ND proxy function.
- In IPv4 co-transmission through backplane interconnection scenarios, if the gateway router reports ALM-25885 IP Address Conflict but the upper-level and lower-level base stations do not report this alarm, check whether the backplane tunnel route is correctly configured on the upper-level base station. After the backplane tunnel route of the upper-level base station is removed, ALM-25885 IP Address Conflict is cleared on the gateway router side.
- In IPv6 co-transmission through backplane interconnection scenarios, when a base station detects that its address conflicts with that of an upper-level or lower-level base station, the peer base station must delete the conflicting address and then delete the backplane tunnel to clear the alarm. If the tunnel is deleted before the address is deleted, the alarm cannot be cleared.
- For a reconstruction from co-transmission through panel interconnection to co-transmission through backplane interconnection, when the IP address of the lower-level base station panel is changed to the IP address of the loopback interface, the lower-level base station reports an address conflict alarm, after the ND proxy function is enabled. The transmission link of the lower-level base station is interrupted. The address conflict alarm is cleared in a maximum of 20 minutes, and the transmission link of the lower-level base station is restored.
- In scenarios of separate-MPT MBTS co-transmission through panel interconnection, the **GTRANSPARA.FORWARDMODE (LTE eNodeB, 5G gNodeB)** parameter must be set to **ROUTE** for the interconnection board of the base station that provides the co-transmission port.

5.4.3 Configuration and Verification

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

5.4.3.1 Main-Control-Board-based Co-transmission Through Panel
Interconnection on the Separate-MPT NL/NT/NM MBTS Side in IPv6 over
FE/GE Mode

In this scenario, the main control board of LTE must be a UMPTb/UMPTe/UMPTg/
UMPTga/UMPTHa/UMPTNb and the main control board of NR must be a UMPTe/
UMPTg/UMPTga/UMPTHa/UMPTNb.

5.4.3.1.1 Data Preparation

Figure 5-11 shows an example of main-control-board-based IP co-transmission
through panel interconnection on the separate-MPT NL MBTS side. The MBTS
connects to the core network through a co-transmission FE/GE port on the
gNodeB UMPT. The eNodeB and the gNodeB are interconnected through ETH
ports. **Table 5-1** describes the IPv6 address plan.

Figure 5-11 Main-control-board-based IPv6 co-transmission through panel
interconnection on the separate-MPT NL MBTS side

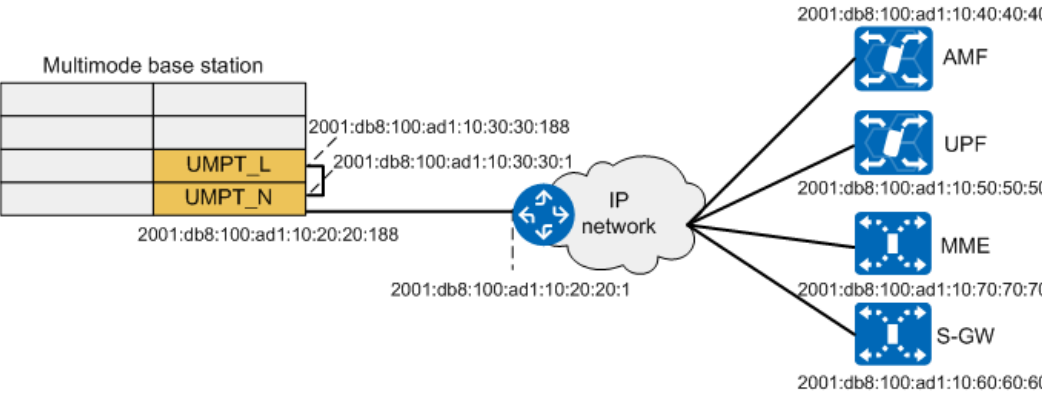


Table 5-1 Data plan

Item	Instance	Remarks
IPv6 address of the AMF	2001:db8:100:ad1:10:40:40/120	None
IPv6 address of the UPF	2001:db8:100:ad1:10:50:50 /120	None
IPv6 address of the S-GW	2001:db8:100:ad1:10:60:60 /120	None
IPv6 address of the MME	2001:db8:100:ad1:10:70:70/120	None
IPv6 address of a port on the gNodeB (for interconnection)	2001:db8:100:ad1:10:30:30:1/120	None
IPv6 address of a port on the gNodeB	2001:db8:100:ad1:10:20:20:188/120	None

Item	Instance	Remarks
IPv6 address of a port on the eNodeB (for interconnection)	2001:db8:100:ad1:10:30:30:188/120	None
IPv6 address of the next-hop router	2001:db8:100:ad1:10:20:20:1/120	None

NOTE

- IPv6 addresses of the two Ethernet ports used for the interconnection between the eNodeB and gNodeB must be on the same network segment.
- Data configurations at the data link layer, such as the duplex mode and rate, must be consistent between the two ETH ports used for connecting the gNodeB to the eNodeB.
- In scenarios of main-control-board-based IPv6 co-transmission through panel interconnection, the ND proxy function cannot be enabled on the co-transmission port for the lower-level base station.

5.4.3.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Only the new transmission configuration model is supported in an IPv6 scenario. That is, the **GTRANSPARA.TRANSCFGMODE (5G gNodeB, LTE eNodeB)** parameter must be set to **NEW**.

LTE Side

When running the eNodeB **ADD IPROUTE6** command to add a route from the eNodeB to the MME/S-GW, change the next-hop address of the route from the IP address of the directly connected router to the IPv6 address of the interconnection port on the UMPT_N.

NR Side

```
//Setting the attributes of the Ethernet port on the UMPT_N that connects to the UMPT_L
SET ETHPORT: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PN=1, PA=COPPER, SPEED=AUTO,
DUPLEX=AUTO,PORTID=71;
//Configuring the interface associated with the Ethernet port on the UMPT_N that connects to the UMPT_L
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFID=0, IPV6SW=ENABLE;
//Setting the IPv6 address of the Ethernet port on the UMPT_N that connects to the UMPT_L. The IPv6
addresses of the two interconnection ports must be on the same network segment.
ADD IPADDR6: IPADDR6ID="NR", ITFID=0, IPV6="2001:db8:100:ad1:10:30:30:1", PFXLEN=120;
//Adding uplink routes from the eNodeB to the MME/S-GW through the gNodeB and with Route Type set
to NEXTHOP and Next Hop IP set to the IPv6 address of the directly connected router
```

```
ADD IPROUTE6: RTIDX=0, DSTIP="2001:db8:100:ad1:10:70:70:00", PFXLEN=120, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:10:20:20:1", PREF=60;
ADD IPROUTE6: RTIDX=1, DSTIP="2001:db8:100:ad1:10:60:60:00", PFXLEN=120, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:10:20:20:1", PREF=60;
//Setting the forwarding mode to route
SET GTRANSPARA: FORWARDMODE=ROUTE;
//(Optional) Enabling DHCP relay. When co-transmission is enabled for the gNodeB and eNodeB, if eNodeB
is to be deployed using DHCP, the gNodeB needs to work as the relay. Therefore, DHCP relay needs to be
enabled for the gNodeB.
ADD INTERFACEPARA6: ITFID=0, DHCPRELAYSW=ENABLE;
//(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the gNodeB
and eNodeB, if eNodeB is to be deployed using DHCP, the gNodeB must work as the relay. Therefore, the
IPv6 address of the DHCP server for the eNodeB needs to be added on the gNodeB.
ADD DHCP6SVRIP6: DHCP6SVRIP="2001:DB8::1";
//(Optional) Configuring VLANs by using the INTERFACE MOs if VLANs are required to isolate LTE data
from NR data
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=22, VRFID=0;
//Configuring the VLAN of the interface for NR data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
```

5.4.3.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.1.4 Activation Verification

After co-transmission is enabled on the MBTS, check whether the feature is enabled by verifying the status of the IP links between the MBTS and the peer devices.

NR Side

After the configuration file is delivered to the gNodeB and activated, perform the following step on the gNodeB to check whether the transmission links of the gNodeB are normal:

On the gNodeB, ping the IPv6 addresses of the AMF/UMF/MME/S-GW. If the IP addresses can be pinged, the transmission links are normal. For example, check whether the transmission link between the gNodeB and the AMF is normal.

```
PING6: DSTIPTYPE=IPADDR6_TYPE_GLOBAL_UNICAST, SRCIP="2001:db8:100:ad1:10:20:20:188",
DSTIP="2001:db8:100:ad1:10:40:40:40", CONTPING=DISABLE;
```

LTE Side

After the configuration file is delivered to the eNodeB and activated, perform the following step on the eNodeB to check whether the transmission links between the eNodeB and the MME/S-GW are normal:

On the eNodeB, ping the IPv6 addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links are normal. For example, check whether the transmission link between the eNodeB and the MME is normal.

```
PING6: DSTIPTYPE=IPADDR6_TYPE_GLOBAL_UNICAST, SRCIP="2001:db8:100:ad1:10:30:30:188",
DSTIP="2001:db8:100:ad1:10:70:70:70", CONTPING=DISABLE;
```

5.4.3.2 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT NL/NT/NM MBTS Side in IPv6 over FE/GE Mode

In this scenario, the main control board of LTE must be a UMPTb/UMPTe/UMPTg/UMPTga/UMPTHa/UMPTNb and the main control board of NR must be a UMPTe/UMPTg/UMPTga/UMPTHa/UMPTNb.

5.4.3.2.1 Data Preparation

Figure 5-12 shows an example of main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT NL MBTS side. The separate-MPT NL MBTS connects to the core network through a co-transmission FE/GE port on the gNodeB UMPT. The eNodeB and the gNodeB are interconnected through the backplane. Figure 5-12 describes the IPv6 address plan.

Figure 5-12 Main-control-board-based IPv6 co-transmission through backplane interconnection on the separate-MPT NL MBTS side

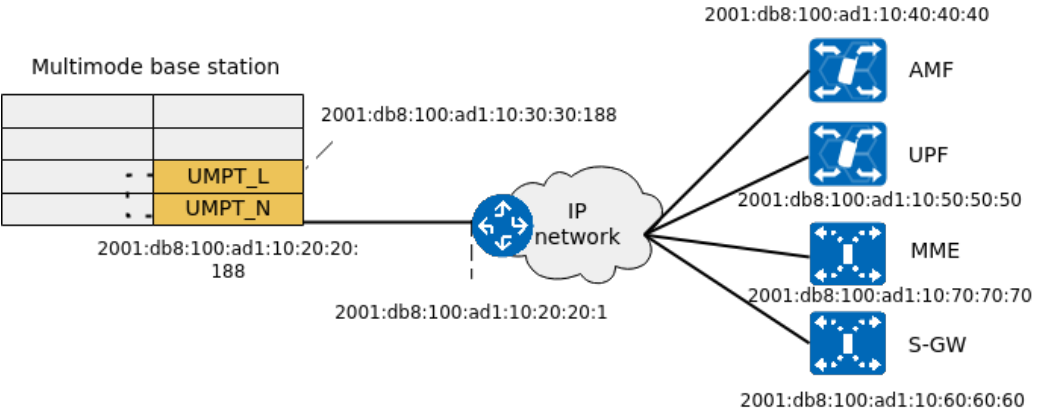


Table 5-2 Data plan

Item	Instance	Remarks
IPv6 address of the AMF	2001:db8:100:ad1:10:40:40/120	None
IPv6 address of the UPF	2001:db8:100:ad1:10:50:50 /120	None
IPv6 address of the MME	2001:db8:100:ad1:10:70:70/120	None
IPv6 address of the S-GW	2001:db8:100:ad1:10:60:60 /120	None
IPv6 address of a port on the gNodeB	2001:db8:100:ad1:10:20:20:188/120	None
IPv6 address of the loopback interface of the eNodeB	2001:db8:100:ad1:10:30:30:188/128	None

Item	Instance	Remarks
IPv6 address of the next-hop router	2001:db8:100:ad1:10:20:20:1/120	None
VRF index	0 to 31	The value is specified by the VRF.VRFIDX (LTE eNodeB, 5G gNodeB) parameter, and the default value is 0. • If VRF-based isolation is not required by the RAT that does not provide the co-transmission port, this parameter does not need to be configured. • If VRF-based isolation is required by the RAT that does not provide the co-transmission port, this parameter must be set to the same value for both RATs. • The VRF index range depends on the board capability. For details, see the "Hardware" section in <i>VRF</i>.

Item	Instance	Remarks
Backplane tunnel number	0 to 7	The value is specified by the BACKTUNNEL.TUNNELNO (LTE eNodeB, 5G gNodeB) parameter, and the default value is 0. • If VRF-based isolation is not required by the RAT that does not provide the co-transmission port, a backplane tunnel is configured for each RAT, and the tunnel number does not need to be modified. • If VRF-based isolation is required by the RAT that does not provide the co-transmission port, backplane tunnels with different tunnel numbers are configured for the two RATs based on the VRF requirements. The tunnel numbers of the backplane tunnels configured for the two RATs that use backplane co-transmission must be the same.

 NOTE

- In this scenario, the UMPT_L does not need to be configured with Ethernet port attributes because the UMPT_L communicates with the UMPT_N through the backplane.
- When the ND proxy function is enabled for the upper RAT, the IP address of the upper RAT must be configured first. Then, configure an IP address in the same network segment as that of the upper RAT for the lower RAT.

5.4.3.2.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Only the new transmission configuration model is supported in an IPv6 scenario. That is, the **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** parameter must be set to **NEW**.

LTE Side

```
//Adding a tunnel from the UMPT_L to the UMPT_N
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING, TUNNELNO=0;
//Adding an interface to the tunnel
ADD INTERFACE: ITFID=2, ITFTYPE=NORMAL, PT=TUNNEL, PORTID=0, VRFIDX=0;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the eNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
eNodeB use the logical IP address.
ADD IPADDR6: IPADDR6ID="0", ITFID=1, IPV6=" 2001:db8:100:ad1:10:30:30:188", PFXLEN=128;
//Adding routes from the eNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Route Type be set to IF and ITFID be set to the ITFID of the tunnel interface.
ADD IPRROUTE6: RTIDX=0, DSTIP=" 2001:db8:100:ad1:10:70:70:00", PFXLEN=120, RTTYPE=IF, ITFID=2;
ADD IPRROUTE6: RTIDX=1, DSTIP=" 2001:db8:100:ad1:10:60:60:00", PFXLEN=120, RTTYPE=IF, ITFID=2;
```

NR Side

```
//Adding an interface to Ethernet port 0 on the UMPT_N (in slot 7)
SET ETHPORT: CN=0, SRN=1, SN=7, PN=0, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO,
DUPLEX=AUTO;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFIDX=0;
//Adding IP address for the gNodeB
ADD IPADDR6: IPADDR6ID="0", ITFID=1, IPV6=" 2001:db8:100:ad1:10:20:20:188", PFXLEN=120;
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSN=7, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=NON_SHAPING,
TUNNELNO=0;
//Adding an interface to the tunnel
ADD INTERFACE: ITFID=2, ITFTYPE=NORMAL, PT=TUNNEL, PORTID=0, VRFIDX=0;
//Adding uplink routes from the eNodeB to the MME/S-GW through the gNodeB with Route Type set to
NEXTHOP and Next Hop IP set to the IPv6 address of the directly connected router
ADD IPRROUTE6: RTIDX=0, DSTIP="2001:db8:100:ad1:10:70:70:00", PFXLEN=120, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:10:20:20:1", PREF=60;
ADD IPRROUTE6: RTIDX=1, DSTIP="2001:db8:100:ad1:10:60:60:00", PFXLEN=120, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:10:20:20:1", PREF=60;
//Adding downlink routes from the MME/S-GW/MAE to the eNodeB through the gNodeB. When IP-based
co-transmission for the eNodeB and the gNodeB is implemented through the backplane tunnel, the
downlink routes from the MME/S-GW/MAE to the eNodeB through the gNodeB must be configured on the
gNodeB. In addition, Route Type must be set to IF and ITFID must be set to the ITFID of the tunnel interface.
ADD IPRROUTE6: RTIDX=0, DSTIP=" 2001:db8:100:ad1:10:30:30:00", PFXLEN=120, RTTYPE=IF, ITFID=2;
//(Optional) Enabling DHCP relay. When co-transmission is enabled for the gNodeB and eNodeB, if the
eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay. Therefore, DHCP relay needs
to be enabled for the gNodeB.
ADD INTERFACEPARA6: ITFID=2, DHCPRELAYSW=ENABLE;
//(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the gNodeB
and eNodeB, if the eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay.
Therefore, the IPv6 address of the DHCP server for the eNodeB needs to be added on the gNodeB.
ADD DHCP6SVRIP6: DHCP6SVRIP="2001:DB8::1";
//(Optional) Configuring VLANs. Differentiated VLAN data can be configured for LTE and NR using the
INTERFACE MO.
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=22, VRFIDX=0;
//Configuring the VLAN of the interface for NR data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFIDX=0;
```

5.4.3.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.2.4 Activation Verification

After co-transmission is enabled on the base station, check whether it is enabled by verifying the status of the IP links between the base station and the peer devices.

NR Side

After the configuration file is delivered to the gNodeB and activated, perform the following step on the gNodeB to check whether the transmission links of the gNodeB are normal:

On the gNodeB, ping the IPv6 addresses of the AMF/UMF/MME/S-GW. If the IP addresses can be pinged, the transmission links are normal. For example, check whether the transmission link to the AMF is normal.

```
PING6: DSTIPTYPE=IPADDR6_TYPE_GLOBAL_UNICAST, SRCIP="2001:db8:100:ad1:10:20:20:188",  
DSTIP="2001:db8:100:ad1:10:40:40:40", CONTPING=DISABLE;
```

LTE Side

After the configuration file is delivered to the eNodeB and activated, perform the following step on the eNodeB to check whether the transmission links between the eNodeB and the MME/S-GW are normal:

On the eNodeB, ping the IPv6 addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links are normal. For example, check whether the transmission link to the MME is normal.

```
PING6: DSTIPTYPE=IPADDR6_TYPE_GLOBAL_UNICAST, SRCIP="2001:db8:100:ad1:10:30:30:188",  
DSTIP="2001:db8:100:ad1:10:70:70:70", CONTPING=DISABLE;
```

5.4.3.3 Main-Control-Board-based Co-transmission Through Panel Interconnection on the Separate-MPT N+GL/GT/GM MBTS Side in IP over FE/GE Mode

In this scenario, the GSM base station can be a GBTS or an eGBTS, and the main control board must be a GTMUc/UMPT. The main control board of NR must be a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga/UMPTga.

5.4.3.3.1 Data Preparation

This section uses GTMU+UMPT_L+UMPT_N as an example to describe main-control-board-based IP co-transmission through panel interconnection on the separate-MPT N+GL/GT/GM MBTS side in NSA networking.

In this scenario, main-control-board-based IP co-transmission through panel interconnection on the separate-MPT N+GL/GT/GM MBTS side is implemented by cascading FE ports on the panels in different BBUs. The GL/GT/GM base stations are cascaded in one BBU through panel FE ports, while the gNodeB is located in the other BBU. The GL/GT/GM base stations connect to the gNodeB UMPT_N

through an FE port on the eNodeB UMPT_L. The N+GL/GT/GM MBTS connects to the BSC, MME, and S-GW through a co-transmission 10GE/25GE/50GE port on the gNodeB UMPT. The GBTS/eGBTS GTMU and the eNodeB UMPT_L are interconnected through FE ports.

Figure 5-13 shows an example of the network topology, and Table 5-3 describes the IP address plan.

Figure 5-13 Main-control-board-based IP co-transmission through panel interconnection on the separate-MPT N+GL/GT/GM MBTS side

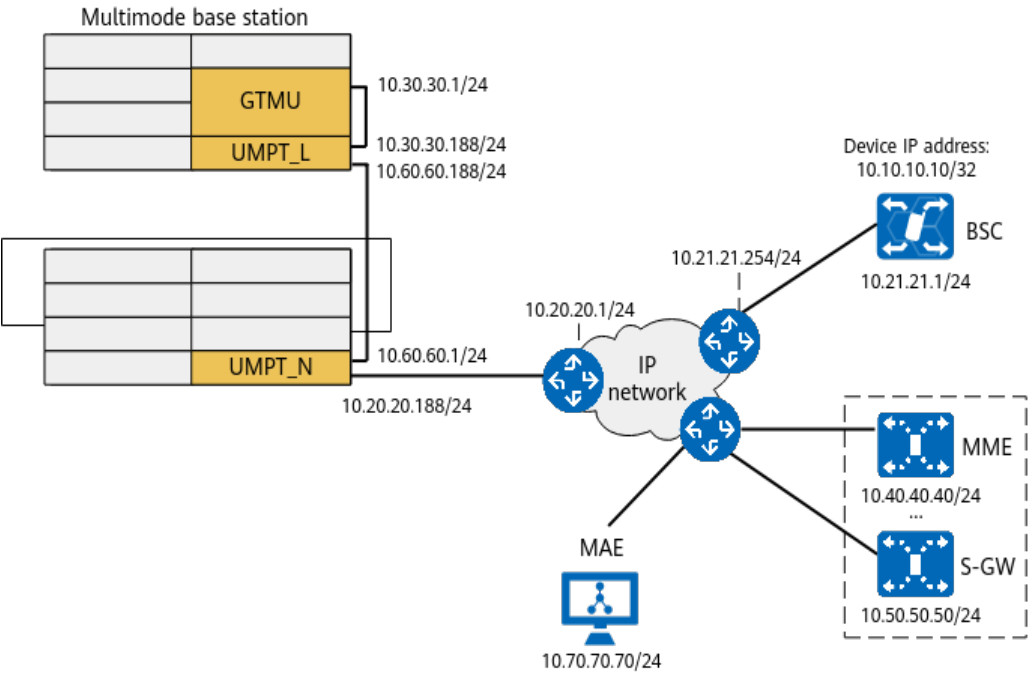


Table 5-3 Data plan

Item	Instance	Remarks
Device IP address of the BSC	10.10.10.10/32	None
Port IP address of the BSC	10.21.21.1/24	None
IP address of the port on the router that connects to the BSC	10.21.21.254/24	None
IP address of FE port 0 on the gNodeB (for interconnection)	10.60.60.1/24	NR: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment
IP address of FE port 1 on the gNodeB	10.20.20.188/24	NR: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment

Item	Instance	Remarks
IP address of the port on the router that connects to the gNodeB	10.20.20.1/24	None
IP address of FE port 0 on the eNodeB (FE port 0 is used for interconnection with the gNodeB.)	10.60.60.188/24	None
IP address of FE port 1 on the eNodeB (FE port 1 is used for interconnection with the GBTS/eGBTS.)	10.30.30.188/24	None
IP address of FE port 0 on the GBTS/eGBTS (FE port 0 is used for interconnection with the eNodeB.)	10.30.30.1/24	None
IP address of the MME	10.40.40.40/24	NR/LTE: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	NR/LTE: peer IP address for the IP path
IP address of the MAE	10.70.70.70/24	The MAE IP address is used as the DHCP server IP address of the eNodeB and gNodeB.
Note: IP addresses of the two Ethernet ports used for interconnecting the eNodeB to the GBTS/eGBTS and interconnecting the eNodeB to the gNodeB must be on the same network segment. Data configurations at the data link layer, such as the duplex mode and rate, must be consistent between the two Ethernet ports used for connecting the eNodeB to the GBTS/eGBTS, and between the two Ethernet ports used for connecting the eNodeB to the gNodeB.		

5.4.3.3.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

MAE Side

On the MAE side, add two routes to the DHCP relay of the eNodeB. In this step, set the destination IP address of one route to the OMCH IP address of the gNodeB and that of the other route to the lower-level transmission port IP address of the gNodeB.

GSM Side (GBTS)

- Step 1** Run the **ADD BTSIPRT** command to add a route from the GBTS to the BSC, and change the next-hop address of the route from the IP address of the directly connected router to the IP address of the interconnection port on the UMPT_L.
- Step 2** (Optional) Run the **ADD IPRT** command to add routes from the BSC to the DHCP relay of the GBTS. In this step, set the destination IP address of one route to the OMCH IP address of the eNodeB and that of the other route to the lower-level transmission port IP address of the eNodeB.

----End

GSM Side (eGBTS)

- Step 1** Run the **ADD IPRT** command (old model)/**ADD IPROUTE4** (new model) on the eGBTS to add a route from the eGBTS to the BSC, and change the next-hop address of the route from the IP address of the directly connected router to the IP address of the interconnection port on the UMPT_L.
- Step 2** (Optional) Run the **ADD IPRT** command on the BSC to add routes from the BSC to the DHCP relay of the eGBTS. In this step, set the destination IP address of one route to the OMCH IP address of the eNodeB and that of the other route to the lower-level transmission port IP address of the eNodeB.

----End

LTE Side

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**:

```
//Setting the attributes of the Ethernet port on the UMPT_L that connects to the GTMU
SET ETHPORT: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PN=1, PA=COPPER, SPEED=AUTO, DUPLEX=AUTO;
//Setting the device IP address of the Ethernet port on the UMPT_L that connects to the GTMU. The IP
addresses of the two interconnection ports must be on the same network segment.
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.30.30.188",
MASK="255.255.255.0", VRFIDX=0;
//Adding an uplink route from the GBTS/eGBTS to the BSC through the eNodeB with Route Type set to
NEXTHOP and Next Hop IP set to the IP address of the interconnection port on the UMPT_N
ADD IPRT: RTIDX=0, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.60.60.1", PREF=60;
//Setting the forwarding mode to route
SET GTRANSPARA: FORWARDMODE=ROUTE;
//((Optional) Adding a downlink route from the BSC to the GBTS/eGBTS through the eNodeB. The downlink
route is required only if the logical IP address is used by the GBTS/eGBTS. If the physical port IP address is
used by the GBTS/eGBTS and the IP addresses of the panel interconnection ports on the eNodeB and the
GBTS/eGBTS are on the same network segment as the port IP address of the GBTS/eGBTS, the downlink
route from the eNodeB to the GBTS/eGBTS does not need to be configured. This example is the latter case.
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS and eNodeB, if the
GBTS/eGBTS is to be deployed using DHCP, the eNodeB needs to work as the relay. Therefore, DHCP relay
needs to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/
```

eGBTS and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS is to be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be added on the eNodeB. For the GBTS, the DHCP server is the BSC. For the eGBTS, the DHCP server is the MAE.

```
//Adding the IP address of the DHCP server for the GBTS
ADD DHCPVRIP: DHCPVRIP="10.10.10.10";
//Adding the IP address of the DHCP server for the eGBTS
ADD DHCPVRIP: DHCPVRIP="10.70.70.70";
```

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**:

```
//Setting the attributes of the Ethernet port on the UMPT_L that connects to the GTMU
SET ETHPORT: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PN=1, PA=COPPER, SPEED=AUTO, DUPLEX=AUTO,
PORTID=71;
//Configuring the interface associated with the Ethernet port on the UMPT_L that connects to the GTMU
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFID=0;
//Setting the IP address of the Ethernet port on the UMPT_L that connects to the GTMU. The IP addresses
of the two interconnection ports must be on the same network segment.
ADD IPADDR4: ITFID=0, IP="10.30.30.188", MASK="255.255.255.0";
//Adding an uplink route from the GBTS/eGBTS to the BSC through the eNodeB with Route Type set to
NEXTHOP and Next Hop IP set to the IP address of the interconnection port on the UMPT_N
ADD IPROUTE4: RTID=0, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.60.60.1", PREF=60;
//Setting the forwarding mode to route
SET GTRANSPARA: FORWARDMODE=ROUTE;
//((Optional) Adding a downlink route from the BSC to the GBTS/eGBTS through the eNodeB. The downlink
route is required only if the logical IP address is used by the GBTS/eGBTS. If the physical port IP address is
used by the GBTS/eGBTS and the IP addresses of the panel interconnection ports on the eNodeB and the
GBTS/eGBTS are on the same network segment as the port IP address of the GBTS/eGBTS, the downlink
route from the eNodeB to the GBTS/eGBTS does not need to be configured. This example is the latter case.
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS and eNodeB, if the
GBTS/eGBTS is to be deployed using DHCP, the eNodeB needs to work as the relay. Therefore, DHCP relay
needs to be enabled for the eNodeB.
SET DHCPRELAYS: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/
eGBTS and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS is to be deployed using
DHCP. Therefore, the IP address of the DHCP server needs to be added on the eNodeB. For the GBTS, the
DHCP server is the BSC. For the eGBTS, the DHCP server is the MAE.
//Adding the IP address of the DHCP server for the GBTS
ADD DHCPVRIP: DHCPVRIP="10.10.10.10";
//Adding the IP address of the DHCP server for the eGBTS
ADD DHCPVRIP: DHCPVRIP="10.70.70.70";
```

NR Side

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**:

```
//Setting the attributes of the Ethernet port on the UMPT that connects to the UMPT_L
SET ETHPORT: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PN=0, PA=COPPER, SPEED=AUTO, DUPLEX=AUTO;
//Setting the device IP address of the Ethernet port on the UMPT that connects to the UMPT_L. The IP
addresses of the two interconnection ports must be on the same network segment.
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=ETH, PN=0, IP="10.60.60.1",
MASK="255.255.255.0", VRFID=0;
//Adding uplink routes from the eNodeB to the MME/S-GW through the gNodeB with Route Type set to
Next Hop and Next Hop IP set to the IP address of the directly connected router
ADD IPRT: RTID=0, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
ADD IPRT: RTID=10, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
//Adding an uplink route from the GBTS/eGBTS to the BSC through the gNodeB with Route Type set to
NEXTHOP and Next Hop IP set to the IP address of the directly connected router
ADD IPRT: RTID=1, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
//Setting the forwarding mode to route
SET GTRANSPARA: FORWARDMODE=ROUTE;
//Adding a downlink route from the BSC to the GBTS/eGBTS through the gNodeB. A downlink route to the
GBTS/eGBTS needs to be configured on the gNodeB side regardless of whether a port IP address or logical
IP address is used by the GBTS/eGBTS. In this step, set DSTIP to the port IP address or logical IP address of
the GBTS/eGBTS and NEXTHOP to the IP address of the port on the eNodeB for interconnecting to the
gNodeB. This step takes the port IP address of the GBTS/eGBTS as an example. When configuring a route to
```

the logical IP address of the GBTS/eGBTS, change the value of DSTIP in the following command:
 ADD IPRT: RTIDX=2, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.30.30.1",
 DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.60.60.188", PREF=60;
 //(Optional) Enabling DHCP relay. When co-transmission is enabled for the eNodeB and gNodeB, if the
 eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay. Therefore, DHCP relay needs
 to be enabled for the gNodeB.
 SET DHCPRELAYSWITCH: ES=ENABLE;
 //(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the eNodeB
 and gNodeB, if the eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay.
 Therefore, the IP address of the DHCP server needs to be added on the gNodeB. For the eNodeB, the DHCP
 server is the MAE. The MAE working as the DHCP server is used as an example.
 ADD DHCPVRIP: DHCPVRIP="10.70.70.70";
 //(Optional) Configuring VLANs. There are two methods of configuring different VLANs for the GBTS/
 eGBTS, eNodeB, and gNodeB data for isolation:
 //Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
 addresses of the uplink routes from the GBTS/eGBTS/eNodeB to the BSC/MME/S-GW through the gNodeB
 must be different from the next-hop address of the uplink route for the gNodeB. For example, you can set
 the next-hop addresses of the uplink routes from the GBTS/eGBTS/eNodeB to the BSC/MME/S-GW through
 the gNodeB to 10.20.20.101 and 10.20.20.201, which are different from the next-hop address (10.20.20.1)
 of the uplink route for the gNodeB.
 ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
 VLANID=2, SETPRIO=DISABLE;
 ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
 VLANID=22, SETPRIO=DISABLE;
 ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
 VLANID=33, SETPRIO=DISABLE;
 //Method 2 (not recommended): Configuring differentiated DSCP values for the GBTS/eGBTS, eNodeB, and
 gNodeB data packets and configuring the mapping between a VLAN group and the next hop
 ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=40, VLANID=22;
 ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
 ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP,
 VLANGROUPNO=1;

If **GTRANSPARA.TRANSFCGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**:

//Setting the attributes of the Ethernet port on the UMPT that connects to the UMPT_L
 SET ETHPORT: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PN=0, PA=COPPER, SPEED=AUTO, DUPLEX=AUTO,
 PORTID=70;
 //Configuring the interface associated with the Ethernet port on the UMPT that connects to the UMPT_L
 ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=70, VRFID=0;
 //Setting the IP address of the Ethernet port on the UMPT that connects to the UMPT_L. The IP addresses
 of the two interconnection ports must be on the same network segment.
 ADD IPADDR4: ITFID=0, IP="10.60.60.1", MASK="255.255.255.0";
 //Adding uplink routes from the eNodeB to the MME/S-GW through the gNodeB with Route Type set to
 Next Hop and Next Hop IP set to the IP address of the directly connected router
 ADD IPROUTE4: RTIDX=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
 NEXTHOP="10.20.20.1", PREF=60;
 ADD IPROUTE4: RTIDX=10, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
 NEXTHOP="10.20.20.1", PREF=60;
 //Adding an uplink route from the GBTS/eGBTS to the BSC through the gNodeB with Route Type set to
 NEXTHOP and Next Hop IP set to the IP address of the directly connected router
 ADD IPROUTE4: RTIDX=1, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
 NEXTHOP="10.20.20.1", PREF=60;
 //Setting the forwarding mode to route
 SET GTRANSPARA: FORWARDMODE=ROUTE;
 //Adding a downlink route from the BSC to the GBTS/eGBTS through the gNodeB. A downlink route to the
 GBTS/eGBTS needs to be configured on the gNodeB side regardless of whether a port IP address or logical
 IP address is used by the GBTS/eGBTS. In this step, set DSTIP to the port IP address or logical IP address of
 the GBTS/eGBTS and NEXTHOP to the IP address of the port on the eNodeB for interconnecting to the
 gNodeB. This step takes the port IP address of the GBTS/eGBTS as an example. When configuring a route to
 the logical IP address of the GBTS/eGBTS, change the value of DSTIP in the following command:
 ADD IPROUTE4: RTIDX=2, DSTIP="10.30.30.1", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
 NEXTHOP="10.60.60.188", PREF=60;
 //(Optional) Enabling DHCP relay. When co-transmission is enabled for the eNodeB and gNodeB, if the
 eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay. Therefore, DHCP relay needs
 to be enabled for the gNodeB.
 SET DHCPRELAYSWITCH: ES=ENABLE;
 //(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the eNodeB
 and gNodeB, if the eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay.
 Therefore, the IP address of the DHCP server needs to be added on the gNodeB. For the eNodeB, the DHCP

```
server is the MAE. The MAE working as the DHCP server is used as an example.
ADD DHCPVSRIP: DHCPVSRIP="10.70.70.70";
//(Optional) Configuring VLANs
//There are two methods of configuring different VLANs for the GBTS/eGBTS, eNodeB, and gNodeB data for
isolation when the VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink routes from the GBTS/eGBTS/eNodeB to the BSC/MME/S-GW through the gNodeB
must be different from the next-hop address of the uplink route for the gNodeB. For example, you can set
the next-hop addresses of the uplink routes from the GBTS/eGBTS/eNodeB to the BSC/MME/S-GW through
the gNodeB to 10.20.20.101 and 10.20.20.201, which are different from the next-hop address (10.20.20.1)
of the uplink route for the gNodeB.
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=33, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GBTS/eGBTS, eNodeB, and
gNodeB data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
//Configuring different VLANs for LTE and GSM data for isolation when the INTERFACE MO is used
//Configuring the VLAN of the interface for GSM data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=22, VRFID=0;
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
```

5.4.3.3.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.3.4 Activation Verification

After co-transmission is enabled on the MBTS, check whether the feature is enabled by verifying the status of the IP links between the MBTS and the peer devices.

NR Side

After the configuration file is delivered to the gNodeB and activated, perform the following step on the gNodeB to check whether the transmission link of the gNodeB is normal:

On the gNodeB, ping the next-hop IP address on the route. If the IP address can be pinged, the IP link is normal.

```
PING: CN=0, SRN=0, SN=7, SRCIP="10.20.20.188", DSTIP="10.20.20.1", CONTPING=DISABLE, APPTIF=NO;
```

LTE Side

After the configuration file is delivered to the eNodeB and activated, perform the following step on the eNodeB to verify that the transmission link of the eNodeB is normal:

On the eNodeB, ping the IP address of the next-hop route. If the IP address can be pinged, the IP link is normal.

```
PING: CN=0, SRN=0, SN=7, SRCIP="10.20.20.188", DSTIP="10.20.20.1", CONTPING=DISABLE, APPTIF=NO;
```

GSM Side

After the configuration file is delivered to the GBTS/eGBTS and activated, perform the following step on the BSC to verify that the transmission link between the BSC and the GBTS/eGBTS is normal:

On the BSC, ping the IP address of the GTMU. If the IP address can be pinged, the IP link is normal.

```
PING IP: SIPADDR="10.10.10.10", DESTIP="10.30.30.1", NEXTHOP="10.21.21.254", CONTPING=NO;
```

5.4.3.4 Main-Control-Board-based Co-transmission Through Panel Interconnection on the Separate-MPT N+LU/TU/MU MBTS Side in IP over FE/GE Mode

5.4.3.4.1 Data Preparation

This section uses UMPT_U+UMPT_L+UMPT_N as an example to describe main-control-board-based IP co-transmission through panel interconnection on the separate-MPT N+LU/TU/MU MBTS side in NSA networking. In this scenario, co-transmission is implemented by cascading FE ports on the main control board panels in different BBUs. The LU/TU/MU base stations are cascaded in one BBU through panel FE ports, while the gNodeB is located in the other BBU. The LU/TU/MU base stations connect to the gNodeB UMPT_N through an FE port on the eNodeB UMPT_L. The N+LU/TU/MU MBTS connects to the RNC, MME, and S-GW through a co-transmission 10GE/25GE/50GE port on the gNodeB UMPT_N. The NodeB UMPT_U and the eNodeB UMPT_L are interconnected through FE ports. Figure 5-14 shows an example of the network topology, and Table 5-4 describes the IP address plan.

Figure 5-14 Main-control-board-based IP co-transmission through panel interconnection on the separate-MPT N+LU/TU/MU MBTS side

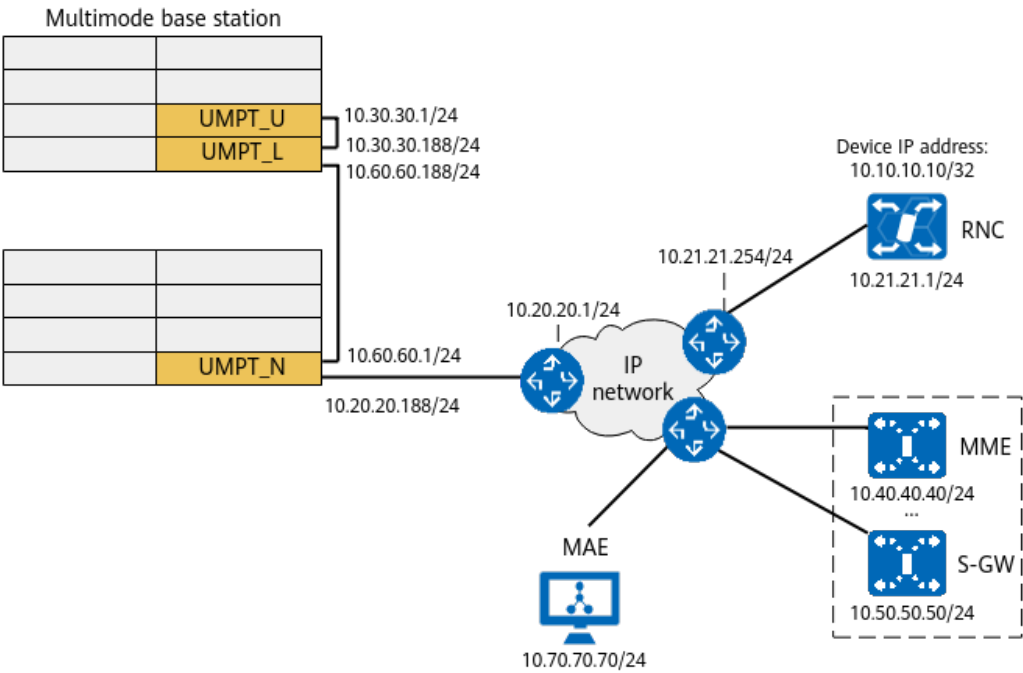


Table 5-4 Data plan

Item	Instance	Remarks
Device IP address of the RNC	10.10.10.10/32	None
Port IP address of the RNC	10.21.21.1/24	None
IP address of the port on the router that connects to the RNC	10.21.21.254/24	UMTS: next hop on the route from the RNC to the NodeB
IP address of the port on the router that connects to the gNodeB	10.20.20.1/24	NR: next hop on the route from the gNodeB to the MME/S-GW
IP address of FE port 0 on the gNodeB (for interconnection)	10.60.60.1/24	NR: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment
IP address of 10GE port 1 on the gNodeB	10.20.20.188/24	NR: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment
IP address of FE port 0 on the eNodeB (FE port 0 is used for interconnection with the gNodeB.)	10.60.60.188/24	None
IP address of FE port 1 on the eNodeB (FE port 1 is used for interconnection with the NodeB.)	10.30.30.188/24	None
IP address of FE port 0 on the NodeB (FE port 0 is used for interconnection with the eNodeB.)	10.30.30.1/24	None
IP address of the MME	10.40.40.40/24	NR/LTE: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	NR/LTE: peer IP address for the IP path
IP address of the MAE	10.70.70.70/24	The MAE IP address is used as the DHCP server IP address of the NodeB, eNodeB, and gNodeB.

Item	Instance	Remarks
Note: IP addresses of the two Ethernet ports used for interconnecting the eNodeB to the NodeB and interconnecting the eNodeB to the gNodeB must be on the same network segment. Data configurations at the data link layer, such as the duplex mode and rate, must be consistent between the two Ethernet ports used for connecting the eNodeB to the NodeB, and between the two Ethernet ports used for connecting the eNodeB to the gNodeB.		

5.4.3.4.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

MAE Side

On the MAE side, add two routes to the DHCP relay of the eNodeB and NodeB with the destination IP address of one route set to the OMCH IP address of the gNodeB and that of the other route set to the lower-level transmission port IP address of the gNodeB.

UMTS Side

- Step 1** When running the **ADD IPRT** command (old model)/**ADD IPROUTE4** (new model) on the NodeB side to add a route from the NodeB to the RNC, change the next-hop address of the route from the IP address of the directly connected router to the IP address of the interconnection port on the UMPT_L.
- Step 2** (Optional) Run the **ADD IPRT** command on the RNC side to add two routes from the RNC to the NodeB's DHCP relay with the destination IP address of one route set to the OMCH IP address of the eNodeB and that of the other route set to the lower-level transmission port IP address of the eNodeB.

----End

LTE Side

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**:

```
//Setting the attributes of the Ethernet port on the UMPT_L that connects to the UMPT_U
SET ETHPORT: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PN=1, PA=COPPER, SPEED=AUTO, DUPLEX=AUTO;
//Setting the device IP address of the Ethernet port on the UMPT_L that connects to the UMPT_U. The IP
addresses of the two interconnection ports must be on the same network segment.
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.30.30.188",
MASK="255.255.255.0", VRFIDX=0;
//Adding an uplink route from the NodeB to the RNC through the eNodeB with Route Type set to NEXTHOP
```

and Next Hop IP set to the IP address of the interconnection port on the UMPT
ADD IPRT: RTIDX=0, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.60.60.1", PREF=60;
//Setting the forwarding mode to route
SET GTRANSPARA: FORWARDMODE=ROUTE;
//((Optional) Adding a downlink route from the RNC to the NodeB through the eNodeB. The downlink route
is required only if a logical IP address is used by the NodeB. If a physical port IP address is used by the
NodeB and the IP addresses of the panel interconnection ports on the eNodeB and the NodeB are on the
same network segment as the port IP address of the NodeB, the downlink route from the eNodeB to the
NodeB does not need to be configured. This example is the latter case.
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the NodeB and eNodeB, if the
NodeB is to be deployed using DHCP, the eNodeB needs to work as the relay. Therefore, DHCP relay needs
to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding a DHCP server IP address. When co-transmission is enabled for the NodeB and
eNodeB, the eNodeB needs to work as the relay if the NodeB is deployed using DHCP. Therefore, the IP
address of the DHCP server needs to be configured on the eNodeB. For NodeBs, the DHCP server is MAE.
ADD DHCPVRIP: DHCPVRIP="10.70.70.70";

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**:

//Setting the attributes of the Ethernet port on the UMPT_L that connects to the UMPT_U
SET ETHPORT: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PN=1, PA=COPPER, SPEED=AUTO, DUPLEX=AUTO,
PORTID=71;
//Configuring the interface associated with the Ethernet port on the UMPT_L that connects to the UMPT_U
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFID=0;
//Setting the IP address of the Ethernet port on the UMPT_L that connects to the UMPT_U. The IP addresses
of the two interconnection ports must be on the same network segment.
ADD IPADDR4: ITFID=0, IP="10.30.30.188", MASK="255.255.255.0";
//Adding an uplink route from the NodeB to the RNC through the eNodeB with Route Type set to NEXTHOP
and Next Hop IP set to the IP address of the interconnection port on the UMPT
ADD IPROUTE4: RTIDX=0, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.60.60.1", PREF=60;
//Setting the forwarding mode to route
SET GTRANSPARA: FORWARDMODE=ROUTE;
//((Optional) Adding a downlink route from the RNC to the NodeB through the eNodeB. The downlink route
is required only if a logical IP address is used by the NodeB. If a physical port IP address is used by the
NodeB and the IP addresses of the panel interconnection ports on the eNodeB and the NodeB are on the
same network segment as the port IP address of the NodeB, the downlink route from the eNodeB to the
NodeB does not need to be configured. This example is the latter case.
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the NodeB and eNodeB, if the
NodeB is to be deployed using DHCP, the eNodeB needs to work as the relay. Therefore, DHCP relay needs
to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding a DHCP server IP address. When co-transmission is enabled for the NodeB and
eNodeB, the eNodeB needs to work as the relay if the NodeB is deployed using DHCP. Therefore, the IP
address of the DHCP server needs to be configured on the eNodeB. For NodeBs, the DHCP server is MAE.
ADD DHCPVRIP: DHCPVRIP="10.70.70.70";

NR Side

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**:

//Setting the attributes of the Ethernet port on the UMPT that connects to the UMPT_L
SET ETHPORT: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PN=0, PA=COPPER, SPEED=AUTO, DUPLEX=AUTO;
//Setting the device IP address of the Ethernet port on the UMPT that connects to the UMPT_L. The IP
addresses of the two interconnection ports must be on the same network segment.
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=ETH, PN=0, IP="10.60.60.1",
MASK="255.255.255.0", VRFID=0;
//Adding uplink routes from the eNodeB to the MME/S-GW through the gNodeB with Route Type set to
Next Hop and Next Hop IP set to the IP address of the directly connected router
ADD IPRT: RTIDX=0, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
ADD IPRT: RTIDX=10, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
//Adding an uplink route from the NodeB to the RNC through the gNodeB with Route Type set to
NEXTHOP and Next Hop IP set to the IP address of the directly connected router
ADD IPRT: RTIDX=1, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;

```
//Adding a downlink route from the RNC to the NodeB through the gNodeB. A downlink route to the
NodeB needs to be configured on the gNodeB side regardless of whether a port IP address or logical IP
address is used by the NodeB. In this step, set DSTIP to the port IP address or logical IP address of the
NodeB and NEXTHOP to the IP address of the port on the eNodeB for interconnecting to the gNodeB. This
step takes the port IP address of the NodeB as an example. When configuring a route to the logical IP
address of the NodeB, change the value of DSTIP in the following command:
ADD IPRT: RTIDX=2, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.30.30.1",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.60.60.188", PREF=60;
//Setting the forwarding mode to route
SET GTRANSPARA: FORWARDMODE=ROUTE;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the eNodeB and gNodeB, if the
eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay. Therefore, DHCP relay needs
to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the eNodeB
and gNodeB, if the eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay.
Therefore, the IP address of the DHCP server needs to be added on the gNodeB. For the eNodeB, the DHCP
server is the RNC or MAE. The MAE working as the DHCP server is used as an example.
ADD DHCPVRIP: DHCPVRIP="10.70.70.70";
//((Optional) Configuring VLANs. There are two methods of configuring different VLANs for the NodeB,
eNodeB, and gNodeB data for isolation:
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink routes from the NodeB/eNodeB to the RNC/MME/S-GW through the gNodeB must
be different from the next-hop address of the uplink route for the gNodeB. For example, you can set the
next-hop addresses of the uplink routes from the NodeB/eNodeB to the RNC/MME/S-GW through the
gNodeB to 10.20.20.101 and 10.20.20.201, which are different from the next-hop address (10.20.20.1) of
the uplink route for the gNodeB.
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=33, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the NodeB, eNodeB, and
gNodeB data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
```

If **GTRANSPARA.TRANSFCGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**:

```
//Setting the attributes of the Ethernet port on the UMPT that connects to the UMPT_L
SET ETHPORT: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PN=0, PA=COPPER, SPEED=AUTO, DUPLEX=AUTO,
PORTID=70;
//Configuring the interface associated with the Ethernet port on the UMPT that connects to the UMPT_L
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFID=0;
//Setting the IP address of the Ethernet port on the UMPT that connects to the UMPT_L. The IP addresses
of the two interconnection ports must be on the same network segment.
ADD IPADDR4: ITFID=0, IP="10.60.60.1", MASK="255.255.255.0";
//Adding uplink routes from the eNodeB to the MME/S-GW through the gNodeB with Route Type set to
Next Hop and Next Hop IP set to the IP address of the directly connected router
ADD IPROUTE4: RTIDX=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1", PREF=60;
ADD IPROUTE4: RTIDX=10, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1", PREF=60;
//Adding an uplink route from the NodeB to the RNC through the gNodeB with Route Type set to
NEXTHOP and Next Hop IP set to the IP address of the directly connected router
ADD IPROUTE4: RTIDX=1, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1", PREF=60;
//Adding a downlink route from the RNC to the NodeB through the gNodeB. A downlink route to the
NodeB needs to be configured on the gNodeB side regardless of whether a port IP address or logical IP
address is used by the NodeB. In this step, set DSTIP to the port IP address or logical IP address of the
NodeB and NEXTHOP to the IP address of the port on the eNodeB for interconnecting to the gNodeB. This
step takes the port IP address of the NodeB as an example. When configuring a route to the logical IP
address of the NodeB, change the value of DSTIP in the following command:
ADD IPROUTE4: RTIDX=2, DSTIP="10.30.30.1", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.60.60.188", PREF=60;
//Setting the forwarding mode to route
SET GTRANSPARA: FORWARDMODE=ROUTE;
```

```

//(Optional) Enabling DHCP relay. When co-transmission is enabled for the eNodeB and gNodeB, if the
eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay. Therefore, DHCP relay needs
to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the eNodeB
and gNodeB, if the eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay.
Therefore, the IP address of the DHCP server needs to be added on the gNodeB. For the eNodeB, the DHCP
server is the RNC or MAE. The MAE working as the DHCP server is used as an example.
ADD DHCPVRIP: DHCPVRIP="10.70.70.70";
//(Optional) Configuring VLANs
//There are two methods of configuring different VLANs for the NodeB, eNodeB, and gNodeB data for
isolation when the VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink routes from the NodeB/eNodeB to the RNC/MME/S-GW through the gNodeB must
be different from the next-hop address of the uplink route for the gNodeB. For example, you can set the
next-hop addresses of the uplink routes from the NodeB/eNodeB to the RNC/MME/S-GW through the
gNodeB to 10.20.20.101 and 10.20.20.201, which are different from the next-hop address (10.20.20.1) of
the uplink route for the gNodeB.
ADD VLANMAP: VRFID=0, NEXTHOP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=33, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the NodeB, eNodeB, and
gNodeB data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
//Configuring different VLANs for LTE and UMTS data for isolation when the INTERFACE MO is used
//Configuring the VLAN of the interface for UMTS data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=22, VRFID=0;
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;

```

5.4.3.4.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.4.4 Activation Verification

After co-transmission is enabled on the MBTS, check whether the feature is enabled by verifying the status of the IP links between the MBTS and the peer devices.

NR Side

After the configuration file is delivered to the gNodeB and activated, perform the following step on the gNodeB to check whether the transmission link of the gNodeB is normal:

On the gNodeB, ping the next-hop IP address on the route. If the IP address can be pinged, the IP link is normal.

```
PING: CN=0, SRN=1, SN=7, SRCIP="10.20.20.188", DSTIP="10.20.20.1", CONTPING=DISABLE, APPTIF=NO;
```

LTE Side

After the configuration file is delivered to the eNodeB and activated, perform the following step on the eNodeB to check whether the transmission link of the eNodeB is normal:

On the eNodeB, ping the IP address of the next-hop route. If the IP address can be pinged, the IP link is normal.

PING: CN=0, SRN=0, SN=7, SRCIP="10.20.20.188", DSTIP="10.20.20.1", CONTPING=DISABLE, APPTIF=NO;

UMTS Side

After the configuration file is delivered to the NodeB and activated, perform the following step on the RNC to verify that the transmission link between the RNC and the NodeB is normal:

On the RNC, ping the IP address of the UMPT_U. If the IP address can be pinged, the IP link is normal.

PING IP: SIPADDR="10.10.10.10", DESTIP="10.30.30.1", NEXTHOP="10.21.21.254", CONTPING=NO;

5.4.3.5 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT NL/NT/NM MBTS Side in IP over FE/GE Mode

5.4.3.5.1 Data Preparation

This section uses UMPT_N+UMPT_L, with UMPT_N providing a co-transmission port in NSA networking, as an example to describe main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT NL/NT/NM MBTS side. The NL/NT/NM MBTS connects to the MME and S-GW through a co-transmission 10GE/25GE/50GE port on the gNodeB UMPT_N. The UMPT_N and UMPT_L are interconnected through the backplane. [Figure 5-15](#) shows an example of the network topology, and [Table 5-5](#) describes the IP address plan.

Figure 5-15 Main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT NL/NT/NM MBTS side

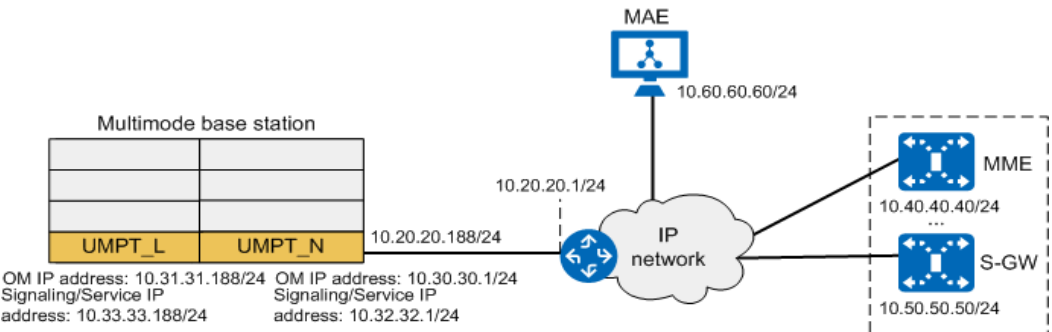


Table 5-5 Data plan

Item	Instance	Remarks
IP address of the port on the router that connects to the UMPT_N	10.20.20.1/24	This IP address is used as the next-hop IP address for transmitting LTE and NR data. If the next-hop IP address for transmitting LTE data needs to be distinguished from that for transmitting NR data, use 10.20.20.1 as the next-hop IP address for transmitting NR data and configure another IP address (for example, 10.21.21.1) as the next-hop IP address for transmitting LTE data. In addition, configure an IP address (for example, 10.21.21.188) for FE port 1 on the UMPT_N board on the same network segment as the next-hop IP address for transmitting LTE data.
IP address of FE port 1 on the UMPT_N of the gNodeB	10.20.20.188/24	None
OMCH IP address of the gNodeB	10.30.30.1/24	NR: management-plane IP address. This is a logical IP address configured on the gNodeB main control board. It can also be a port IP address.
Signaling/Service IP address of the gNodeB	10.32.32.1/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the gNodeB main control board. It can also be a port IP address.
Signaling/Service IP address of the eNodeB	10.33.33.188/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the eNodeB main control board.
OMCH IP address of the eNodeB	10.31.31.188/24	LTE: management-plane IP address. This is a logical IP address configured on the eNodeB main control board.
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
IP address of the MAE	10.60.60.60/24	LTE/NR: peer IP address for the OMCH

Item	Instance	Remarks
VRF index	0 to 31	The value is specified by the VRF.VRFIDX (LTE eNodeB, 5G gNodeB) parameter, and the default value is 0. • If VRF-based isolation is not required by the RAT that does not provide the co-transmission port, this parameter does not need to be configured. • If VRF-based isolation is required by the RAT that does not provide the co-transmission port, this parameter must be set to the same value for both RATs. • The VRF index range depends on the board capability. For details, see the "Hardware" section in <i>VRF</i>.
Backplane tunnel number	0 to 7	The value is specified by the BACKTUNNEL.TUNNELNO (LTE eNodeB, 5G gNodeB) parameter, and the default value is 0. • If VRF-based isolation is not required by the RAT that does not provide the co-transmission port, a backplane tunnel is configured for each RAT, and the tunnel number does not need to be modified. • If VRF-based isolation is required by the RAT that does not provide the co-transmission port, backplane tunnels with different tunnel numbers are configured for the two RATs based on the VRF requirements. The tunnel numbers of the backplane tunnels configured for the two RATs that use backplane co-transmission must be the same.

 NOTE

In this scenario, the UMPT_L does not need to be configured with Ethernet port attributes because the UMPT_N communicates with the UMPT_L through the backplane.

5.4.3.5.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

MAE Side

On the MAE side, add two routes to the eNodeB's DHCP relay with the destination IP address of one route set to the OMCH IP address of the gNodeB and that of the other route set to the lower-level transmission port IP address of the gNodeB.

NR Side

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**:

```
//Adding IP addresses for the gNodeB
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.20.20.188",
MASK="255.255.255.0", VRFIDX=0;
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=1, IP="10.32.32.1",
MASK="255.255.255.255", VRFIDX=0;
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=6, DCN=00, DSRN=0, DSN=6,
TUNNELTYPE=NON_SHAPING;
//Adding uplink routes from the eNodeB to the MME/S-GW/MAE through the gNodeB with Route Type set
to NEXTHOP and Next Hop IP set to the IP address of the directly connected router
ADD IPRT: RTIDX=0, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=1, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=2, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
//Adding downlink routes from the MME/S-GW/MAE to the eNodeB through the gNodeB. When IP-based
co-transmission for the eNodeB and the gNodeB is implemented through the backplane tunnel, the
downlink routes from the MME/S-GW/MAE to the eNodeB through the gNodeB must be configured on the
gNodeB. In addition, Subboard Type must be set to BACK_BOARD and Port Type must be set to TUNNEL.
ADD IPRT: RTIDX=3, SN=7, SBT=BACK_BOARD, DSTIP="10.31.31.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=4, SN=7, SBT=BACK_BOARD, DSTIP="10.33.33.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the eNodeB and gNodeB, if the
eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay. Therefore, DHCP relay needs
to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the eNodeB
and gNodeB, if the eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay.
Therefore, the IP address of the DHCP server needs to be added on the gNodeB. For the eNodeB, the DHCP
server is the MAE.
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//((Optional) Configuring VLANs. There are two methods of configuring different VLANs for LTE and NR
data for isolation:
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
address of the uplink route from the eNodeB to the MME/S-GW through the gNodeB must be different
from the next-hop address of the uplink route for the gNodeB. For example, you can set the next-hop
address of the uplink route from the eNodeB to the MME/S-GW through the gNodeB to 10.20.20.201,
which is different from the next-hop address (10.20.20.1) of the uplink route for the gNodeB.
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the eNodeB and gNodeB data
packets and configuring the mapping between a VLAN group and the next hop
```

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
```

If **GTRANSPARA.TRANSFCGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**:

```
//Adding an interface to Ethernet port 0 on the UMPT_N (in slot 7)
SET ETHPORT: CN=0, SRN=1, SN=7, PN=0, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO,
DUPLEX=AUTO;
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFID=0;
//Adding the INTERFACE setting to the loopback interface of the UMPT_N (in slot 7)
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
//Adding IP addresses for the gNodeB
ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSN=7, DCN=00, DSRN=0, DSN=6, TUNNELTYPE=NON_SHAPING;
//Adding uplink routes from the eNodeB to the MME/S-GW/MAE through the gNodeB with Route Type set
to NEXTHOP and Next Hop IP set to the IP address of the directly connected router
ADD IPRROUTE4: RTID=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTID=1, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTID=2, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
//Adding downlink routes from the MME/S-GW/MAE to the eNodeB through the gNodeB. When IP-based
co-transmission for the eNodeB and the gNodeB is implemented through the backplane tunnel, the
downlink routes from the MME/S-GW/MAE to the eNodeB through the gNodeB must be configured on the
gNodeB. In addition, Route Type must be set to IF and Port Type must be set to TUNNEL.
ADD IPRROUTE4: RTID=3, DSTIP="10.31.31.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTID=4, DSTIP="10.33.33.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the eNodeB and gNodeB, if the
eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay. Therefore, DHCP relay needs
to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the eNodeB
and gNodeB, if the eNodeB is to be deployed using DHCP, the gNodeB needs to work as the relay.
Therefore, the IP address of the DHCP server needs to be added on the gNodeB. For the eNodeB, the DHCP
server is the MAE.
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//((Optional) Configuring VLANs
//There are two methods of configuring different VLANs for LTE and NR data for isolation when the
VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
address of the uplink route from the eNodeB to the MME/S-GW through the gNodeB must be different
from the next-hop address of the uplink route for the gNodeB. For example, you can set the next-hop
address of the uplink route from the eNodeB to the MME/S-GW through the gNodeB to 10.20.20.201,
which is different from the next-hop address (10.20.20.1) of the uplink route for the gNodeB.
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the eNodeB and gNodeB data
packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
ADD VLANMAP: NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
//When the INTERFACE MO is used to configure different VLANs for LTE and NR data for isolation, add
interfaces with ITFTYPE set to VLAN to replace the preceding configuration with ITFTYPE set to NORMAL.
//Configuring the VLAN of the interface for LTE data (After the configuration, add an interface IP address
for the eNodeB.)
ADD INTERFACE: ITFID=2, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for NR data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;
```

LTE Side

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_L to the UMPT_N
ADD TUNNEL: SCN=0, SSRN=0, SSN=6, TUNNELID=6, DCN=00, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the eNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
eNodeB use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the eNodeB can be set as required.
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.33.33.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the eNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set
to TUNNEL.
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=2, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=3000, RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000,
TXPBS=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, SRN=0, SN=6, SBT=BACK_BOARD, PT=TUNNEL, PN=0, JNRSCGRP=DISABLE,
LOCALIP="10.33.33.188", PEERIP="10.50.50.50", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_L to the UMPT_N
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=00, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the eNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
eNodeB use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.33.33.188", MASK="255.255.255.255";
//Adding routes from the eNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set
to TUNNEL.
ADD IPROUTE4: RTIDX=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=1, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=2, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=3000,
RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000, TXPBS=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, TRANS CFGMODE=NEW, JNIPRSCGRP=DISABLE, BPT=TUNNEL, PORTID=0,
LOCALIP="10.33.33.188", PEERIP="10.50.50.50", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

5.4.3.5.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.5.4 Activation Verification

After co-transmission is enabled on the MBTS, check whether the feature is enabled by verifying the status of the IP links between the MBTS and the peer devices.

NR Side

After the configuration file is delivered to the gNodeB and activated, perform the following step on the gNodeB to check whether the transmission links of the gNodeB are normal:

On the gNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links are normal.

```
PING: CN=0, SRN=0, SN=7, SRCIP="10.20.20.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;  
PING: CN=0, SRN=0, SN=7, SRCIP="10.20.20.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

LTE Side

After the configuration file is delivered to the eNodeB and activated, perform the following step on the eNodeB to check whether the transmission links of the eNodeB are normal:

On the eNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links are normal.

```
PING: CN=0, SRN=0, SN=6, SRCIP="10.33.33.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;  
PING: CN=0, SRN=0, SN=6, SRCIP="10.33.33.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

5.4.3.6 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN/TN/MN MBTS Side in IP over FE/GE Mode

5.4.3.6.1 Data Preparation

This section uses UMPT_N+UMPT_L, with UMPT_L providing a co-transmission port in NSA networking, as an example to describe main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LN/TN/MN MBTS side. The LN/TN/MN MBTS connects to the MME and S-GW through a co-transmission 10GE/25GE/50GE port on the eNodeB UMPT_L. The UMPT_N and UMPT_L are interconnected through the backplane. [Figure 5-16](#) shows an example of the network topology, and [Table 5-6](#) describes the IP address plan.

Figure 5-16 Main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LN/TN/MN MBTS side

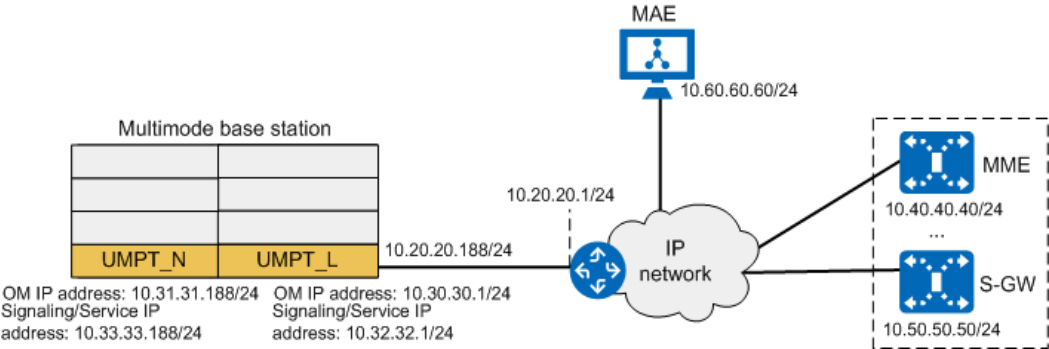


Table 5-6 Data plan

Item	Instance	Remarks
IP address of the port on the router that connects to the UMPT_L	10.20.20.1/24	None
IP address of FE port 1 on the UMPT_L of the eNodeB	10.20.20.188/24	None
OMCH IP address of the eNodeB	10.30.30.1/24	LTE: management-plane IP address. This is a logical IP address configured on the eNodeB main control board. It can also be a port IP address.
Signaling/Service IP address of the eNodeB	10.32.32.1/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the eNodeB main control board. It can also be a port IP address.
Signaling/Service IP address of the gNodeB	10.33.33.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the gNodeB main control board.
OMCH IP address of the gNodeB	10.31.31.188/24	NR: management-plane IP address. This is a logical IP address configured on the gNodeB main control board.
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
IP address of the MAE	10.60.60.60/24	LTE/NR: peer IP address for the OMCH

Item	Instance	Remarks
VRF index	0 to 31	The value is specified by the VRF.VRFIDX (LTE eNodeB, 5G gNodeB) parameter, and the default value is 0. • If VRF-based isolation is not required by the RAT that does not provide the co-transmission port, this parameter does not need to be configured. • If VRF-based isolation is required by the RAT that does not provide the co-transmission port, this parameter must be set to the same value for both RATs. • The VRF index range depends on the board capability. For details, see the "Hardware" section in <i>VRF</i>.
Backplane tunnel number	0 to 7	The value is specified by the BACKTUNNEL.TUNNELNO (LTE eNodeB, 5G gNodeB) parameter, and the default value is 0. • If VRF-based isolation is not required by the RAT that does not provide the co-transmission port, a backplane tunnel is configured for each RAT, and the tunnel number does not need to be modified. • If VRF-based isolation is required by the RAT that does not provide the co-transmission port, backplane tunnels with different tunnel numbers are configured for the two RATs based on the VRF requirements. The tunnel numbers of the backplane tunnels configured for the two RATs that use backplane co-transmission must be the same.

 NOTE

In this scenario, the UMPT_N does not need to be configured with Ethernet port attributes because the UMPT_N communicates with the UMPT_L through the backplane.

5.4.3.6.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

MAE Side

On the MAE side, add two routes to the gNodeB's DHCP relay with the destination IP address of one route set to the OMCH IP address of the eNodeB and that of the other route set to the lower-level transmission port IP address of the eNodeB.

LTE Side

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**:

```
//Adding the IP address of the eNodeB
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.20.20.188",
MASK="255.255.255.0", VRFIDX=0;
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=1, IP="10.32.32.1",
MASK="255.255.255.255", VRFIDX=0;
//Adding a tunnel from the UMPT_L to the UMPT_N
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=6,
TUNNELTYPE=NON_SHAPING;
//Adding uplink routes from the gNodeB to the MME/S-GW/MAE through the eNodeB with Route Type set
to NEXTHOP and Next Hop IP set to the IP address of the directly connected router
ADD IPRT: RTIDX=0, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=1, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=2, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
//Adding downlink routes from the MME/S-GW/MAE to the gNodeB through the eNodeB. When IP-based
co-transmission for the gNodeB and the eNodeB is implemented through the backplane tunnel, the
downlink routes from the MME/S-GW/MAE to the gNodeB through the eNodeB must be configured on the
eNodeB. In addition, Subboard Type must be set to BACK_BOARD and Port Type must be set to TUNNEL.
ADD IPRT: RTIDX=3, SN=7, SBT=BACK_BOARD, DSTIP="10.31.31.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=4, SN=7, SBT=BACK_BOARD, DSTIP="10.33.33.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the eNodeB and gNodeB, if the
gNodeB is to be deployed using DHCP, the eNodeB needs to work as the relay. Therefore, DHCP relay needs
to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the eNodeB
and gNodeB, if the gNodeB is to be deployed using DHCP, the eNodeB needs to work as the relay.
Therefore, the IP address of the DHCP server needs to be added on the eNodeB. For the gNodeB, the DHCP
server is the MAE.
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//((Optional) Configuring VLANs. There are two methods of configuring different VLANs for LTE and NR
data for isolation:
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
address of the uplink route from the gNodeB to the MME/S-GW through the eNodeB must be different
from the next-hop address of the uplink route for the eNodeB. For example, you can set the next-hop
address of the uplink route from the gNodeB to the MME/S-GW through the eNodeB to 10.20.20.201,
which is different from the next-hop address (10.20.20.1) of the uplink route for the eNodeB.
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the gNodeB and eNodeB data
packets and configuring the mapping between a VLAN group and the next hop
```

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIOR=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIOR=0, VLANID=22;
ADD VLANMAP: NEXTHOPIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
```

If **GTRANSPARA.TRANSFCGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**:

```
//Adding an interface to Ethernet port 0 on the UMPT_L (in slot 7)
SET ETHPORT: CN=0, SRN=1, SN=7, PN=0, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO,
DUPLEX=AUTO;
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFIDX=0;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Adding the IP address of the eNodeB
ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
//Adding a tunnel from the UMPT_L to the UMPT_N
ADD BACKTUNNEL: PORTID=0, SSN=7, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=NON_SHAPING;
//Adding uplink routes from the gNodeB to the MME/S-GW/MAE through the eNodeB with Route Type set
to NEXTHOP and Next Hop IP set to the IP address of the directly connected router
ADD IPRROUTE4: RTIDX=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTIDX=1, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTIDX=2, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
//Adding downlink routes from the MME/S-GW/MAE to the gNodeB through the eNodeB. When IP-based
co-transmission for the gNodeB and the eNodeB is implemented through the backplane tunnel, the
downlink routes from the MME/S-GW/MAE to the gNodeB through the eNodeB must be configured on the
eNodeB. In addition, Port Type must be set to TUNNEL.
ADD IPRROUTE4: RTIDX=3, DSTIP="10.31.31.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=4, DSTIP="10.33.33.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the eNodeB and gNodeB, if the
gNodeB is to be deployed using DHCP, the eNodeB needs to work as the relay. Therefore, DHCP relay needs
to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the eNodeB
and gNodeB, if the gNodeB is to be deployed using DHCP, the eNodeB needs to work as the relay.
Therefore, the IP address of the DHCP server needs to be added on the eNodeB. For the gNodeB, the DHCP
server is the MAE.
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//((Optional) Configuring VLANs. There are two methods of configuring different VLANs for LTE and NR
data for isolation when the VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
address of the uplink route from the gNodeB to the MME/S-GW through the eNodeB must be different
from the next-hop address of the uplink route for the eNodeB. For example, you can set the next-hop
address of the uplink route from the gNodeB to the MME/S-GW through the eNodeB to 10.20.20.201,
which is different from the next-hop address (10.20.20.1) of the uplink route for the eNodeB.
ADD VLANMAP: VRFIDX=0, NEXTHOPIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOPIP="10.20.20.201", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the gNodeB and eNodeB data
packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIOR=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIOR=0, VLANID=22;
ADD VLANMAP: NEXTHOPIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
//Configuring different VLANs for NR and LTE data for isolation when the INTERFACE MO is used
//Configuring the VLAN of the interface for NR data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=41, VLANID=22, VRFIDX=0;
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=41, VLANID=33, VRFIDX=0;
```


NR Side

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=0, SSN=6, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting an IP address for the gNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
gNodeB use the logical IP address.
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.33.33.188",
MASK="255.255.255.255", VRFID=0;
//Adding routes from the gNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set
to TUNNEL.
ADD IPRT: RTID=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTID=1, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTID=2, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=3000, RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000,
TXPBS=4000;
//Adding a user-plane link
//Adding an S1-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="10.33.33.188", IPSECSWITCH=DISABLE;
//Adding an S1-U peer
ADD USERPLANEPEER: UPEERID=0, IPVERSION=IPv4, PEERIPV4="10.50.50.50", IPSECSWITCH=DISABLE;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPv4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
//Setting an IP address for the gNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
gNodeB use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.33.33.188", MASK="255.255.255.255";
//Adding routes from the gNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPROUTE4: RTID=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTID=1, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTID=2, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=3000,
RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000, TXPBS=4000;
//Adding a user-plane link
//Adding an S1-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="10.33.33.188", IPSECSWITCH=DISABLE;
//Adding an S1-U peer
ADD USERPLANEPEER: UPEERID=0, IPVERSION=IPv4, PEERIPV4="10.50.50.50", IPSECSWITCH=DISABLE;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPv4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

5.4.3.6.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.6.4 Activation Verification

After co-transmission is enabled on the MBTS, check whether the feature is enabled by verifying the status of the IP links between the MBTS and the peer devices.

LTE Side

After the configuration file is delivered to the eNodeB and activated, perform the following step on the eNodeB to check whether the transmission links of the eNodeB are normal:

On the eNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links are normal.

```
PING: CN=0, SRN=0, SN=7, SRCIP="10.20.20.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;  
PING: CN=0, SRN=0, SN=7, SRCIP="10.20.20.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

NR Side

After the configuration file is delivered to the gNodeB and activated, perform the following step on the gNodeB to check whether the transmission links of the gNodeB are normal:

On the gNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links are normal.

```
PING: CN=0, SRN=0, SN=6, SRCIP="10.33.33.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;  
PING: CN=0, SRN=0, SN=6, SRCIP="10.33.33.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

5.4.3.7 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LG+N/TG+N/MG+N MBTS Side in IP over FE/GE Mode

In this scenario, the GSM base station can be a GBTS or an eGBTS, and the main control board must be a GTMUc/UMPT. The main control board of NR must be a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga/UMPTga.

5.4.3.7.1 Data Preparation

This section uses GTMU+UMPT_L+UMPT_N, with UMPT_L providing a co-transmission port in NSA networking, as an example to describe main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LG+N/TG+N/MG+N MBTS side. In this scenario, a 10GE/25GE/50GE port on the eNodeB UMPT_L serves as the co-transmission port of the separate-MPT LG+N/TG+N/MG+N MBTS and connects to the BSC/MME/S-GW. The GTMU and UMPT_L in one BBU are interconnected through the backplane. The UMPT_N and UMPT_L are interconnected by using interconnection cables. [Figure 5-17](#) shows an example of the network topology, and [Table 5-7](#) describes the IP address plan.

Figure 5-17 Main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LG+N/TG+N/MG+N MBTS side

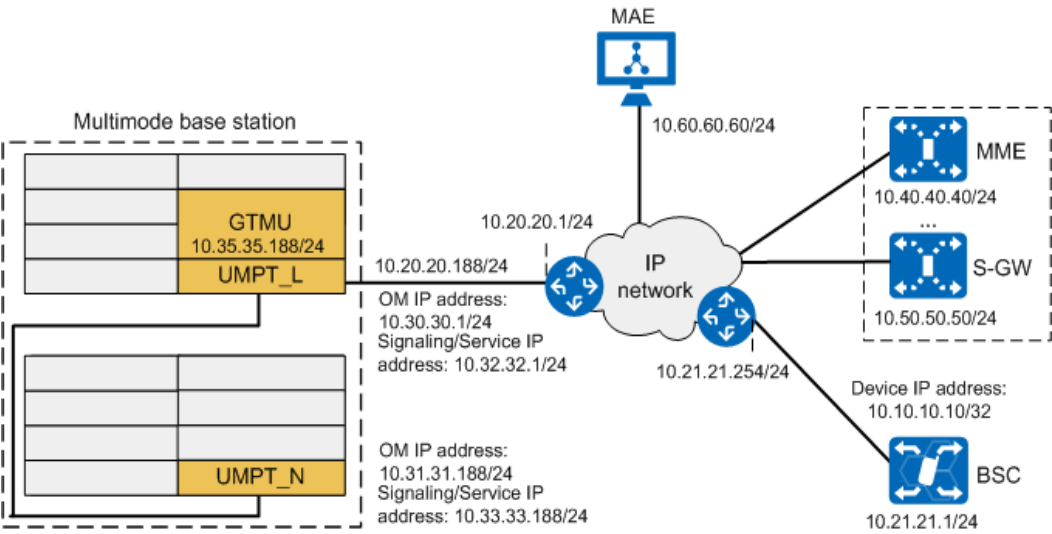


Table 5-7 Data plan

Item	Instance	Remarks
Device IP address of the BSC	10.10.10.10/32	None
Port IP address of the BSC	10.21.21.1/24	None
IP address of the port on the router that connects to the BSC	10.21.21.254/24	None
IP address of the port on the router that connects to the UMPT_L	10.20.20.1/24	None
OMCH IP address of the eNodeB	10.30.30.1/24	LTE: management-plane IP address. This IP address is configured on the main control board.
Signaling/Service IP address of the eNodeB	10.32.32.1/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of 10GE port 1 on the eNodeB UMPT_L	10.20.20.188/24	LTE: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment This IP address is configured on the co-transmission port.
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link

Item	Instance	Remarks
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
IP address of the MAE	10.60.60.60/24	LTE/NR: peer IP address for the OMCH
OMCH IP address of the gNodeB	10.31.31.188/24	NR: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the gNodeB	10.33.33.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of the GBTS	10.35.35.188/24	GSM: communication IP address of the GBTS. This is a logical IP address configured on the GBTS main control board.
OM/Signaling/Service IP address of the eGBTS	10.35.35.188/24	The same IP address is used for OM, signaling, and services for the eGBTS.
ESN of the GBTS	abcdefghijklm n	For the GBTS, the ESN needs to be planned. For the eGBTS, the ESN does not need to be planned.
VRF index	0 to 31	The value is specified by the VRF.VRFIDX (LTE eNodeB, 5G gNodeB) parameter, and the default value is 0. • If VRF-based isolation is not required by the RAT that does not provide the co-transmission port, this parameter does not need to be configured. • If VRF-based isolation is required by the RAT that does not provide the co-transmission port, this parameter must be set to the same value for both RATs. • The VRF index range depends on the board capability. For details, see the "Hardware" section in <i>VRF</i>.

Item	Instance	Remarks
Backplane tunnel number	0 to 7	The value is specified by the BACKTUNNEL.TUNNELNO (LTE eNodeB, 5G gNodeB) parameter, and the default value is 0. - If VRF-based isolation is not required by the RAT that does not provide the co-transmission port, a backplane tunnel is configured for each RAT, and the tunnel number does not need to be modified. - If VRF-based isolation is required by the RAT that does not provide the co-transmission port, backplane tunnels with different tunnel numbers are configured for the two RATs based on the VRF requirements. The tunnel numbers of the backplane tunnels configured for the two RATs that use backplane co-transmission must be the same.

NOTE

- In this scenario, the GTMU does not need to be configured with Ethernet port attributes because the GTMU communicates with the UMPT_L through the backplane. The UMPT_N does not need to be configured with Ethernet port attributes because the UMPT_N communicates with the UMPT_L through the CI port.
- Unlike non-co-transmission scenarios, a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga must be configured on the eNodeB when a separate-MPT LG+N MBTS uses main-control-board-based IP co-transmission through backplane interconnection.

5.4.3.7.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

(Optional) MAE Side

On the MAE side, add two routes to the gNodeB's DHCP relay with the destination IP address of one route set to the OMCH IP address of the eNodeB and that of the other route set to the lower-level transmission port IP address of the eNodeB.

GSM Side (GBTS)

```
//Adding a tunnel from the GTMU to the UMPT_L
ADD BTSTUNNEL: IDTYPE=BYID, BTSID=10, SRCN=0, SRCSRN=0, SRCSN=6, TN=0, DSTCN=0, DSTSRN=0,
DSTSN=7;
//Adding the IP address of the GTMU
ADD BTSDEVIP: IDTYPE=BYID, BTSID=10, PT=LOOPINTERFACE, PN=0, CN=0, SRN=0, SN=6,
IP="10.35.35.188", MASK="255.255.255.255";
//Setting the communication IP address of the GBTS. Unlike non-co-transmission scenarios, this scenario
requires that the GBTS use the logical IP address, that is, BTS Communication Type must be set to LOGICIP.
In non-co-transmission scenarios, the IP address of the GBTS can be set as required.
SET BTSIP: IDTYPE=BYID, BTSID=10, BTSCOMTYPE=LOGICIP, BTSIP="10.35.35.188", BSCIP="10.10.10.10",
BTSMUTIP=NO;
//Adding a route from the GBTS to the BSC. Unlike non-co-transmission scenarios, this scenario requires
that Route Type be set to OUTIF, Interface Type be set to TUNNEL, and Outgoing Interface No. be set to the
number of the added tunnel.
ADD BTSIPRT: IDTYPE=BYID, BTSID=10, RTIDX=1, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=OUTIF, CN=0, SRN=0, SN=6, ITFType=TUNNEL, IFNO=0;
//Adding the ESN of the GBTS
ADD BTSESN: IDTYPE=BYID, BTSID=10, MAINDEVTAB="abcdefghijklmn", OMBEARBOARD=BACKBOARD;
//Optional Adding two routes from the BSC to the GBTS's DHCP relay with the destination IP address of
one route set to the OMCH IP address of the eNodeB and that of the other route set to the lower-level
transmission port IP address of the eNodeB
ADD IPRT: SRN=0, SN=16, DSTIP="10.20.20.188", DSTMASK="255.255.255.255", NEXTHOP="10.21.21.254",
PRIORITY=HIGH, REMARK="relay";
```

GSM Side (eGBTS)

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the GTMU to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=0, SSN=6, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the eGBTS can be set as required.
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.35.35.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to
TUNNEL.
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.21.21.1", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the GTMU to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=6;
ADD INTERFACE: ITFID=1, ITFType=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.35.35.188", MASK="255.255.255.255";
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRUTE4: RTIDX=0, DSTIP="10.21.21.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

NR Side

NOTE

In this scenario, the gNodeB uses the CI port for communication. Therefore, Ethernet ports are not required.

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the gNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
gNodeB use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the gNodeB can be set as required.
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.33.33.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the gNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set
to TUNNEL.
ADD IPRT: RTIDX=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=2, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=1, SN=7, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=3000, RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000,
TXPBS=4000;
//Adding a user-plane link
//Adding an S1-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="10.33.33.188", IPSECSWITCH=DISABLE;
//Adding an S1-U peer
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="10.50.50.50", IPSECSWITCH=DISABLE;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the gNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
gNodeB use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.33.33.188", MASK="255.255.255.255";
//Adding routes from the gNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPROUTE4: RTIDX=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=1, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=2, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=3000,
RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000, TXPBS=4000;
//Adding a user-plane link
//Adding an S1-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="10.33.33.188", IPSECSWITCH=DISABLE;
//Adding an S1-U peer
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="10.50.50.50", IPSECSWITCH=DISABLE;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

LTE Side

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding the IP address of the eNodeB
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.20.20.188",
MASK="255.255.255.0", VRFIDX=0;
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=1, IP="10.32.32.1",
MASK="255.255.255.255", VRFIDX=0;
//Adding tunnels from the UMPT_L to the GTMU and UMPT_N
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=SHAPING;
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=1, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Adding an uplink route from the GBTS/eGBTS to the BSC through the eNodeB and uplink routes from the
gNodeB to the MME/S-GW/MAE through the eNodeB. In this step, set Route Type to NEXTHOP and Next
Hop IP to the IP address of the directly connected router.
ADD IPRT: RTIDX=0, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=1, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=2, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=3, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
//Adding a downlink route from the BSC to the GBTS/eGBTS through the eNodeB and downlink routes from
the MME/S-GW/MAE to the gNodeB through the eNodeB. When IP-based co-transmission for the GBTS/
eGBTS, gNodeB, and eNodeB is implemented through backplane tunnels, the downlink route from the BSC
to the GBTS/eGBTS through the eNodeB and the downlink routes from the MME/S-GW/MAE to the gNodeB
through the eNodeB must be configured on the eNodeB. In addition, Subboard Type must be set to
BACK_BOARD and Port Type must be set to TUNNEL.
ADD IPRT: RTIDX=4, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.35.35.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=5, SN=7, SBT=BACK_BOARD, DSTIP="10.31.31.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=1;
ADD IPRT: RTIDX=6, SN=7, SBT=BACK_BOARD, DSTIP="10.33.33.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=1;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, gNodeB, and
eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS and gNodeB are to be deployed using
DHCP. Therefore, DHCP relay needs to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/
eGBTS, gNodeB, and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS and gNodeB are to
be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be added on the eNodeB.
For the GBTS, the DHCP server is the BSC. For the eGBTS, the DHCP server is the MAE. For the gNodeB, the
DHCP server is the MAE.
//Adding the IP address of the DHCP server for the eGBTS/gNodeB
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//Adding the IP address of the DHCP server for the GBTS
ADD DHCPVRIP: DHCPVRIP="10.10.10.10";
//((Optional) Configuring VLANs. There are two methods of configuring different VLANs for GSM and LTE
data for isolation:
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink routes from the GBTS/eGBTS/gNodeB to the BSC/MME/S-GW through the eNodeB
must be different from the next-hop address of the uplink route for the eNodeB. For example, you can set
the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through the eNodeB to
10.20.20.101, and set the next-hop address of the uplink route from the gNodeB to the MME/S-GW through
the eNodeB to 10.20.20.201, which are different from the next-hop address (10.20.20.1) of the uplink route
for the eNodeB.
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.101",
MASK="255.255.255.255", VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=32, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GBTS/eGBTS, gNodeB, and
eNodeB data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
```



```
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=
VLANGROUP, VLANGROUPNO=1, SETPRIO=DISABLE;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding an interface to Ethernet port 1 on the UMPT_L (in slot 7)
SET ETHPORT: CN=0, SRN=1, SN=7, PN=1, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO,
DUPLEX=AUTO;
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFID=0;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
//Adding the IP address of the eNodeB
ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
//Adding tunnels from the UMPT_L to the GTMU and UMPT_N
ADD BACKTUNNEL: PORTID=0, SSN=7, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=NON_SHAPING;
ADD BACKTUNNEL: PORTID=1, SSN=7, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=NON_SHAPING;
//Adding an uplink route from the GBTS/eGBTS to the BSC through the eNodeB and uplink routes from the
gNodeB to the MME/S-GW/MAE through the eNodeB. In this step, set Route Type to NEXTHOP and Next
Hop IP to the IP address of the directly connected router.
ADD IPRROUTE4: RTID=0, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTID=1, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTID=2, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTID=3, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
//Adding a downlink route from the BSC to the GBTS/eGBTS through the eNodeB and downlink routes from
the MME/S-GW/MAE to the gNodeB through the eNodeB. When IP-based co-transmission for the GBTS/
eGBTS, gNodeB, and eNodeB is implemented through backplane tunnels, the downlink route from the BSC
to the GBTS/eGBTS through the eNodeB and the downlink routes from the MME/S-GW/MAE to the gNodeB
through the eNodeB must be configured on the eNodeB. In addition, Port Type must be set to TUNNEL.
ADD IPRROUTE4: RTID=4, DSTIP="10.35.35.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTID=5, DSTIP="10.31.31.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=1;
ADD IPRROUTE4: RTID=6, DSTIP="10.33.33.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=1;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, gNodeB, and
eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS and gNodeB are to be deployed using
DHCP. Therefore, DHCP relay needs to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/
eGBTS, gNodeB, and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS and gNodeB are to
be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be added on the eNodeB.
For the GBTS, the DHCP server is the BSC. For the eGBTS, the DHCP server is the MAE. For the gNodeB, the
DHCP server is the MAE.
//Adding the IP address of the DHCP server for the eGBTS/gNodeB
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//Adding the IP address of the DHCP server for the GBTS
ADD DHCPVRIP: DHCPVRIP="10.10.10.10";
//((Optional) Configuring VLANs
//There are two methods of configuring different VLANs for GSM and LTE data for isolation when the
VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink routes from the GBTS/eGBTS/gNodeB to the BSC/MME/S-GW through the eNodeB
must be different from the next-hop address of the uplink route for the eNodeB. For example, you can set
the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through the eNodeB to
10.20.20.101, and set the next-hop address of the uplink route from the gNodeB to the MME/S-GW through
the eNodeB to 10.20.20.201, which are different from the next-hop address (10.20.20.1) of the uplink route
for the eNodeB.
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101",
MASK="255.255.255.255",VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=32, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GBTS/eGBTS, gNodeB, and
```

```
eNodeB data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOP="10.20.20.1", MASK="255.255.255.255", VLANMODE=
VLANGROUP, VLANGROUPNO=1, SETPRI=DISABLE;
//Configuring different VLANs for GSM, NR, and LTE data for isolation when the INTERFACE MO is used
//Configuring the VLAN of the interface for GSM data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=22, VRFID=0;
//Configuring the VLAN of the interface for NR data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=34, VRFID=0;
```

5.4.3.7.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.7.4 Activation Verification

After co-transmission is enabled on the base station, check whether it is enabled by verifying the status of the IP links between the base station and the peer devices.

LTE Side

After the configuration file is delivered to the eNodeB and activated, perform the following step on the eNodeB to check whether the transmission links between the eNodeB and the MME/S-GW are normal:

On the eNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the eNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

GSM Side

After the configuration file is delivered to the GBTS/eGBTS and activated, perform the following step on the BSC to verify that the transmission link between the BSC and the GBTS/eGBTS is normal:

On the BSC, ping the IP address of the GTMUc. If the ping operations succeed, the transmission links are normal.

```
PING IP: SIPADDR="10.10.10.10", DESTIP="10.35.35.188", NEXTHOP="10.21.21.254", CONTPING=NO;
```

NR Side

On the eNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the eNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=1, SN=7, SRCIP="10.33.33.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
PING: CN=0, SRN=1, SN=7, SRCIP="10.33.33.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

5.4.3.8 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT N+GL/N+GT/N+GM MBTS Side in IP over FE/GE Mode

In this scenario, the GSM base station can be a GBTS or an eGBTS, and the main control board must be a GTMUc/UMPT. The main control board of NR must be a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga/UMPTga.

5.4.3.8.1 Data Preparation

Figure 5-18 shows an example of main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT N+GL/N+GT/N+GM MBTS side. In this scenario, a 10GE/25GE/50GE port on the gNodeB UMPT_N serves as the co-transmission port of the separate-MPT MBTS and connects to the BSC/MME/S-GW. The UMPT_N connects to the GTMU and UMPT_L through the UCIU. Table 5-8 describes the IP address plan.

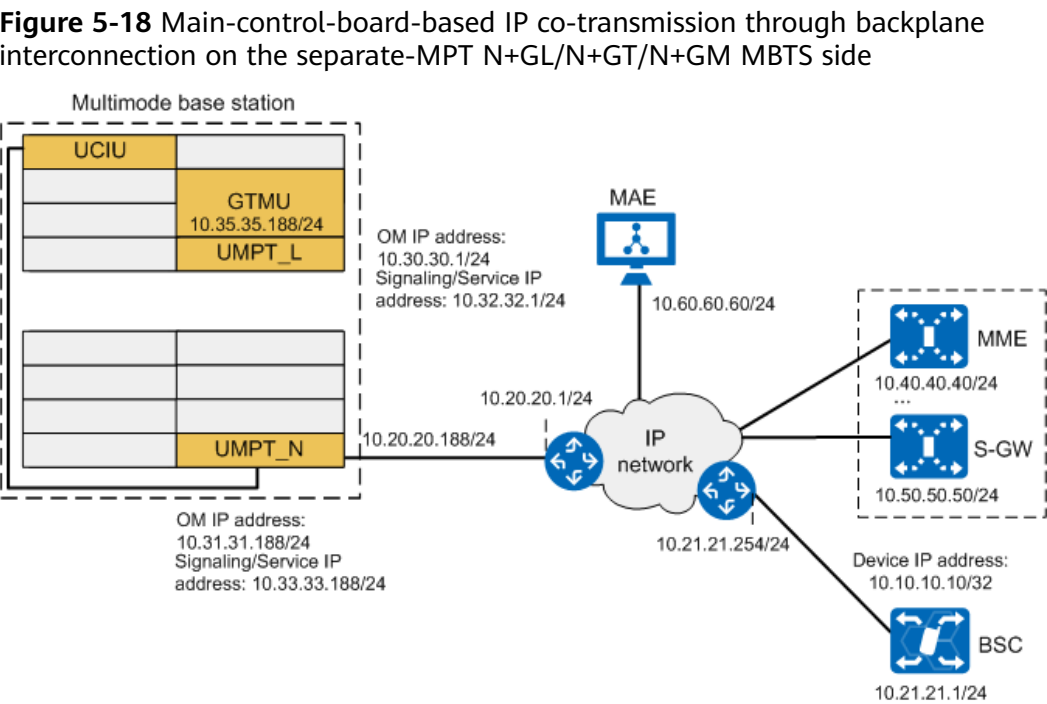


Table 5-8 Data plan

Item	Instance	Remarks
Device IP address of the BSC	10.10.10.10/32	None
Port IP address of the BSC	10.21.21.1/24	None
IP address of the port on the router that connects to the BSC	10.21.21.254/24	None

Item	Instance	Remarks
IP address of the port on the router that connects to the UMPT_N	10.20.20.1/24	None
OMCH IP address of the eNodeB	10.30.30.1/24	LTE: management-plane IP address. This IP address is configured on the main control board.
Signaling/Service IP address of the eNodeB	10.32.32.1/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of 10GE port 1 on the gNodeB UMPT_N	10.20.20.188/24	NR: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment This IP address is configured on the co-transmission port.
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
IP address of the MAE	10.60.60.60/24	NR/LTE: peer IP address for the OMCH
OMCH IP address of the gNodeB	10.31.31.188/24	NR: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the gNodeB	10.33.33.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of the GBTS	10.35.35.188/24	GSM: communication IP address of the GBTS. This is a logical IP address configured on the GBTS main control board.
OM/Signaling/Service IP address of the eGBTS	10.35.35.188/24	The same IP address is used for OM, signaling, and services for the eGBTS.
ESN of the GBTS	abcdefghijklmn	For the GBTS, the ESN needs to be planned. For the eGBTS, the ESN does not need to be planned.

NOTE

In this scenario, the GTMU and UMPT_L do not need to be configured with Ethernet port attributes because the GTMU and UMPT_L communicate with the UCIU through the backplane and the UCIU communicates with the UMPT_N through the M interface.

5.4.3.8.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

(Optional) MAE Side

On the MAE side, add two routes to the eNodeB's DHCP relay with the destination IP address of one route set to the OMCH IP address of the gNodeB and that of the other route set to the lower-level transmission port IP address of the gNodeB.

GSM Side (GBTS)

```
//Adding a tunnel from the GTMU to the UMPT_N
ADD BTSTUNNEL: IDTYPE=BYID, BTSID=10, SRCN=0, SRCRN=0, SRCSN=6, TN=0, DSTCN=0, DSTSRN=1,
DSTSN=7;
//Adding the IP address of the GTMU
ADD BTSDEVIP: IDTYPE=BYID, BTSID=10, PT=LOOPINTERFACE, PN=0, CN=0, SRN=0, SN=6,
IP="10.35.35.188", MASK="255.255.255.255";
//Setting the communication IP address of the GBTS. Unlike non-co-transmission scenarios, this scenario
requires that the GBTS use the logical IP address, that is, BTS Communication Type must be set to LOGICIP.
In non-co-transmission scenarios, the IP address of the GBTS can be set as required.
SET BTSIP: IDTYPE=BYID, BTSID=10, BTSCOMTYPE=LOGICIP, BTSIP="10.35.35.188", BSCIP="10.10.10.10",
BTSMUTIP=NO;
//Adding a route from the GBTS to the BSC. Unlike non-co-transmission scenarios, this scenario requires
that Route Type be set to OUTIF, Interface Type be set to TUNNEL, and Outgoing Interface No. be set to the
number of the added tunnel.
ADD BTSIPRT: IDTYPE=BYID, BTSID=10, RTIDX=1, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=OUTIF, CN=0, SRN=0, SN=6, ITFType=TUNNEL, IFNO=0;
//Adding the ESN of the GBTS
ADD BTSESN: IDTYPE=BYID, BTSID=10, MAINDEVTAB="abcdefghijklmn", OMBEARBOARD=BACKBOARD;
//Optional Adding two routes from the BSC to the GBTS's DHCP relay with the destination IP address of
one route set to the OMCH IP address of the gNodeB and that of the other route set to the lower-level
transmission port IP address of the gNodeB
ADD IPRT: SRN=0, SN=16, DSTIP="10.20.20.188", DSTMASK="255.255.255.255", NEXTHOP="10.21.21.254",
PRIORITY=HIGH, REMARK="relay";
```

GSM Side (eGBTS)

If **GTRANSPARA.TRANSCFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the GTMU to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=0, SSN=6, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the eGBTS can be set as required.
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.35.35.188",
MASK="255.255.255.255", VRFIDX=0;
```

```
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to
TUNNEL.
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.21.21.1", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the GTMU to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.35.35.188", MASK="255.255.255.255";
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTIDX=0, DSTIP="10.21.21.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=0, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

LTE Side

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_L to the UMPT_N
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Adding the IP address of the eNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
eNodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the eNodeB can be
set as required.
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.32.32.1",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the eNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set
to TUNNEL.
ADD IPRT: RTIDX=0, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=10, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=0, SN=7, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=4000, RXBW=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, SN=7, SBT=BACK_BOARD, PT=TUNNEL, JNRSCGRP=DISABLE,
LOCALIP="10.32.32.1", PEERIP="10.50.50.50", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_L to the UMPT_N
ADD BACKTUNNEL: PORTID=0, SSN=7, DCN=0, DSRN=1, DSN=7;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Adding the IP address of the eNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
```

```
eNodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the eNodeB can be
set as required.
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
//Adding routes from the eNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPROUTE4: RTIDX=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=10, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=4000,
RXBW=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, TRANSCFGMODE=NEW, JNIPRSCGRP=DISABLE, BPT=TUNNEL, PORTID=0,
LOCALIP="10.32.32.1", PEERIP="10.50.50.50", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

NR Side

NOTE

In this scenario, the gNodeB uses the CI port for communication. Therefore, Ethernet ports are not required.

If **GTRANSPARA.TRANSCFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding tunnels from the UMPT_N to the GTMU and UMPT_L
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=1, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=SHAPING;
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting IP addresses for the gNodeB
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.33.33.188",
MASK="255.255.255.255", VRFID=0;
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.20.20.188",
MASK="255.255.255.0", VRFID=0;
//Adding uplink routes from the eNodeB to the MME/S-GW/MAE through the gNodeB and an uplink route
from the GBTS/eGBTS to the BSC through the gNodeB
ADD IPRT: RTIDX=0, CN=0, SRN=1, SN=7, SBT= BASE_BOARD, DSTIP="10.40.40.40",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=10, CN=0, SRN=1, SN=7, SBT= BASE_BOARD, DSTIP="10.50.50.50",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=1, CN=0, SRN=1, SN=7, SBT= BASE_BOARD, DSTIP="10.60.60.60",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=2, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
//Adding downlink routes from the MAE/MME/S-GW to the eNodeB through the gNodeB and a downlink
route from the BSC to the GBTS/eGBTS through the gNodeB. When IP-based co-transmission for the GBTS/
eGBTS, eNodeB, and gNodeB is implemented through backplane tunnels, the downlink routes to the
MME/S-GW/MAE to the eNodeB through the gNodeB and the downlink route from the BSC to the GBTS/
eGBTS through the gNodeB must be configured on the gNodeB. In addition, Subboard Type must be set to
BACK_BOARD and Port Type must be set to TUNNEL.
ADD IPRT: RTIDX=3, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.30.30.1",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=4, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.32.32.1",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=5, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.35.35.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, gNodeB, and
eNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS and eNodeB are to be deployed using
DHCP. Therefore, DHCP relay needs to be enabled for the gNodeB.
SET DHCPRELAYSITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/
eGBTS, gNodeB, and eNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS and eNodeB are to
be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be added on the gNodeB.
For the eNodeB, the DHCP server is the MAE. For the GBTS, the DHCP server is the BSC. For the eGBTS, the
```

DHCP server is the MAE.
 //Adding the IP address of the DHCP server for the eGBTS/eNodeB
 ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
 //Adding the IP address of the DHCP server for the GBTS
 ADD DHCPVRIP: DHCPVRIP="10.10.10.10";
 //(Optional) Configuring VLANs. There are two methods of configuring different VLANs for GSM, NR, and LTE data for isolation:
 //Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop addresses of the uplink route from the GBTS/eGBTS to the BSC through the gNodeB and the uplink route from the eNodeB to the MME/S-GW through the gNodeB must be different from the next-hop address of the uplink route for the gNodeB. For example, you can set the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through the gNodeB to 10.20.20.101, and set the next-hop address of the uplink route from the eNodeB to the MME/S-GW through the gNodeB to 10.20.20.201, which are different from the next-hop address (10.20.20.1) of the uplink route for the gNodeB.
 ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=2, SETPRIO=DISABLE;
 ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
 ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=32, SETPRIO=DISABLE;
 //Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, NR, and LTE data packets and configuring the mapping between a VLAN group and the next hop
 ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=2;
 ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=32;
 ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
 ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP, VLANGROUPNO=1;

If **GTRANSPARA.TRANSFCGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

//Adding tunnels from the UMPT_N to the GTMU and UMPT_L
 ADD BACKTUNNEL: PORTID=1, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=6;
 ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=7;
 //Adding an interface to an Ethernet port on the UMPT_N (in slot 7 of subrack 1)
 SET ETHPORT: CN=0, SRN=1, SN=7, PN=1, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO, DUPLEX=AUTO;
 ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFID=0;
 //Adding the INTERFACE setting to a loopback interface
 ADD LOOPBACK: PORTID=700, SRN=1, SN=7;
 ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
 //Setting IP addresses for the gNodeB
 ADD IPADDR4: ITFID=1, IP="10.33.33.188", MASK="255.255.255.255";
 ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
 //Adding uplink routes from the eNodeB to the MME/S-GW/MAE through the gNodeB and an uplink route from the GBTS/eGBTS to the BSC through the gNodeB
 ADD IPRROUTE4: RTID=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
 ADD IPRROUTE4: RTID=10, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
 ADD IPRROUTE4: RTID=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
 ADD IPRROUTE4: RTID=2, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
 //Adding downlink routes from the MAE/MME/S-GW to the eNodeB through the gNodeB and a downlink route from the BSC to the GBTS/eGBTS through the gNodeB. When IP-based co-transmission for the GBTS/eGBTS, eNodeB, and gNodeB is implemented through backplane tunnels, the downlink routes from the MME/S-GW/MAE to the eNodeB through the gNodeB and the downlink route from the BSC to the GBTS/eGBTS through the gNodeB must be configured on the gNodeB. In addition, Port Type must be set to TUNNEL.
 ADD IPRROUTE4: RTID=3, DSTIP="10.30.30.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
 ADD IPRROUTE4: RTID=4, DSTIP="10.32.32.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
 ADD IPRROUTE4: RTID=5, DSTIP="10.35.35.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
 //(Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, gNodeB, and eNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS and eNodeB are to be deployed using DHCP. Therefore, DHCP relay needs to be enabled for the gNodeB.
 SET DHCPRELAYSWITCH: ES=ENABLE;
 //(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/

eGBTS, gNodeB, and eNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS and eNodeB are to be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be added on the gNodeB. For the eNodeB, the DHCP server is the MAE. For the GBTS, the DHCP server is the BSC. For the eGBTS, the DHCP server is the MAE.

//Adding the IP address of the DHCP server for the eGBTS/eNodeB

ADD DHCPVRIP: DHCPVRIP="10.60.60.60";

//Adding the IP address of the DHCP server for the GBTS

ADD DHCPVRIP: DHCPVRIP="10.10.10.10";

//(Optional) Configuring VLANs. There are two methods of configuring different VLANs for GSM, NR, and LTE data for isolation when the VLANMAP MO is used.

//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop addresses of the uplink route from the GBTS/eGBTS to the BSC through the gNodeB and the uplink route from the eNodeB to the MME/S-GW through the gNodeB must be different from the next-hop address of the uplink route for the gNodeB. For example, you can set the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through the gNodeB to 10.20.20.101, and set the next-hop address of the uplink route from the eNodeB to the MME/S-GW through the gNodeB to 10.20.20.201, which are different from the next-hop address (10.20.20.1) of the uplink route for the gNodeB.

ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=2, SETPRIO=DISABLE;

ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;

ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=32, SETPRIO=DISABLE;

//Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, NR, and LTE data packets and configuring the mapping between a VLAN group and the next hop

ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=2;

ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=32;

ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;

ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP, VLANGROUPNO=1;

//Configuring different VLANs for GSM, NR, and LTE data for isolation when the INTERFACE MO is used

//Configuring the VLAN of the interface for NR data

ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=41, VLANID=2, VRFID=0;

//Configuring the VLAN of the interface for LTE data

ADD INTERFACE: ITFID=1, ITFTYPE=VLAN, PT=ETH, PORTID=41, VLANID=32, VRFID=0;

//Configuring the VLAN of the interface for GSM data

ADD INTERFACE: ITFID=2, ITFTYPE=VLAN, PT=ETH, PORTID=41, VLANID=22, VRFID=0;

5.4.3.8.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.8.4 Activation Verification

After co-transmission is enabled on the base station, check whether it is enabled by verifying the status of the IP links between the base station and the peer devices.

NR Side

On the gNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the gNodeB and the MME/S-GW are normal.

PING: CN=0, SRN=1, SN=7, SRCIP="10.20.20.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
PING: CN=0, SRN=1, SN=7, SRCIP="10.20.20.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;

LTE Side

After the configuration file is delivered to the eNodeB and activated, perform the following step on the eNodeB to check whether the transmission links between the eNodeB and the MME/S-GW are normal:

On the eNodeB, ping the IP addresses of the MME/S-GW. If the ping operations succeed, the transmission links are normal.

```
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;  
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

GSM Side

After the configuration file is delivered to the GBTS/eGBTS and activated, perform the following step on the BSC to verify that the transmission link between the BSC and the GBTS/eGBTS is normal:

On the BSC, ping the IP address of the GTMUc. If the ping operations succeed, the transmission links are normal.

```
PING IP: SIPADDR="10.10.10.10", DESTIP="10.35.35.188", NEXTHOP="10.21.21.254", CONTPING=NO;
```

5.4.3.9 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LU+N/TU+N/MU+N MBTS Side in IP over FE/GE Mode

5.4.3.9.1 Data Preparation

This section uses UMPT_U+UMPT_L+UMPT_N, with UMPT_L providing a co-transmission port in NSA networking, as an example to describe main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LU+N/TU+N/MU+N MBTS side. In this scenario, a 10GE/25GE/50GE port on the eNodeB UMPT_L serves as the co-transmission port of the separate-MPT LU+N/TU+N/MU+N MBTS and connects to the RNC/MME/S-GW. The UMPT_U and UMPT_L in one BBU are interconnected through the backplane. The UMPT_N and UMPT_L are interconnected by using interconnection cables. [Figure 5-19](#) shows an example of the network topology, and [Table 5-9](#) describes the IP address plan.

Figure 5-19 Main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LU+N/TU+N/MU+N MBTS side

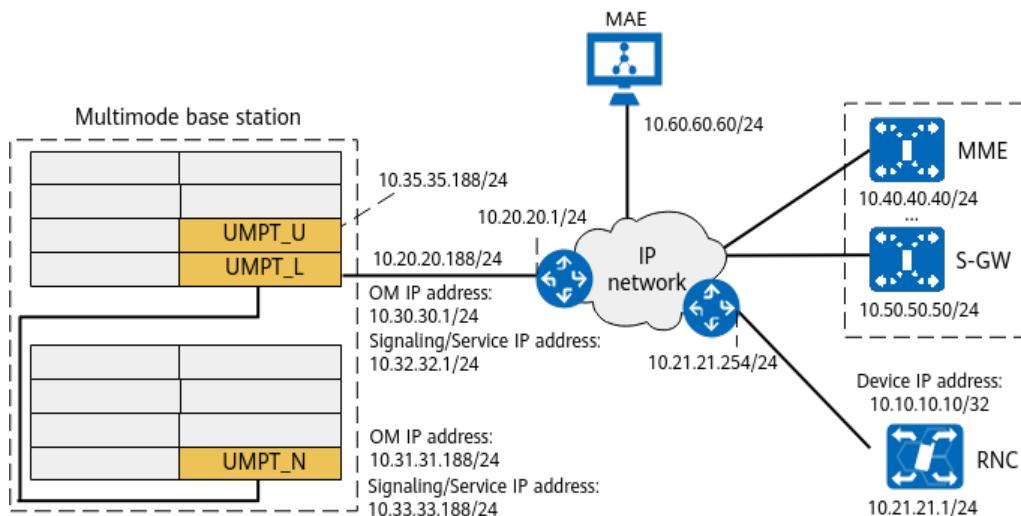


Table 5-9 Data plan

Item	Instance	Remarks
Device IP address of the RNC	10.10.10.10/32	None
Port IP address of the RNC	10.21.21.1/24	None
IP address of the port on the router that connects to the RNC	10.21.21.254/24	None
IP address of the port on the router that connects to the UMPT_L	10.20.20.1/24	None
OMCH IP address of the eNodeB	10.30.30.1/24	LTE: management-plane IP address. This IP address is configured on the main control board.
Signaling/Service IP address of the eNodeB	10.32.32.1/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of 10GE port 1 on the eNodeB UMPT_L	10.20.20.188/24	LTE: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment. This IP address is configured on the co-transmission port.
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link

Item	Instance	Remarks
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
IP address of the MAE	10.60.60.60/24	LTE/NR: peer IP address for the OMCH
OMCH IP address of the gNodeB	10.31.31.188/24	NR: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the gNodeB	10.33.33.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of the NodeB	10.35.35.188/24	UMTS: communication IP address of the NodeB. This is a logical IP address configured on the NodeB main control board.
OM/Signaling/Service IP address of the NodeB	10.35.35.188/24	The same IP address is used for OM, signaling, and services for the NodeB
VRF index	0 to 31	The value is specified by the VRF.VRFIDX (LTE eNodeB, 5G gNodeB) parameter, and the default value is 0. • If VRF-based isolation is not required by the RAT that does not provide the co-transmission port, this parameter does not need to be configured. • If VRF-based isolation is required by the RAT that does not provide the co-transmission port, this parameter must be set to the same value for both RATs. • The VRF index range depends on the board capability. For details, see the "Hardware" section in <i>VRF</i>.

Item	Instance	Remarks
Backplane tunnel number	0 to 7	The value is specified by the BACKTUNNEL.TUNNELNO (LTE eNodeB, 5G gNodeB) parameter, and the default value is 0. - If VRF-based isolation is not required by the RAT that does not provide the co-transmission port, a backplane tunnel is configured for each RAT, and the tunnel number does not need to be modified. - If VRF-based isolation is required by the RAT that does not provide the co-transmission port, backplane tunnels with different tunnel numbers are configured for the two RATs based on the VRF requirements. The tunnel numbers of the backplane tunnels configured for the two RATs that use backplane co-transmission must be the same.

NOTE

- In this scenario, the UMPT_U does not need to be configured with Ethernet port attributes because the UMPT_U communicates with the UMPT_L through the backplane. The UMPT_N does not need to be configured with Ethernet port attributes because the UMPT_N communicates with the UMPT_L through the CI port.
- Unlike non-co-transmission scenarios, a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga must be configured on the eNodeB when a separate-MPT LU+N MBTS uses main-control-board-based IP co-transmission through backplane interconnection.

5.4.3.9.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

(Optional) MAE Side

On the MAE side, add two routes to the gNodeB's DHCP relay with the destination IP address of one route set to the OMCH IP address of the eNodeB and that of the other route set to the lower-level transmission port IP address of the eNodeB.

UMTS Side

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_U to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=0, SSN=6, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the
NodeB use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the NodeB can be set as required.
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.35.35.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to
TUNNEL.
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.21.21.1", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_U to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=0, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the
NodeB use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.35.35.188", MASK="255.255.255.255";
//Adding routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTIDX=0, DSTIP="10.21.21.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

NR Side

NOTE

In this scenario, the gNodeB uses the CI port for communication. Therefore, Ethernet ports are not required.

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the gNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
gNodeB use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the gNodeB can be set as required.
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.33.33.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the gNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set
to TUNNEL.
ADD IPRT: RTIDX=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=2, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
```

```

ADD RSCGRP: CN=0, SRN=1, SN=7, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=3000, RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000,
TXPBS=4000;
//Adding a user-plane link
//Adding an S1-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="10.33.33.188", IPSECSWITCH=DISABLE;
//Adding an S1-U peer
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="10.50.50.50", IPSECSWITCH=DISABLE;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;

```

If **GTRANSPARA.TRANSFCGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```

//Adding a tunnel from the UMPT_N to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=1, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the gNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
gNodeB use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.33.33.188", MASK="255.255.255.255";
//Adding routes from the gNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTIDX=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=1, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=2, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=3000,
RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000, TXPBS=4000;
//Adding a user-plane link
//Adding an S1-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="10.33.33.188", IPSECSWITCH=DISABLE;
//Adding an S1-U peer
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="10.50.50.50", IPSECSWITCH=DISABLE;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;

```

LTE Side

If **GTRANSPARA.TRANSFCGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```

//Adding the IP address of the eNodeB
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.20.20.188",
MASK="255.255.255.0", VRFIDX=0;
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=1, IP="10.32.32.1",
MASK="255.255.255.255", VRFIDX=0;
//Adding tunnels from the UMPT_L to the UMPT_U and UMPT_N
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=SHAPING;
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=1, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Adding an uplink route from the NodeB to the BSC through the eNodeB and uplink routes from the
gNodeB to the MME/S-GW/MAE through the eNodeB. In this step, set Route Type to NEXTHOP and Next
Hop IP to the IP address of the directly connected router.
ADD IPRT: RTIDX=0, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=1, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=2, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=3, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
//Adding a downlink route from the RNC to the NodeB through the eNodeB and downlink routes from the
MME/S-GW/MAE to the gNodeB through the eNodeB. When IP-based co-transmission for the NodeB,
gNodeB, and eNodeB is implemented through backplane tunnels, the downlink route from the RNC to the
NodeB through the eNodeB and the downlink routes from the MME/S-GW/MAE to the gNodeB through the

```

eNodeB must be configured on the eNodeB. In addition, Subboard Type must be set to BACK_BOARD and Port Type must be set to TUNNEL.

```
ADD IPRT: RTIDX=4, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.35.35.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=5, SN=7, SBT=BACK_BOARD, DSTIP="10.31.31.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=6, SN=7, SBT=BACK_BOARD, DSTIP="10.33.33.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//(Optional) Enabling DHCP relay. When co-transmission is enabled for the NodeB, gNodeB, and eNodeB,
the eNodeB needs to work as the relay if the NodeB and gNodeB are to be deployed using DHCP. Therefore,
DHCP relay needs to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the NodeB,
gNodeB, and eNodeB, the eNodeB needs to work as the relay if the NodeB and gNodeB are to be deployed
using DHCP. Therefore, the IP address of the DHCP server needs to be added on the eNodeB. For the
NodeB, the DHCP server is the MAE. For the gNodeB, the DHCP server is the MAE.
ADD DHCPVSRIP: DHCPVSRIP="10.60.60.60";
//(Optional) Configuring VLANs. There are two methods of configuring different VLANs for UMTS and LTE
data for isolation:
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink routes from the NodeB/gNodeB to the RNC/MME/S-GW through the eNodeB must
be different from the next-hop address of the uplink route for the eNodeB. For example, you can set the
next-hop address of the uplink route from the NodeB to the RNC through the eNodeB to 10.20.20.101, and
set the next-hop address of the uplink route from the gNodeB to the MME/S-GW through the eNodeB to
10.20.20.201, which are different from the next-hop address (10.20.20.1) of the uplink route for the eNodeB.
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101",
MASK="255.255.255.255", VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=32, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the NodeB, gNodeB and
eNodeB data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=
VLANGROUP, VLANGROUPNO=1, SETPRIO=DISABLE;
```

If **GTRANSPARA.TRANSFCGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding an interface to Ethernet port 1 on the UMPT_L (in slot 7)
SET ETHPORT: CN=0, SRN=1, SN=7, PN=1, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO,
DUPLEX=AUTO;
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFID=0;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
//Adding the IP address of the eNodeB
ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
//Adding tunnels from the UMPT_L to the UMPT_U and UMPT_N
ADD BACKTUNNEL: PORTID=0, SSN=7, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=NON_SHAPING;
ADD BACKTUNNEL: PORTID=1, SSN=7, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Adding an uplink route from the NodeB to the BSC through the eNodeB and uplink routes from the
gNodeB to the MME/S-GW/MAE through the eNodeB. In this step, set Route Type to NEXTHOP and Next
Hop IP to the IP address of the directly connected router.
ADD IPROUTE4: RTIDX=0, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPROUTE4: RTIDX=1, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPROUTE4: RTIDX=2, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPROUTE4: RTIDX=3, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
//Adding a downlink route from the RNC to the NodeB through the eNodeB and downlink routes from the
MME/S-GW/MAE to the gNodeB through the eNodeB. When IP-based co-transmission for the NodeB,
gNodeB, and eNodeB is implemented through backplane tunnels, the downlink route from the RNC to the
NodeB through the eNodeB and the downlink routes from the MME/S-GW/MAE to the gNodeB through the
eNodeB must be configured on the eNodeB. In addition, Port Type must be set to TUNNEL.
```



```
ADD IPROUTE4: RTIDX=4, DSTIP="10.35.35.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=5, DSTIP="10.31.31.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=1;
ADD IPROUTE4: RTIDX=6, DSTIP="10.33.33.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=1;
//(Optional) Enabling DHCP relay. When co-transmission is enabled for the NodeB, gNodeB, and eNodeB,
the eNodeB needs to work as the relay if the NodeB and gNodeB are to be deployed using DHCP. Therefore,
DHCP relay needs to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the NodeB,
gNodeB, and eNodeB, the eNodeB needs to work as the relay if the NodeB and gNodeB are to be deployed
using DHCP. Therefore, the IP address of the DHCP server needs to be added on the eNodeB. For the
NodeB, the DHCP server is the MAE. For the gNodeB, the DHCP server is the MAE.
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//(Optional) Configuring VLANs
//There are two methods of configuring different VLANs for UMTS and LTE data for isolation when the
VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink routes from the NodeB/gNodeB to the RNC/MME/S-GW through the eNodeB must
be different from the next-hop address of the uplink route for the eNodeB. For example, you can set the
next-hop address of the uplink route from the NodeB to the RNC through the eNodeB to 10.20.20.101, and
set the next-hop address of the uplink route from the gNodeB to the MME/S-GW through the eNodeB to
10.20.20.201, which are different from the next-hop address (10.20.20.1) of the uplink route for the eNodeB.
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101",
MASK="255.255.255.255",VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=32, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the NodeB, gNodeB and
eNodeB data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=
VLANGROUP, VLANGROUPNO=1, SETPRIO=DISABLE;
//Configuring different VLANs for UMTS and LTE data for isolation when the INTERFACE MO is used
//Configuring the VLAN of the interface for UMTS data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;
```

5.4.3.9.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.9.4 Activation Verification

After co-transmission is enabled on the MBTS, check whether the feature is enabled by verifying the status of the IP links between the MBTS and the peer devices.

LTE Side

After the configuration file is delivered to the eNodeB and activated, perform the following step on the eNodeB to check whether the transmission links between the eNodeB and the MME/S-GW are normal:

On the eNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the eNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;  
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

UMTS Side

After the configuration file is delivered to the NodeB and activated, perform the following step on the RNC to verify that the transmission link between the RNC and the NodeB is normal:

On the RNC, ping the IP address of the UMPT_U. If the IP address can be pinged, the IP link is normal.

```
PING IP: SIPADDR="10.10.10.10", DESTIP="10.35.35.188", NEXTHOP="10.21.21.254", CONTPING=NO;
```

NR Side

On the gNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the gNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=1, SN=7, SRCIP="10.33.33.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;  
PING: CN=0, SRN=1, SN=7, SRCIP="10.33.33.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

5.4.3.10 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT N+UL/N+UT/N+UM MBTS Side in IP over FE/GE Mode

5.4.3.10.1 Data Preparation

Figure 5-20 shows an example of main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT N+UL/N+UT/N+UM MBTS side. In this scenario, a 10GE/25GE/50GE port on the gNodeB UMPT_N serves as the co-transmission port of the separate-MPT MBTS and connects to the RNC/MME/S-GW. The UMPT_N connects to the UMPT_U and UMPT_L through the UCIU. **Table 5-10** describes the IP address plan.

Figure 5-20 Main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT N+UL/N+UT/N+UM MBTS side

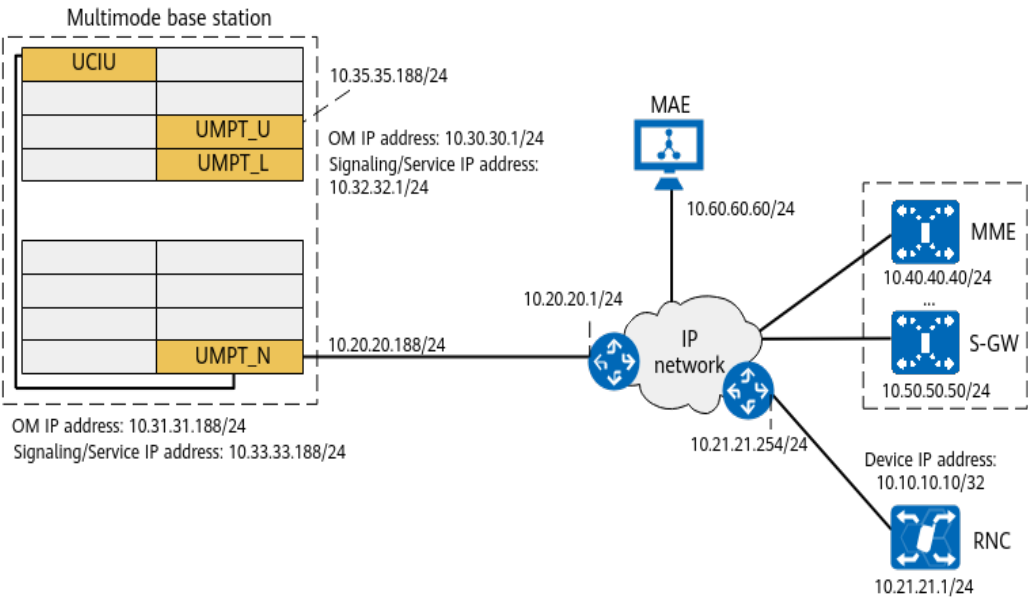


Table 5-10 Data plan

Item	Instance	Remarks
Device IP address of the RNC	10.10.10.10/32	None
Port IP address of the RNC	10.21.21.1/24	None
IP address of the port on the router that connects to the RNC	10.21.21.254/24	None
IP address of the port on the router that connects to the UMPT_N	10.20.20.1/24	None
OMCH IP address of the eNodeB	10.30.30.1/24	LTE: management-plane IP address. This IP address is configured on the main control board.
Signaling/Service IP address of the eNodeB	10.32.32.1/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of 10GE port 1 on the gNodeB UMPT_N	10.20.20.188/24	NR: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment This IP address is configured on the co-transmission port.

Item	Instance	Remarks
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
IP address of the MAE	10.60.60.60/24	NR/LTE: peer IP address for the OMCH
OMCH IP address of the gNodeB	10.31.31.188/24	NR: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the gNodeB	10.33.33.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
OM/Signaling/Service IP address of the NodeB	10.35.35.188/24	The same IP address is used for OM, signaling, and services for the NodeB

NOTE

In this scenario, the UMPT_U and UMPT_L do not need to be configured with Ethernet port attributes because the UMPT_U and UMPT_L communicate with the UCIU through the backplane and the UCIU communicates with the UMPT_N through the M interface.

5.4.3.10.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

MAE Side

On the MAE side, add two routes to the eNodeB's DHCP relay with the destination IP address of one route set to the OMCH IP address of the gNodeB and that of the other route set to the lower-level transmission port IP address of the gNodeB.

UMTS Side

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_U to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=0, SSN=6, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the
```

NodeB use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission scenarios, the IP address of the NodeB can be set as required.
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.35.35.188", MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to TUNNEL.
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.21.21.1", DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60", PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

//Adding a tunnel from the UMPT_U to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=0, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the NodeB use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.35.35.188", MASK="255.255.255.255";
//Adding routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTIDX=0, DSTIP="10.21.21.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
ADD IPRROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60", PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;

LTE Side

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

//Adding a tunnel from the UMPT_L to the UMPT_N
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Adding the IP address of the eNodeB. Unlike non-co-transmission scenarios, this scenario requires that the eNodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the eNodeB can be set as required.
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.32.32.1", MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the eNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to TUNNEL.
ADD IPRT: RTIDX=0, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=10, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=0, SN=7, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS, TXBW=4000, RXBW=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, SN=7, SBT=BACK_BOARD, PT=TUNNEL, JNRSCGRP=DISABLE, LOCALIP="10.32.32.1", PEERIP="10.50.50.50", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60", PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

//Adding a tunnel from the UMPT_L to the UMPT_N
ADD BACKTUNNEL: PORTID=0, SSN=7, DCN=0, DSRN=1, DSN=7;

```
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Adding the IP address of the eNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
eNodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the eNodeB can be
set as required.
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
//Adding routes from the eNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTIDX=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=10, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=4000,
RXBW=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, TRANSCFGMODE=NEW, JNIPRSCGRP=DISABLE, BPT=TUNNEL, PORTID=0,
LOCALIP="10.32.32.1", PEERIP="10.50.50.50", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

NR Side

NOTE

In this scenario, the gNodeB uses the CI port for communication. Therefore, Ethernet ports are not required.

If **GTRANSPARA**. **TRANSCFGMODE** (**5G gNodeB**, **LTE eNodeB**) is set to **OLD**:

```
//Adding tunnels from the UMPT_N to the UMPT_U and UMPT_L
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=1, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=SHAPING;
//Setting IP addresses for the gNodeB
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.33.33.188",
MASK="255.255.255.255", VRFIDX=0;
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.20.20.188",
MASK="255.255.255.0", VRFIDX=0;
//Adding uplink routes from the eNodeB to the MME/S-GW/MAE through the gNodeB and an uplink route
from the NodeB to the RNC through the gNodeB
ADD IPRT: RTIDX=0, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=10, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=1, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.60.60.60",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=2, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
//Adding downlink routes from the MAE/MME/S-GW to the eNodeB through the gNodeB and a downlink
route from the RNC to the NodeB through the gNodeB. When IP-based co-transmission for the NodeB,
eNodeB, and gNodeB is implemented through backplane tunnels, the downlink routes from the MME/S-
GW/MAE to the eNodeB through the gNodeB and the downlink route from the RNC to the NodeB through
the gNodeB must be configured on the gNodeB. In addition, Subboard Type must be set to BACK_BOARD
and Port Type must be set to TUNNEL.
ADD IPRT: RTIDX=3, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.30.30.1",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=4, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.32.32.1",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=5, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.35.35.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the NodeB, gNodeB, and eNodeB, if
the NodeB and eNodeB are to be deployed using DHCP, the gNodeB needs to work as the relay. Therefore,
DHCP relay needs to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
```

```

//(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the NodeB,
gNodeB, and eNodeB, the gNodeB needs to work as the relay if the NodeB and eNodeB are to be deployed
using DHCP. Therefore, the IP address of the DHCP server needs to be added on the gNodeB. For the
eNodeB, the DHCP server is the MAE. For the NodeB, the DHCP server is the MAE.
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//(Optional) Configuring VLANs. There are two methods of configuring different VLANs for UMTS, NR, and
LTE data for isolation:
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink route from the NodeB to the RNC through the gNodeB and the uplink route from
the eNodeB to the MME/S-GW through the gNodeB must be different from the next-hop address of the
uplink route for the gNodeB. For example, you can set the next-hop address of the uplink route from the
NodeB to the RNC through the gNodeB to 10.20.20.101, and set the next-hop address of the uplink route
from the eNodeB to the MME/S-GW through the gNodeB to 10.20.20.201, which are different from the
next-hop address (10.20.20.1) of the uplink route for the gNodeB.
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=32, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the UMTS, NR, and LTE data
packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP,
VLANGROUPNO=1;

```

If **GTRANSPARA.TRANSFCGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```

//Adding tunnels from the UMPT_N to the UMPT_U and UMPT_L
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=7;
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=6;
//Adding an interface to Ethernet port 1 on the UMPT_L (in slot 6)
SET ETHPORT: CN=0, SRN=1, SN=7, PN=1, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO,
DUPLEX=AUTO;
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFID=0;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=1, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
//Setting IP addresses for the gNodeB
ADD IPADDR4: ITFID=1, IP="10.33.33.188", MASK="255.255.255.255";
ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
//Adding uplink routes from the eNodeB to the MME/S-GW/MAE through the gNodeB and an uplink route
from the NodeB to the RNC through the gNodeB
ADD IPROUTE4: RTID=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPROUTE4: RTID=10, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPROUTE4: RTID=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPROUTE4: RTID=2, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1", PREF=60;
//Adding downlink routes from the MAE/MME/S-GW to the eNodeB through the gNodeB and a downlink
route from the RNC to the NodeB through the gNodeB. When IP-based co-transmission for the NodeB,
eNodeB, and gNodeB is implemented through backplane tunnels, the downlink routes from the MME/S-
GW/MAE to the eNodeB through the gNodeB and the downlink route from the RNC to the NodeB through
the gNodeB must be configured on the gNodeB. In addition, Port Type must be set to TUNNEL.
ADD IPROUTE4: RTID=3, DSTIP="10.30.30.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTID=4, DSTIP="10.32.32.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTID=5, DSTIP="10.35.35.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//(Optional) Enabling DHCP relay. When co-transmission is enabled for the NodeB, gNodeB, and eNodeB, if
the NodeB and eNodeB are to be deployed using DHCP, the gNodeB needs to work as the relay. Therefore,
DHCP relay needs to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the NodeB,
gNodeB, and eNodeB, the gNodeB needs to work as the relay if the NodeB and eNodeB are to be deployed

```

using DHCP. Therefore, the IP address of the DHCP server needs to be added on the gNodeB. For the eNodeB, the DHCP server is the MAE. For the NodeB, the DHCP server is the MAE.

```
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//(Optional) Configuring VLANs. There are two methods of configuring different VLANs for UMTS, NR, and
LTE data for isolation when the VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink route from the NodeB to the RNC through the gNodeB and the uplink route from
the eNodeB to the MME/S-GW through the gNodeB must be different from the next-hop address of the
uplink route for the gNodeB. For example, you can set the next-hop address of the uplink route from the
NodeB to the RNC through the gNodeB to 10.20.20.101, and set the next-hop address of the uplink route
from the eNodeB to the MME/S-GW through the gNodeB to 10.20.20.201, which are different from the
next-hop address (10.20.20.1) of the uplink route for the gNodeB.
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=32, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the UMTS, NR, and LTE data
packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
```

5.4.3.10.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.10.4 Activation Verification

After co-transmission is enabled on the MBTS, check whether the feature is enabled by verifying the status of the IP links between the MBTS and the peer devices.

NR Side

On the gNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the gNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=1, SN=7, SRCIP="10.20.20.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
PING: CN=0, SRN=1, SN=7, SRCIP="10.20.20.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

LTE Side

After the configuration file is delivered to the eNodeB and activated, perform the following step on the eNodeB to check whether the transmission links between the eNodeB and the MME/S-GW are normal:

On the eNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links are normal.

```
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```


UMTS Side

After the configuration file is delivered to the NodeB and activated, perform the following step on the RNC to verify that the transmission link between the RNC and the NodeB is normal:

On the RNC, ping the IP address of the UMPT_U. If the IP address can be pinged, the transmission link is normal.

PING IP: SIPADDR="10.10.10.10", DESTIP="10.35.35.188", NEXTHOP="10.21.21.254", CONTPING=NO;

5.4.3.11 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT NL/NT/NM+GU MBTS Side in IP over FE/GE Mode

In this scenario, the GSM base station can be a GBTS or an eGBTS, and the main control board must be a GTMUc/UMPT. The main control board of NR must be a UMPTe/UMPTg/UMPTga/UMPTha/UMPThb.

5.4.3.11.1 Data Preparation

Figure 5-21 shows an example of main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT NL/NT/NM+GU MBTS side. In this scenario, a 10GE/25GE/50GE port on the gNodeB UMPT_N serves as the co-transmission port of the separate-MPT MBTS and connects to the RNC/BSC/MME/S-GW. The UMPT_L and the UMPT_N in one BBU are interconnected through the backplane. The UMPT_N connects to the GTMU and UMPT_U through the UCIU. Table 5-11 describes the IP address plan.

Figure 5-21 Main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT NL/NT/NM+GU MBTS side

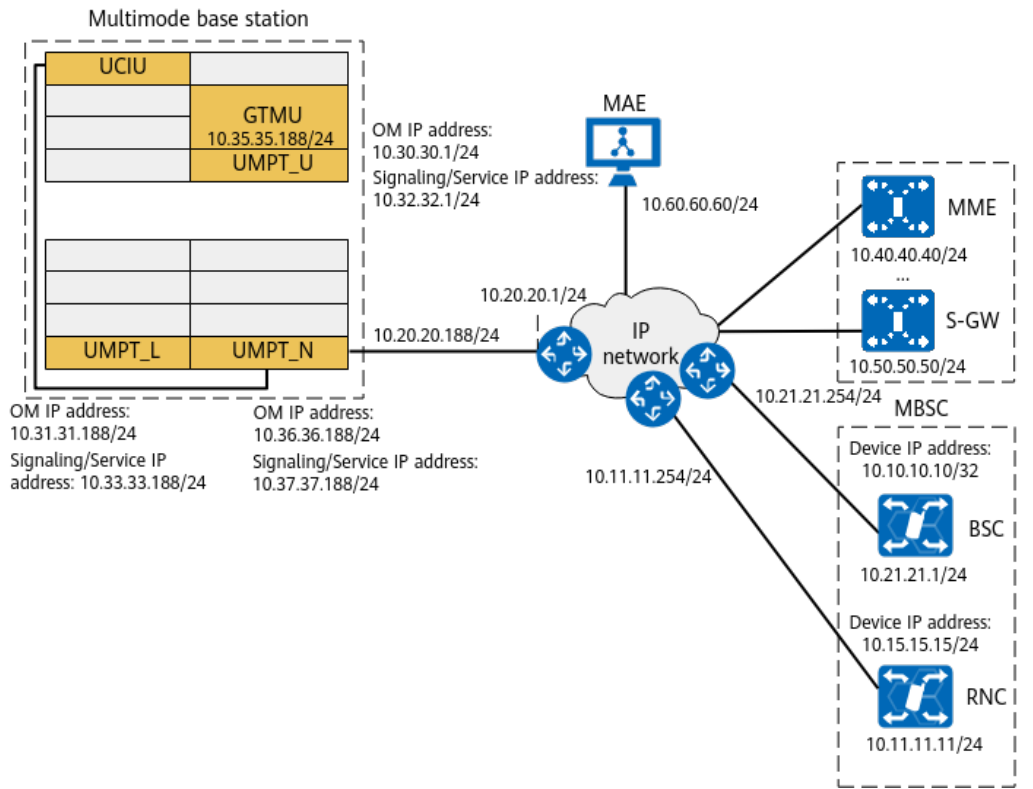


Table 5-11 Data plan

Item	Instance	Remarks
Device IP address of the BSC	10.10.10.10/32	None
Port IP address of the BSC	10.21.21.1/24	None
Device IP address of the RNC	10.15.15.15/24	None
Port IP address of the RNC	10.11.11.11/24	None
IP address of the port on the router that connects to the RNC	10.11.11.254/24	None
IP address of the port on the router that connects to the BSC	10.21.21.254/24	None
IP address of the port on the router that connects to the UMPT_N	10.20.20.1/24	None
OMCH IP address of the NodeB	10.30.30.1/24	UMTS: management-plane IP address. This IP address is configured on the main control board.
Signaling/Service IP address of the NodeB	10.32.32.1/24	UMTS: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of 10GE port 1 on the gNodeB UMPT_N	10.20.20.188/24	NR: device IP address during configuration on the MAE-Deployment This IP address is configured on the co-transmission port.
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
IP address of the MAE	10.60.60.60/24	UMTS/LTE/NR: peer IP address for the OMCH
OMCH IP address of the gNodeB	10.36.36.188/24	NR: management-plane IP address. This is a logical IP address configured on the main control board.

Item	Instance	Remarks
Signaling/Service IP address of the gNodeB	10.37.37.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
OMCH IP address of the eNodeB	10.31.31.188/24	LTE: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the eNodeB	10.33.33.188/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of the GBTS	10.35.35.188/24	GSM: communication IP address of the GBTS. This is a logical IP address configured on the GBTS main control board.
OM/Signaling/Service IP address of the eGBTS	10.35.35.188/24	The same IP address is used for OM, signaling, and services for the eGBTS.
ESN of the GBTS	abcdefghijklmn	For the GBTS, the ESN needs to be planned. For the eGBTS, the ESN does not need to be planned.

NOTE

In this scenario, the GTMU, UMPT_U, and UMPT_L do not need to be configured with Ethernet port attributes because the GTMU and UMPT_U communicate with the UCIU through the backplane, the UCIU communicates with the UMPT_N through the M interface, and the UMPT_L communicates with the UMPT_N through the backplane.

5.4.3.11.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

MAE Side

On the MAE side, add two routes to the DHCP relay of the NodeB and eNodeB with the destination IP address of one route set to the OMCH IP address of the gNodeB and that of the other route set to the lower-level transmission port IP address of the gNodeB.

GSM Side (GBTS)

```
//Adding a tunnel from the GTMU to the UMPT_N
ADD BTSTUNNEL: IDTYPE=BYID, BTSID=10, SRCN=0, SRCSRN=0, SRCN=6, TN=0, DSTCN=0, DSTSRN=1,
DSTSN=7;
//Adding the IP address of the GTMU
ADD BTSDEVIP: IDTYPE=BYID, BTSID=10, PT=LOOPINTERFACE, PN=0, CN=0, SRN=0, SN=6,
IP="10.35.35.188", MASK="255.255.255.255";
//Setting the communication IP address of the GBTS. Unlike non-co-transmission scenarios, this scenario
requires that the GBTS use the logical IP address, that is, BTS Communication Type must be set to LOGICIP.
In non-co-transmission scenarios, the IP address of the GBTS can be set as required.
SET BTSIP: IDTYPE=BYID, BTSID=10, BTSCOMTYPE=LOGICIP, BTSIP="10.35.35.188", BSCIP="10.10.10.10",
BTSMUTIP=NO;
//Adding a route from the GBTS to the BSC. Unlike non-co-transmission scenarios, this scenario requires
that Route Type be set to OUTIF, Interface Type be set to TUNNEL, and Outgoing Interface No. be set to the
number of the added tunnel.
ADD BTSIPRT: IDTYPE=BYID, BTSID=10, RTIDX=1, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=OUTIF, CN=0, SRN=0, SN=6, ITFType=TUNNEL, IFNO=0;
//Adding the ESN of the GBTS
ADD BTSESN: IDTYPE=BYID, BTSID=10, MAINDEVTAB="abcdefghijklmn", OMBEARBOARD=BACKBOARD;
//Optional Adding two routes from the BSC to the GBTS's DHCP relay with the destination IP address of
one route set to the OMCH IP address of the gNodeB and that of the other route set to the lower-level
transmission port IP address of the gNodeB
ADD IPRT: SRN=0, SN=16, DSTIP="10.20.20.188", DSTMASK="255.255.255.255", NEXTHOP="10.21.21.254",
PRIORITY=HIGH, REMARK="relay";
```

GSM Side (eGBTS)

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the GTMU to the UMPT_U
ADD TUNNEL: SCN=0, SSRN=0, SSN=6, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the eGBTS can be set as required.
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.35.35.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to
TUNNEL.
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.21.21.1", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the GTMU to the UMPT_U
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=0, SN=6;
ADD INTERFACE: ITFID=1, ITFType=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.35.35.188", MASK="255.255.255.255";
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPROUTE4: RTIDX=0, DSTIP="10.21.21.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

UMTS Side

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_U to the UMPT_N
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Adding the IP address of the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the
NodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the NodeB can be set
as required.
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.32.32.1",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to
TUNNEL.
ADD IPRT: RTIDX=0, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.15.15.15", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=0, SN=7, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=4000, RXBW=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, SN=7, SBT=BACK_BOARD, PT=TUNNEL, JNRSCGRP=DISABLE,
LOCALIP="10.32.32.1", PEERIP="10.15.15.15", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_U to the UMPT_N
ADD BACKTUNNEL: PORTID=0, SSN=7, DCN=0, DSRN=1, DSN=7;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Adding the IP address of the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the
NodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the NodeB can be set
as required.
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
//Setting routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTIDX=0, DSTIP="10.15.15.15", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=4000,
RXBW=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, TRANS CFGMODE=NEW, JNIPRSCGRP=DISABLE, BPT=TUNNEL, PORTID=0,
LOCALIP="10.32.32.1", PEERIP="10.15.15.15", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

LTE Side

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_L to the UMPT_N
ADD TUNNEL: SCN=0, SSRN=1, SSN=6, TUNNELID=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the eNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
eNodeB use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the eNodeB can be set as required.
ADD DEVIP: CN=0, SRN=1, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.33.33.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the eNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set
```

```
to TUNNEL.
ADD IPRT: RTIDX=0, SRN=1, SN=6, SBT=BACK_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, SRN=1, SN=6, SBT=BACK_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=2, SRN=1, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=1, SN=6, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=3000, RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000,
TXPBS=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, SRN=1, SN=6, SBT=BACK_BOARD, PT=TUNNEL, PN=0, JNRSCGRP=DISABLE,
LOCALIP="10.33.33.188", PEERIP="10.50.50.50", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_L to the UMPT_N
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=1, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
//Setting an IP address for the eNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
eNodeB use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.33.33.188", MASK="255.255.255.255";
//Adding routes from the eNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPROUTE4: RTIDX=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=1, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=2, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=3000,
RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000, TXPBS=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, TRANS CFGMODE=NEW, JNIPRSCGRP=DISABLE, BPT=TUNNEL, PORTID=0,
LOCALIP="10.33.33.188", PEERIP="10.50.50.50", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

NR Side

NOTE

In this scenario, the gNodeB uses the CI port for communication. Therefore, Ethernet ports are not required.

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding tunnels from the UMPT_N to the GTMU, UMPT_U, and UMPT_L
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=SHAPING;
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=1, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=10, DCN=0, DSRN=1, DSN=6, TUNNELTYPE=SHAPING;
//Setting IP addresses for the gNodeB
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.37.37.188",
MASK="255.255.255.255", VRFID=0;
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.20.20.188",
MASK="255.255.255.0", VRFID=0;
//Adding uplink routes from the NodeB to the RNC/MAE through the gNodeB, an uplink route from the
GBTS/eGBTS to the BSC through the gNodeB, and uplink routes from the eNodeB to the MME/S-GW/MAE
through the gNodeB
ADD IPRT: RTIDX=0, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.15.15.15",
```

```
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=1, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.60.60.60",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=2, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
ADD IPRT: RTIDX=10, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
ADD IPRT: RTIDX=11, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
//Adding downlink routes from the MAE/RNC to the NodeB through the gNodeB, from the BSC to the GBTS/
eGBTS through the gNodeB, and from the MAE/MME/S-GW to the eNodeB through the gNodeB. When IP-
based co-transmission for the GBTS/eGBTS, NodeB, eNodeB, and gNodeB is implemented through
backplane tunnels, the downlink routes from the RNC/MAE to the NodeB through the gNodeB, from the
MME/S-GW/MAE to the eNodeB through the gNodeB, and from the BSC to the GBTS/eGBTS through the
gNodeB must be configured on the gNodeB. In addition, Subboard Type must be set to BACK_BOARD and
Port Type must be set to TUNNEL.
ADD IPRT: RTIDX=3, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.30.30.1",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=4, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.32.32.1",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=5, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.35.35.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=3;
ADD IPRT: RTIDX=12, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.33.33.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=10;
ADD IPRT: RTIDX=13, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.31.31.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=11;
//Adding a downlink route from the MAE to the eNodeB through the gNodeB on the UMPT_N (in slot 7 of
subrack 1)
//(Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, eNodeB, NodeB,
and gNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS, NodeB, and eNodeB are to be
deployed using DHCP. Therefore, DHCP relay needs to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/
eGBTS, eNodeB, NodeB, and gNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS, NodeB,
and eNodeB are to be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be
added on the gNodeB. For the NodeB, the DHCP server is the RNC or MAE. For the GBTS, the DHCP server
is the BSC. For the eGBTS and eNodeB, the DHCP server is the MAE.
//Adding the IP address of the DHCP server for the eGBTS/NodeB/eNodeB
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//Adding the IP address of the DHCP server for the GBTS
ADD DHCPVRIP: DHCPVRIP="10.10.10.10";
//(Optional) Configuring VLANs. There are two methods of configuring different VLANs for GSM, UMTS,
LTE, and NR data for isolation:
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink route from the GBTS/eGBTS to the BSC through the gNodeB, the uplink route from
the NodeB to the RNC through the gNodeB, and the uplink route from the eNodeB to the MME/S-GW
through the gNodeB must be different from the next-hop address of the uplink route for the gNodeB. For
example, you can set the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through
the gNodeB to 10.20.20.101, the next-hop address of the uplink route from the NodeB to the RNC through
the gNodeB to 10.20.20.201, and the next-hop address of the uplink route from the eNodeB to the MME/S-
GW through the gNodeB to 10.20.20.202, which are different from the next-hop address (10.20.20.1) of the
uplink route for the gNodeB.
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=32, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.202", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=42, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, LTE, and NR
data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=42;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
```

If **GTRANSPARA.TRANSFCGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding tunnels from the UMPT_N to the GTMU, UMPT_U, and UMPT_L
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=6;
ADD BACKTUNNEL: PORTID=1, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=7;
ADD BACKTUNNEL: PORTID=10, SSRN=1, SSN=7, DCN=0, DSRN=1, DSN=6;
//Adding an interface for an Ethernet port
SET ETHPORT: CN=0, SRN=1, SN=7, PN=1, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO,
DUPLEX=AUTO;
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFIDX=0;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=1, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting IP addresses for the gNodeB
ADD IPADDR4: ITFID=1, IP="10.37.37.188", MASK="255.255.255.255";
ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
//Adding uplink routes from the NodeB to the RNC/MAE through the gNodeB, an uplink route from the
GBTS/eGBTS to the BSC through the gNodeB, and uplink routes from the eNodeB to the MME/S-GW/MAE
through the gNodeB
ADD IPRROUTE4: RTIDX=0, DSTIP="10.15.15.15", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTIDX=2, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1", PREF=60;
ADD IPRROUTE4: RTIDX=10, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1", PREF=60;
ADD IPRROUTE4: RTIDX=11, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1", PREF=60;
//Adding downlink routes from the MAE/RNC to the NodeB through the gNodeB, from the BSC to the GBTS/
eGBTS through the gNodeB, and from the MAE/MME/S-GW to the eNodeB through the gNodeB. When IP-
based co-transmission for the GBTS/eGBTS, NodeB, eNodeB, and gNodeB is implemented through
backplane tunnels, the downlink routes from the RNC/MAE to the NodeB through the gNodeB, from the
MME/S-GW/MAE to the eNodeB through the gNodeB, and from the BSC to the GBTS/eGBTS through the
gNodeB must be configured on the gNodeB. In addition, Port Type must be set to TUNNEL.
ADD IPRROUTE4: RTIDX=3, DSTIP="10.30.30.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=4, DSTIP="10.32.32.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=5, DSTIP="10.35.35.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=1;
ADD IPRROUTE4: RTIDX=12, DSTIP="10.33.33.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=10;
ADD IPRROUTE4: RTIDX=13, DSTIP="10.31.31.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=11;
//(Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, eNodeB, NodeB,
and gNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS, NodeB, and eNodeB are to be
deployed using DHCP. Therefore, DHCP relay needs to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/
eGBTS, eNodeB, NodeB, and gNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS, NodeB,
and eNodeB are to be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be
added on the gNodeB. For the NodeB, the DHCP server is the RNC or MAE. For the GBTS, the DHCP server
is the BSC. For the eGBTS and eNodeB, the DHCP server is the MAE.
//Adding the IP address of the DHCP server for the eGBTS/NodeB/eNodeB
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//Adding the IP address of the DHCP server for the GBTS
ADD DHCPVRIP: DHCPVRIP="10.10.10.10";
//(Optional) Configuring VLANs
//There are two methods of configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation
when the VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink route from the GBTS/eGBTS to the BSC through the gNodeB, the uplink route from
the NodeB to the RNC through the gNodeB, and the uplink route from the eNodeB to the MME/S-GW
through the gNodeB must be different from the next-hop address of the uplink route for the gNodeB. For
example, you can set the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through
the gNodeB to 10.20.20.101, the next-hop address of the uplink route from the NodeB to the RNC through
the gNodeB to 10.20.20.201, and the next-hop address of the uplink route from the eNodeB to the MME/S-
GW through the gNodeB to 10.20.20.202, which are different from the next-hop address (10.20.20.1) of the
uplink route for the gNodeB.
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
```



```
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=32, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.202", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=42, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, LTE, and NR
data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=40, VLANID=42;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
//Configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation when the INTERFACE MO is
used
//Configuring the VLAN of the interface for GSM data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for UMTS data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;
```

5.4.3.11.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.11.4 Activation Verification

After co-transmission is enabled on the base station, check whether it is enabled by verifying the status of the IP links between the base station and the peer devices.

NR Side

On the gNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the gNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=1, SN=7, SRCIP="10.20.20.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
PING: CN=0, SRN=1, SN=7, SRCIP="10.20.20.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

LTE Side

On the eNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the eNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=1, SN=6, SRCIP="10.33.33.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
PING: CN=0, SRN=1, SN=6, SRCIP="10.33.33.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

UMTS Side

After the configuration file is delivered to the NodeB and activated, perform the following step on the NodeB to verify that the transmission link between the NodeB and the RNC is normal:

On the NodeB, ping the IP address of the RNC. If the ping operations succeed, the transmission links are normal.
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.15.15.15", CONTPING=DISABLE, APPTIF=NO;

GSM Side

After the configuration file is delivered to the GBTS/eGBTS and activated, perform the following step on the BSC to verify that the transmission link between the BSC and the GBTS/eGBTS is normal:

On the BSC, ping the IP address of the GTMUc. If the ping operations succeed, the transmission links are normal.
PING IP: SIPADDR="10.10.10.10", DESTIP="10.35.35.188", NEXTHOP="10.21.21.254", CONTPING=NO;

5.4.3.12 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN/TN/MN+GU MBTS Side in IP over FE/GE Mode

In this scenario, the GSM base station can be a GBTS or an eGBTS, and the main control board must be a GTMUc/UMPT. The main control board of NR and LTE must be a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga/UMPTga.

5.4.3.12.1 Data Preparation

Figure 5-22 shows an example of main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LN/TN/MN+GU MBTS side. In this scenario, a 10GE/25GE/50GE port on the eNodeB UMPT_L serves as the co-transmission port of the separate-MPT LN/TN/MN+GU MBTS and connects to the RNC/BSC/MME/S-GW. The UMPT_L and the UMPT_N in one BBU are interconnected through the backplane. The UMPT_L connects to the GTMU and UMPT_U through the UCIU. Table 5-12 describes the IP address plan.

Figure 5-22 Main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LN/TN/MN+GU MBTS side

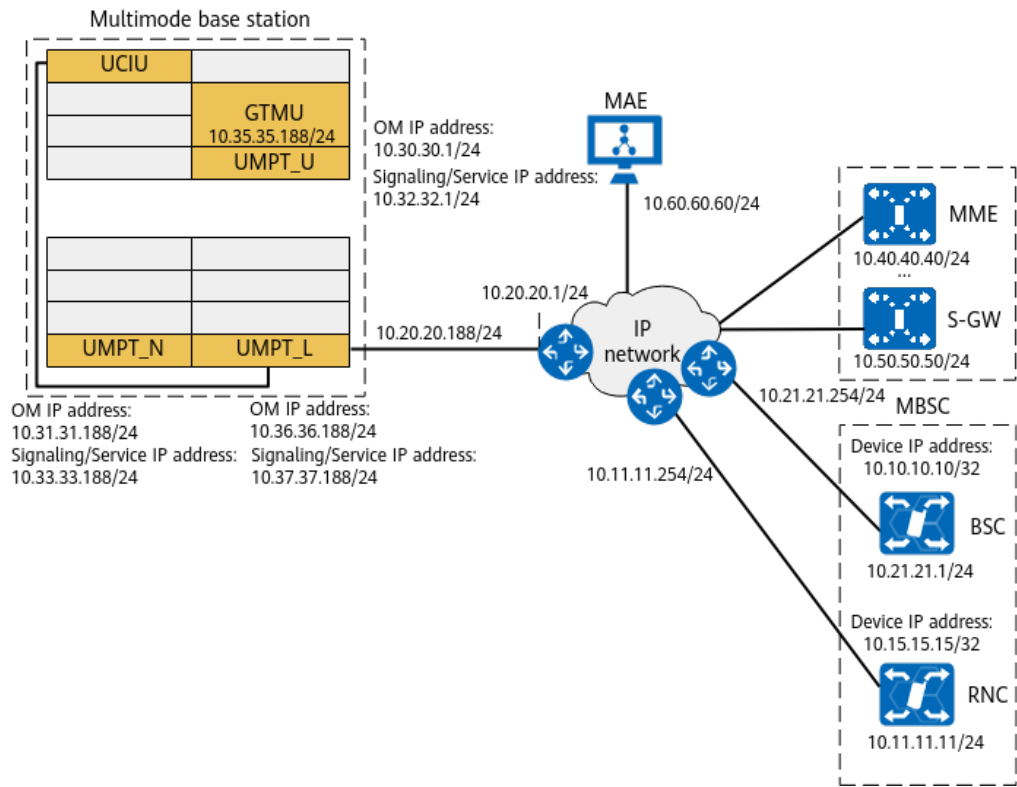


Table 5-12 Data plan

Item	Instance	Remarks
Device IP address of the BSC	10.10.10.10/32	None
Port IP address of the BSC	10.21.21.1/24	None
Device IP address of the RNC	10.15.15.15/32	None
Port IP address of the RNC	10.11.11.11/24	None
IP address of the port on the router that connects to the RNC	10.11.11.254/24	None
IP address of the port on the router that connects to the BSC	10.21.21.254/24	None
IP address of the port on the router that connects to the UMPT_L	10.20.20.1/24	None

Item	Instance	Remarks
OMCH IP address of the NodeB	10.30.30.1/24	UMTS: management-plane IP address. This IP address is configured on the main control board.
Signaling/Service IP address of the NodeB	10.32.32.1/24	UMTS: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of 10GE port 1 on the eNodeB UMPT_L	10.20.20.188/24	LTE: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment This IP address is configured on the co-transmission port.
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
IP address of the MAE	10.60.60.60/24	UMTS/LTE/NR: peer IP address for the OMCH
OMCH IP address of the eNodeB	10.36.36.188/24	LTE: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the eNodeB	10.37.37.188/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
OMCH IP address of the gNodeB	10.31.31.188/24	NR: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the gNodeB	10.33.33.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of the GBTS	10.35.35.188/24	GSM: communication IP address of the GBTS. This is a logical IP address configured on the GBTS main control board.
OM/Signaling/Service IP address of the eGBTS	10.35.35.188/24	The same IP address is used for OM, signaling, and services for the eGBTS.
ESN of the GBTS	abcdefghijklmn	For the GBTS, the ESN needs to be planned. For the eGBTS, the ESN does not need to be planned.

NOTE

In this scenario, the GTMU, UMPT_U, and UMPT_N do not need to be configured with Ethernet port attributes because the GTMU and UMPT_U communicate with the UCIU through the backplane, the UCIU communicates with the UMPT_L through the M interface, and the UMPT_N communicates with the UMPT_L through the backplane.

5.4.3.12.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

MAE Side

On the MAE side, add two routes to the DHCP relay of the NodeB and gNodeB with the destination IP address of one route set to the OMCH IP address of the eNodeB and that of the other route set to the lower-level transmission port IP address of the eNodeB.

GSM Side (GBTS)

```
//Adding a tunnel from the GTMUc to the UMPT_L
ADD BTSTUNNEL: IDTYPE=BYID, BTSID=10, SRCN=0, SRCN=6, TN=0, DSTCN=0, DSTSRN=1,
DSTSN=7;
//Adding an IP address for the GTMUc
ADD BTSDEVIP: IDTYPE=BYID, BTSID=10, PT=LOOPINTERFACE, PN=0, CN=0, SRN=0, SN=6,
IP="10.35.35.188", MASK="255.255.255.255";
//Setting the communication IP address of the GBTS. Unlike non-co-transmission scenarios, this scenario
requires that the GBTS use the logical IP address, that is, BTS Communication Type must be set to LOGICIP.
In non-co-transmission scenarios, the IP address of the GBTS can be set as required.
SET BTSIP: IDTYPE=BYID, BTSID=10, BTSCOMTYPE=LOGICIP, BTSIP="10.35.35.188", BSCIP="10.10.10.10",
BTSMUTIP=NO;
//Adding a route from the GBTS to the BSC. Unlike non-co-transmission scenarios, this scenario requires
that Route Type be set to OUTIF, Interface Type be set to TUNNEL, and Outgoing Interface No. be set to the
number of the added tunnel.
ADD BTSIPRT: IDTYPE=BYID, BTSID=10, RTIDX=1, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=OUTIF, CN=0, SRN=0, SN=6, ITFTYPE=TUNNEL, IFNO=0;
//Adding the ESN of the GBTS
ADD BTSESIN: IDTYPE=BYID, BTSID=10, MAINDEVTAB="abcdefghijklmn", OMBEARBOARD=BACKBOARD;
//((Optional) Adding two routes from the BSC to the GBTS's DHCP relay with the destination IP address of
one route set to the OMCH IP address of the eNodeB and that of the other route set to the lower-level
transmission port IP address of the eNodeB
ADD IPRT: SRN=0, SN=16, DSTIP="10.20.20.188", DSTMASK="255.255.255.255", NEXTHOP="10.21.21.254",
PRIORITY=HIGH, REMARK="relay";
```

GSM Side (eGBTS)

If **GTRANSPARA.TRANSCFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the GTMUc to the UMPT_U
ADD TUNNEL: SCN=0, SSRN=0, SSN=6, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the eGBTS can be set as required.
```

```

ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.35.35.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to
TUNNEL.
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.21.21.1", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;

```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```

//Adding a tunnel from the GTMUc to the UMPT_U
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=0, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.35.35.188", MASK="255.255.255.255";
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPROUTE4: RTIDX=0, DSTIP="10.21.21.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=0, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;

```

UMTS Side

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```

//Adding a tunnel from the UMPT_U to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Adding the IP address of the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the
NodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the NodeB can be set
as required.
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.32.32.1",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to
TUNNEL.
ADD IPRT: RTIDX=0, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.15.15.15", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=0, SN=7, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=4000, RXBW=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, SN=7, SBT=BACK_BOARD, PT=TUNNEL, JNRSCGRP=DISABLE,
LOCALIP="10.32.32.1", PEERIP="10.15.15.15", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;

```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```

//Adding a tunnel from the UMPT_U to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSN=7, DCN=0, DSRN=1, DSN=7;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Adding the IP address of the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the

```

NodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the NodeB can be set as required.

```
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
//Setting routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTIDX=0, DSTIP="10.15.15.15", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
ADD IPRROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=4000, RXBW=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, TRANSCFGMODE=NEW, JNIPRSCGRP=DISABLE, BPT=TUNNEL, PORTID=0, LOCALIP="10.32.32.1", PEERIP="10.15.15.15", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60", PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

NR Side

If **GTRANSPARA.TRANSCFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=1, SSN=6, TUNNELID=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the gNodeB. Unlike non-co-transmission scenarios, this scenario requires that the gNodeB use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission scenarios, the IP address of the gNodeB can be set as required.
ADD DEVIP: CN=0, SRN=1, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.33.33.188", MASK="255.255.255.255", VRFID=0;
//Adding routes from the gNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to TUNNEL.
ADD IPRT: RTIDX=0, SRN=1, SN=6, SBT=BACK_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=1;
ADD IPRT: RTIDX=1, SRN=1, SN=6, SBT=BACK_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=2;
ADD IPRT: RTIDX=2, SRN=1, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=3;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=1, SN=6, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS, TXBW=3000, RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000, TXPBS=4000;
//Adding a user-plane link
//Adding an S1-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="10.33.33.188", IPSECSWITCH=DISABLE;
//Adding an S1-U peer
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="10.50.50.50", IPSECSWITCH=DISABLE;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60", PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANSCFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=1, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
//Setting an IP address for the gNodeB. Unlike non-co-transmission scenarios, this scenario requires that the gNodeB use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.33.33.188", MASK="255.255.255.255";
//Adding routes from the gNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTIDX=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
ADD IPRROUTE4: RTIDX=1, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
ADD IPRROUTE4: RTIDX=2, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
```

```
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=3000,
RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000, TXPBS=4000;
//Adding a user-plane link
//Adding an S1-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="10.33.33.188", IPSECSWITCH=DISABLE;
//Adding an S1-U peer
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="10.50.50.50", IPSECSWITCH=DISABLE;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

LTE Side

NOTE

In this scenario, the eNodeB uses the CI port for communication. Therefore, Ethernet ports are not required.

If **GTRANSPARA.TRANSFCGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding tunnels from the UMPT_L to the UMPT_U, GTMUc, and UMPT_N
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=1, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=SHAPING;
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=10, DCN=0, DSRN=1, DSN=6, TUNNELTYPE=SHAPING;
//Setting IP addresses for the gNodeB
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.37.37.188",
MASK="255.255.255.255", VRFID=0;
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.20.20.188",
MASK="255.255.255.0", VRFID=0;
//Adding uplink routes from the NodeB to the RNC/MAE through the eNodeB, from the GBTS/eGBTS to the
BSC through the eNodeB, and from the gNodeB to the MME/S-GW/MAE through the eNodeB
ADD IPRT: RTID=0, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.15.15.15",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTID=1, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.60.60.60",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTID=2, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
ADD IPRT: RTID=10, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
ADD IPRT: RTID=11, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
//Adding downlink routes from the MAE/RNC to the NodeB through the eNodeB, from the BSC to the GBTS/
eGBTS through the eNodeB, and from the MAE/MME/S-GW to the gNodeB through the eNodeB. When IP-
based co-transmission for the GBTS/eGBTS, NodeB, gNodeB, and eNodeB is implemented through
backplane tunnels, the downlink routes from the RNC/MAE to the NodeB through the eNodeB, from the
MME/S-GW/MAE to the gNodeB through the eNodeB, and from the BSC to the GBTS/eGBTS through the
eNodeB must be configured on the eNodeB. In addition, Subboard Type must be set to BACK_BOARD and
Port Type must be set to TUNNEL.
ADD IPRT: RTID=3, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.30.30.1",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=1;
ADD IPRT: RTID=4, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.32.32.1",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=2;
ADD IPRT: RTID=5, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.35.35.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=3;
ADD IPRT: RTID=12, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.33.33.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=10;
ADD IPRT: RTID=13, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.31.31.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=11;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, gNodeB, NodeB,
and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS, NodeB, and gNodeB are to be
deployed using DHCP. Therefore, DHCP relay needs to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/
eGBTS, gNodeB, NodeB, and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS, NodeB,
and gNodeB are to be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be
added on the eNodeB. For the NodeB, the DHCP server is the RNC or MAE. For the GBTS, the DHCP server
```


is the BSC. For the eGBTS and gNodeB, the DHCP server is the MAE.

```
ADD DHCPVSRIP: DHCPVSRIP="10.60.60.60";
```

```
ADD DHCPVSRIP: DHCPVSRIP="10.10.10.10";
```

// (Optional) Configuring VLANs. There are two methods of configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation:

// Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop addresses of the uplink route from the GBTS/eGBTS to the BSC through the eNodeB, the uplink route from the NodeB to the RNC through the eNodeB, and the uplink routes from the gNodeB to the MME/S-GW through the eNodeB must be different from the next-hop address of the uplink route for the eNodeB. For example, you can set the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through the eNodeB to 10.20.20.101, the next-hop address of the uplink route from the NodeB to the RNC through the eNodeB to 10.20.20.201, and the next-hop address of the uplink route from the gNodeB to the MME/S-GW through the eNodeB to 10.20.20.202, which are different from the next-hop address (10.20.20.1) of the uplink route for the eNodeB.

```
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=2, SETPRIO=DISABLE;
```

```
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
```

```
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=32, SETPRIO=DISABLE;
```

```
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.202", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=42, SETPRIO=DISABLE;
```

// Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, LTE, and NR data packets and configuring the mapping between a VLAN group and the next hop

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
```

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=32;
```

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=42;
```

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
```

```
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP, VLANGROUPNO=1;
```

If **GTRANSPARA.TRANSCEGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
// Adding tunnels from the UMPT_L to the UMPT_U, GTMUC, and UMPT_N
```

```
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=7;
```

```
ADD BACKTUNNEL: PORTID=1, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=6;
```

```
ADD BACKTUNNEL: PORTID=10, SSRN=1, SSN=7, DCN=0, DSRN=1, DSN=6;
```

```
// Adding an interface to Ethernet port 1 on the UMPT_L (in slot 6)
```

```
SET ETHPORT: CN=0, SRN=1, SN=7, PN=1, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO, DUPLEX=AUTO;
```

```
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFID=0;
```

```
// Adding the INTERFACE setting to a loopback interface
```

```
ADD LOOPBACK: PORTID=700, SRN=1, SN=7;
```

```
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
```

```
// Setting IP addresses for the gNodeB
```

```
ADD IPADDR4: ITFID=1, IP="10.37.37.188", MASK="255.255.255.255";
```

```
ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
```

// Adding uplink routes from the NodeB to the RNC/MAE through the eNodeB, from the GBTS/eGBTS to the BSC through the eNodeB, and from the gNodeB to the MME/S-GW/MAE through the eNodeB

```
ADD IPROUTE4: RTID=0, DSTIP="10.15.15.15", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
```

```
ADD IPROUTE4: RTID=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
```

```
ADD IPROUTE4: RTID=2, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
```

```
ADD IPROUTE4: RTID=10, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
```

```
ADD IPROUTE4: RTID=11, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
```

// Adding downlink routes from the MAE/RNC to the NodeB through the eNodeB, from the BSC to the GBTS/eGBTS through the eNodeB, and from the MAE/MME/S-GW to the gNodeB through the eNodeB. When IP-based co-transmission for the GBTS/eGBTS, NodeB, gNodeB, and eNodeB is implemented through backplane tunnels, the downlink routes from the RNC/MAE to the NodeB through the eNodeB, from the MME/S-GW/MAE to the gNodeB through the eNodeB, and from the BSC to the GBTS/eGBTS through the eNodeB must be configured on the eNodeB. In addition, Port Type must be set to TUNNEL.

```
ADD IPROUTE4: RTID=3, DSTIP="10.30.30.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
```

```
ADD IPROUTE4: RTID=4, DSTIP="10.32.32.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
```

```
ADD IPROUTE4: RTID=5, DSTIP="10.35.35.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
```

```

PORTID=1;
ADD IPROUTE4: RTIDX=12, DSTIP="10.33.33.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=10;
ADD IPROUTE4: RTIDX=13, DSTIP="10.31.31.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=11;
//(Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, gNodeB, NodeB,
and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS, NodeB, and gNodeB are to be
deployed using DHCP. Therefore, DHCP relay needs to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/
eGBTS, gNodeB, NodeB, and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS, NodeB,
and gNodeB are to be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be
added on the eNodeB. For the NodeB, the DHCP server is the RNC or MAE. For the GBTS, the DHCP server
is the BSC. For the eGBTS and gNodeB, the DHCP server is the MAE.
//Adding the IP address of the DHCP server for the eGBTS/NodeB/gNodeB
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//Adding the IP address of the DHCP server for the GBTS
ADD DHCPVRIP: DHCPVRIP="10.10.10.10";
//(Optional) Configuring VLANs
//There are two methods of configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation
when the VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink route from the GBTS/eGBTS to the BSC through the eNodeB, the uplink route from
the NodeB to the RNC through the eNodeB, and the uplink routes from the gNodeB to the MME/S-GW
through the eNodeB must be different from the next-hop address of the uplink route for the eNodeB. For
example, you can set the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through
the eNodeB to 10.20.20.101, the next-hop address of the uplink route from the NodeB to the RNC through
the eNodeB to 10.20.20.201, and the next-hop address of the uplink route from the gNodeB to the MME/S-
GW through the eNodeB to 10.20.20.202, which are different from the next-hop address (10.20.20.1) of the
uplink route for the eNodeB.
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=32, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.202", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=42, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, LTE, and NR
data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=34, VLANID=42;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
//Configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation when the INTERFACE MO is
used
//Configuring the VLAN of the interface for GSM data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for UMTS data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;

```

5.4.3.12.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.12.4 Activation Verification

After co-transmission is enabled on the base station, check whether it is enabled by verifying the status of the IP links between the base station and the peer devices.

LTE Side

On the eNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the eNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=1, SN=7, SRCIP="10.20.20.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;  
PING: CN=0, SRN=1, SN=7, SRCIP="10.20.20.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

NR Side

On the gNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the gNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=1, SN=6, SRCIP="10.33.33.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;  
PING: CN=0, SRN=1, SN=6, SRCIP="10.33.33.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

UMTS Side

After the configuration file is delivered to the NodeB and activated, perform the following step on the NodeB to verify that the transmission link between the NodeB and the RNC is normal:

On the NodeB, ping the IP address of the RNC. If the ping operations succeed, the transmission links are normal.

```
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.15.15.15", CONTPING=DISABLE, APPTIF=NO;
```

GSM Side

After the configuration file is delivered to the GBTS/eGBTS and activated, perform the following step on the BSC to verify that the transmission link between the BSC and the GBTS/eGBTS is normal:

On the BSC, ping the IP address of the GTMUc. If the ping operations succeed, the transmission links are normal.

```
PING IP: SIPADDR="10.10.10.10", DESTIP="10.35.35.188", NEXTHOP="10.21.21.254", CONTPING=NO;
```

5.4.3.13 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LG/TG/MG+UN MBTS Side in IP over FE/GE Mode

In this scenario, the GSM base station can be a GBTS or an eGBTS, and the main control board must be a GTMUc/UMPT. The main control board of NR must be a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga/UMPTga.

5.4.3.13.1 Data Preparation

Figure 5-23 shows an example of main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LG/TG/MG+UN MBTS side. In this scenario, a 10GE/25GE/50GE port on the eNodeB UMPT_L serves as

the co-transmission port of the separate-MPT LG/TG/MG+UN MBTS and connects to the RNC/BSC/MME/S-GW. The GTMU and UMPT_L in one BBU are interconnected through the backplane. The UMPT_L connects to the UMPT_U and UMPT_N by using interconnection cables. [Table 5-13](#) describes the IP address plan.

Figure 5-23 Main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LG/TG/MG+UN MBTS side

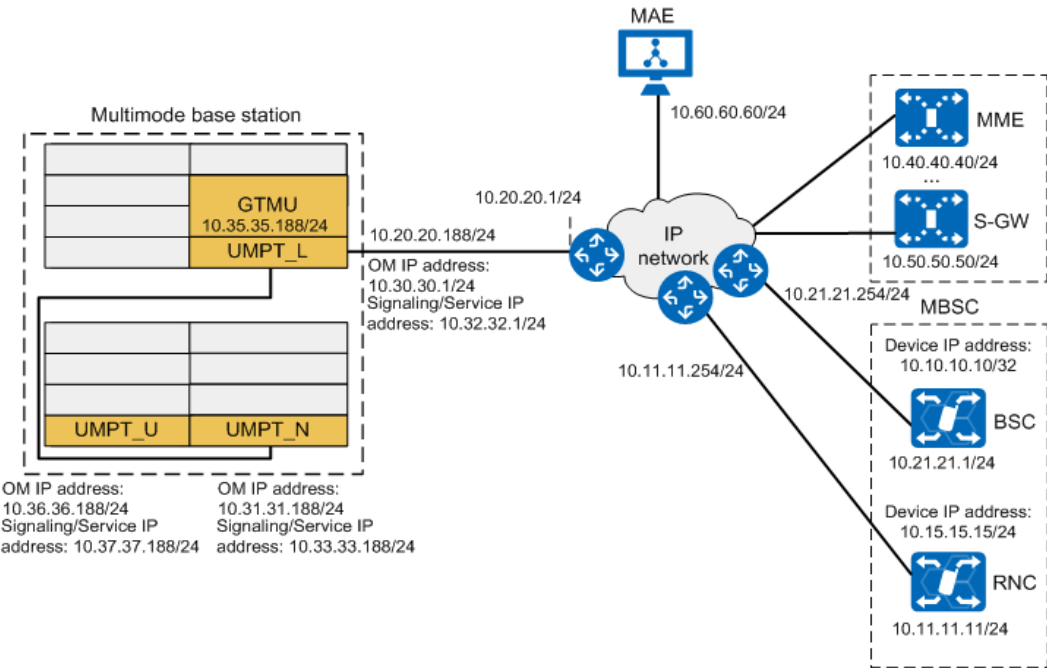


Table 5-13 Data plan

Item	Instance	Remarks
Device IP address of the BSC	10.10.10.10/32	None
Port IP address of the BSC	10.21.21.1/24	None
Device IP address of the RNC	10.15.15.15/24	None
Port IP address of the RNC	10.11.11.11/24	None
IP address of the port on the router that connects to the RNC	10.11.11.254/24	None
IP address of the port on the router that connects to the BSC	10.21.21.254/24	None

Item	Instance	Remarks
IP address of the port on the router that connects to the UMPT_L	10.20.20.1/24	None
OMCH IP address of the eNodeB	10.30.30.1/24	LTE: management-plane IP address. This IP address is configured on the main control board.
Signaling/Service IP address of the eNodeB	10.32.32.1/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of 10GE port 1 on the eNodeB UMPT_L	10.20.20.188/24	LTE: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment. This IP address is configured on the co-transmission port.
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
IP address of the MAE	10.60.60.60/24	UMTS/LTE/NR: peer IP address for the OMCH
OMCH IP address of the gNodeB	10.31.31.188/24	NR: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the gNodeB	10.33.33.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
OMCH IP address of the NodeB	10.36.36.188/24	UMTS: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the NodeB	10.37.37.188/24	UMTS: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of the GBTS	10.35.35.188/24	GSM: communication IP address of the GBTS. This is a logical IP address configured on the GBTS main control board.
OM/Signaling/Service IP address of the eGBTS	10.35.35.188/24	The same IP address is used for OM, signaling, and services for the eGBTS.

Item	Instance	Remarks
ESN of the GBTS	abcdefghijklmn	For the GBTS, the ESN needs to be planned. For the eGBTS, the ESN does not need to be planned.

NOTE

In this scenario, the GTMU, UMPT_U, and UMPT_N do not need to be configured with Ethernet port attributes because the GTMU communicates with the UMPT_L through the backplane, the UMPT_U communicates with the UMPT_N through the backplane, and the UMPT_N communicates with the UMPT_L through the CI port.

5.4.3.13.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

MAE Side

On the MAE side, add two routes to the DHCP relay of the NodeB and gNodeB with the destination IP address of one route set to the OMCH IP address of the eNodeB and that of the other route set to the lower-level transmission port IP address of the eNodeB.

GSM Side (GBTS)

```
//Adding a tunnel from the GTMU to the UMPT_L
ADD BTSTUNNEL: IDTYPE=BYID, BTSID=10, SRCCN=0, SRCN=6, TN=0, DSTCN=0, DSTSRN=0,
DSTSN=7;
//Adding the IP address of the GTMU
ADD BTSDEVIP: IDTYPE=BYID, BTSID=10, PT=LOOPINTERFACE, PN=0, CN=0, SRN=0, SN=6,
IP="10.35.35.188", MASK="255.255.255.255";
//Setting the communication IP address of the GBTS. Unlike non-co-transmission scenarios, this scenario
requires that the GBTS use the logical IP address, that is, BTS Communication Type must be set to LOGICIP.
In non-co-transmission scenarios, the IP address of the GBTS can be set as required.
SET BTSIP: IDTYPE=BYID, BTSID=10, BTSCOMTYPE=LOGICIP, BTSIP="10.35.35.188", BSCIP="10.10.10.10",
BTSMUTIP=NO;
//Adding a route from the GBTS to the BSC. Unlike non-co-transmission scenarios, this scenario requires
that Route Type be set to OUTIF, Interface Type be set to TUNNEL, and Outgoing Interface No. be set to the
number of the added tunnel.
ADD BTSIPRT: IDTYPE=BYID, BTSID=10, RTIDX=1, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=OUTIF, CN=0, SRN=0, SN=6, ITFTYPE=TUNNEL, IFNO=0;
//Adding the ESN of the GBTS
ADD BTSESN: IDTYPE=BYID, BTSID=10, MAINDEVTAB="abcdefghijklmn", OMBEARBOARD=BACKBOARD;
//Optional Adding two routes from the BSC to the GBTS's DHCP relay with the destination IP address of
one route set to the OMCH IP address of the eNodeB and that of the other route set to the lower-level
transmission port IP address of the eNodeB
ADD IPRT: SRN=0, SN=6, DSTIP="10.20.20.188", DSTMASK="255.255.255.255", NEXTHOP="10.21.21.254",
PRIORITY=HIGH, REMARK="relay";
```

GSM Side (eGBTS)

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the GTMU to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=0, SSN=6, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the eGBTS can be set as required.
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.35.35.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to
TUNNEL.
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.21.21.1", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the GTMU to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=0, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.35.35.188", MASK="255.255.255.255";
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTIDX=0, DSTIP="10.21.21.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

NR Side

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the IP address of the gNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
gNodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the gNodeB can be
set as required.
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.33.33.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the gNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set
to TUNNEL.
ADD IPRT: RTIDX=0, SRN=1, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.40.40.40",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=10, SRN=1, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.50.50.50",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, SRN=1, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.60.60.60",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=1, SN=7, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=4000, RXBW=4000;
//Adding a user-plane link
//Adding an S1-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPV4, LOCIPV4="10.33.33.188", IPSECSWITCH=DISABLE;
```

```
//Adding an S1-U peer
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="10.50.50.50", IPSECSWITCH=DISABLE;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_N to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=7;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
//Adding the IP address of the gNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
gNodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the gNodeB can be
set as required.
ADD IPADDR4: ITFID=1, IP="10.33.33.188", MASK="255.255.255.255";
//Adding routes from the gNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTID=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTID=10, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTID=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=4000,
RXBW=4000;
//Adding a user-plane link
//Adding an S1-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="10.33.33.188", IPSECSWITCH=DISABLE;
//Adding an S1-U peer
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="10.50.50.50", IPSECSWITCH=DISABLE;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

UMTS Side

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_U to the UMPT_N
ADD TUNNEL: SCN=0, SSRN=1, SSN=6, TUNNELID=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the
NodeB use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the NodeB can be set as required.
ADD DEVIP: CN=0, SRN=1, SN=6, SBT=BACK_BOARD, PT=LOOPINT, PN=0, IP="10.37.37.188",
MASK="255.255.255.255", VRFID=0;
//Adding routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to
TUNNEL.
ADD IPRT: RTID=0, SRN=1, SN=6, SBT=BACK_BOARD, DSTIP="10.15.15.15", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTID=2, SRN=1, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=1, SN=6, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=3000, RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000,
TXPBS=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, SRN=1, SN=6, SBT=BACK_BOARD, PT=TUNNEL, PN=0, JNRSCGRP=DISABLE,
LOCALIP="10.37.37.188", PEERIP="10.11.11.11", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.36.36.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_U to the UMPT_N
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
```



```
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=1, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
//Setting an IP address for the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the
NodeB use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.37.37.188", MASK="255.255.255.255";
//Adding routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTID=0, DSTIP="10.15.15.15", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTID=2, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=3000,
RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000, TXPBS=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, TRANSCFGMODE=NEW, JNIPRSCGRP=DISABLE, BPT=TUNNEL, PORTID=0,
LOCALIP="10.37.37.188", PEERIP="10.11.11.11", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.36.36.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

LTE Side

If **GTRANSPARA**. **TRANSCFGMODE** (**5G gNodeB**, **LTE eNodeB**) is set to **OLD**:

```
//Adding tunnels from the UMPT_L to the UMPT_N, GTMU, and UMPT_U
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=6, DCN=0, DSRN=1, DSN=7,
TUNNELTYPE=NON_SHAPING;
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=1, DCN=0, DSRN=0, DSN=6,
TUNNELTYPE=NON_SHAPING;
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=10, DCN=0, DSRN=1, DSN=6,
TUNNELTYPE=NON_SHAPING;
//Setting IP addresses for the eNodeB
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.32.32.1",
MASK="255.255.255.255", VRFID=0;
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.20.20.188",
MASK="255.255.255.0", VRFID=0;
//Adding uplink routes from the NodeB to the RNC/MAE through the eNodeB, from the GBTS/eGBTS to the
BSC through the eNodeB, and from the gNodeB to the MME/S-GW through the eNodeB
ADD IPRT: RTID=0, CN=0, SRN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.15.15.15",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTID=1, CN=0, SRN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.60.60.60",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTID=2, CN=0, SRN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
ADD IPRT: RTID=10, CN=0, SRN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
ADD IPRT: RTID=11, CN=0, SRN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
//Adding downlink routes from the MAE/RNC to the NodeB through the eNodeB, from the BSC to the GBTS/
eGBTS through the eNodeB, and from the MAE/MME/S-GW to the gNodeB through the eNodeB. When IP-
based co-transmission for the GBTS/eGBTS, NodeB, gNodeB, and eNodeB is implemented through
backplane tunnels, the downlink routes from the RNC/MAE to the NodeB through the eNodeB, from the
MME/S-GW/MAE to the gNodeB through the eNodeB, and from the BSC to the GBTS/eGBTS through the
eNodeB must be configured on the eNodeB. In addition, Subboard Type must be set to BACK_BOARD and
Port Type must be set to TUNNEL.
ADD IPRT: RTID=3, CN=0, SRN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.36.36.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=1;
ADD IPRT: RTID=4, CN=0, SRN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.37.37.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=2;
ADD IPRT: RTID=5, CN=0, SRN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.35.35.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=3;
ADD IPRT: RTID=10, CN=0, SRN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.33.33.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=10;
ADD IPRT: RTID=11, CN=0, SRN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.31.31.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=11;
//(Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, gNodeB, NodeB,
and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS, NodeB, and gNodeB are to be
```

deployed using DHCP. Therefore, DHCP relay needs to be enabled for the eNodeB.

```
SET DHCPRELAYSWITCH: ES=ENABLE;
```

// (Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/eGBTS, gNodeB, NodeB, and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS, NodeB, and gNodeB are to be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be added on the eNodeB. For the NodeB, the DHCP server is the RNC or MAE. For the GBTS, the DHCP server is the BSC. For the eGBTS and gNodeB, the DHCP server is the MAE.

```
// Adding the IP address of the DHCP server for the eGBTS/NodeB/gNodeB
```

```
ADD DHCPVSRIP: DHCPVSRIP="10.60.60.60";
```

```
// Adding the IP address of the DHCP server for the GBTS
```

```
ADD DHCPVSRIP: DHCPVSRIP="10.10.10.10";
```

// (Optional) Configuring VLANs. There are two methods of configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation:

// Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop addresses of the uplink route from the GBTS/eGBTS to the BSC through the eNodeB, the uplink route from the NodeB to the RNC through the eNodeB, and the uplink routes from the gNodeB to the MME/S-GW through the eNodeB must be different from the next-hop address of the uplink route for the eNodeB. For example, you can set the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through the eNodeB to 10.20.20.101, the next-hop address of the uplink route from the NodeB to the RNC through the eNodeB to 10.20.20.201, and the next-hop address of the uplink route from the gNodeB to the MME/S-GW through the eNodeB to 10.20.20.202, which are different from the next-hop address (10.20.20.1) of the uplink route for the eNodeB.

```
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=2, SETPRIO=DISABLE;
```

```
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
```

```
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=32, SETPRIO=DISABLE;
```

```
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.202", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=42, SETPRIO=DISABLE;
```

// Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, LTE, and NR data packets and configuring the mapping between a VLAN group and the next hop

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=40, VLANID=22;
```

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=26, VLANID=32;
```

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=34, VLANID=42;
```

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
```

```
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP, VLANGROUPNO=1;
```

If **GTRANSPARA.TRANSFCGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
// Adding tunnels from the UMPT_L to the UMPT_N, GTMU, and UMPT_U
```

```
ADD BACKTUNNEL: PORTID=0, SSRN=0, SSN=7, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=NON_SHAPING;
```

```
ADD BACKTUNNEL: PORTID=1, SSRN=0, SSN=7, DCN=0, DSRN=1, DSN=6, TUNNELTYPE=NON_SHAPING;
```

```
ADD BACKTUNNEL: PORTID=10, SSRN=0, SSN=7, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=NON_SHAPING;
```

```
// Adding an interface to Ethernet port 1 on the UMPT_L (in slot 7)
```

```
SET ETHPORT: CN=0, SRN=0, SN=7, PN=1, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO, DUPLEX=AUTO;
```

```
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFID=0;
```

```
// Adding the INTERFACE setting to a loopback interface
```

```
ADD LOOPBACK: PORTID=700, SRN=0, SN=7;
```

```
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
```

```
// Setting IP addresses for the eNodeB
```

```
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
```

```
ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
```

// Adding uplink routes from the NodeB to the RNC/MAE through the eNodeB, from the GBTS/eGBTS to the BSC through the eNodeB, and from the gNodeB to the MME/S-GW through the eNodeB

```
ADD IPROUTE4: RTID=0, DSTIP="10.15.15.15", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
```

```
ADD IPROUTE4: RTID=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
```

```
ADD IPROUTE4: RTID=2, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
```

```
ADD IPROUTE4: RTID=10, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
```

```
ADD IPROUTE4: RTID=11, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
```

// Adding downlink routes from the MAE/RNC to the NodeB through the eNodeB, from the BSC to the GBTS/eGBTS through the eNodeB, and from the MAE/MME/S-GW to the gNodeB through the eNodeB. When IP-based co-transmission for the GBTS/eGBTS, NodeB, gNodeB, and eNodeB is implemented through

backplane tunnels, the downlink routes from the RNC/MAE to the NodeB through the eNodeB, from the MME/S-GW/MAE to the gNodeB through the eNodeB, and from the BSC to the GBTS/eGBTS through the eNodeB must be configured on the eNodeB. In addition, Port Type must be set to TUNNEL.

```
ADD IPROUTE4: RTIDX=3, DSTIP="10.36.36.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
ADD IPROUTE4: RTIDX=4, DSTIP="10.37.37.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=0;
ADD IPROUTE4: RTIDX=5, DSTIP="10.35.35.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=1;
ADD IPROUTE4: RTIDX=10, DSTIP="10.33.33.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=10;
ADD IPROUTE4: RTIDX=11, DSTIP="10.31.31.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=11;
//(Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, gNodeB, NodeB, and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS, NodeB, and gNodeB are to be deployed using DHCP. Therefore, DHCP relay needs to be enabled for the eNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//(Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/eGBTS, gNodeB, NodeB, and eNodeB, the eNodeB needs to work as the relay if the GBTS/eGBTS, NodeB, and gNodeB are to be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be added on the eNodeB. For the NodeB, the DHCP server is the RNC or MAE. For the GBTS, the DHCP server is the BSC. For the eGBTS and gNodeB, the DHCP server is the MAE.
//Adding the IP address of the DHCP server for the eGBTS/NodeB/gNodeB
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//Adding the IP address of the DHCP server for the GBTS
ADD DHCPVRIP: DHCPVRIP="10.10.10.10";
//(Optional) Configuring VLANs
//There are two methods of configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation when the VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop addresses of the uplink route from the GBTS/eGBTS to the BSC through the eNodeB, the uplink route from the NodeB to the RNC through the eNodeB, and the uplink routes from the gNodeB to the MME/S-GW through the eNodeB must be different from the next-hop address of the uplink route for the eNodeB. For example, you can set the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through the eNodeB to 10.20.20.101, the next-hop address of the uplink route from the NodeB to the RNC through the eNodeB to 10.20.20.201, and the next-hop address of the uplink route from the gNodeB to the MME/S-GW through the eNodeB to 10.20.20.202, which are different from the next-hop address (10.20.20.1) of the uplink route for the eNodeB.
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=32, SETPRIO=DISABLE;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.202", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=42, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, LTE, and NR data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=34, VLANID=42;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP, VLANGROUPNO=1;
//Configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation when the INTERFACE MO is used
//Configuring the VLAN of the interface for GSM data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for UMTS data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;
//Configuring the VLAN of the interface for NR data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;
```

5.4.3.13.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.13.4 Activation Verification

After co-transmission is enabled on the base station, check whether it is enabled by verifying the status of the IP links between the base station and the peer devices.

NR Side

On the gNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the gNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=1, SN=7, SRCIP="10.33.33.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;  
PING: CN=0, SRN=1, SN=7, SRCIP="10.33.33.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

LTE Side

On the eNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the eNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=0, SN=7, SRCIP="10.20.20.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;  
PING: CN=0, SRN=0, SN=7, SRCIP="10.20.20.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

UMTS Side

After the configuration file is delivered to the NodeB and activated, perform the following step on the NodeB to verify that the transmission link between the NodeB and the RNC is normal:

On the NodeB, ping the IP address of the RNC. If the ping operations succeed, the transmission links are normal.

```
PING: CN=0, SRN=1, SN=6, SRCIP="10.37.37.188", DSTIP="10.15.15.15", CONTPING=DISABLE, APPTIF=NO;
```

GSM Side

After the configuration file is delivered to the GBTS/eGBTS and activated, perform the following step on the BSC to verify that the transmission link between the BSC and the GBTS/eGBTS is normal:

On the BSC, ping the IP address of the GTMUc. If the ping operations succeed, the transmission links are normal.

```
PING IP: SIPADDR="10.10.10.10", DESTIP="10.35.35.188", NEXTHOP="10.21.21.254", CONTPING=NO;
```

5.4.3.14 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT NU+GL/GT/GM MBTS Side in IP over FE/GE Mode

In this scenario, the GSM base station can be a GBTS or an eGBTS, and the main control board must be a GTMUc/UMPT. The main control board of NR must be a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga/UMPTga.

5.4.3.14.1 Data Preparation

Figure 5-24 shows an example of main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT NU+GL/GT/GM MBTS side. In this scenario, a 10GE/25GE/50GE port on the gNodeB UMPT_N serves as the co-transmission port of the separate-MPT NU+GL/GT/GM MBTS and connects to the RNC/BSC/MME/S-GW. The UMPT_U and the UMPT_N in one BBU are interconnected through the backplane. The UMPT_N connects to the GTMUc and UMPT_L through the UCIU. Table 5-14 describes the IP address plan.

Figure 5-24 Main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT NU+GL/GT/GM MBTS side

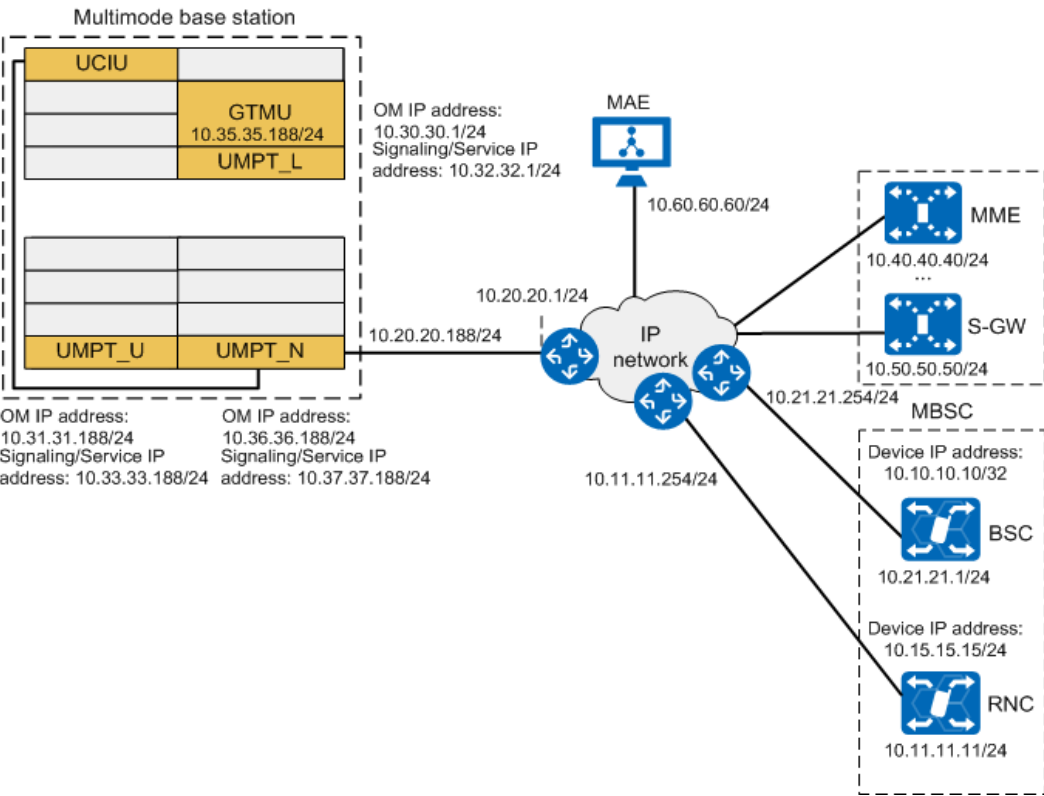


Table 5-14 Data plan

Item	Instance	Remarks
Device IP address of the BSC	10.10.10.10/32	None

Item	Instance	Remarks
Port IP address of the BSC	10.21.21.1/24	None
Device IP address of the RNC	10.15.15.15/24	None
Port IP address of the RNC	10.11.11.11/24	None
IP address of the port on the router that connects to the RNC	10.11.11.254/24	None
IP address of the port on the router that connects to the BSC	10.21.21.254/24	None
IP address of the port on the router that connects to the UMPT_N	10.20.20.1/24	None
OMCH IP address of the eNodeB	10.30.30.1/24	LTE: management-plane IP address. This IP address is configured on the main control board.
Signaling/Service IP address of the eNodeB	10.32.32.1/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of 10GE port 1 on the gNodeB UMPT_N	10.20.20.188/24	NR: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment This IP address is configured on the co-transmission port.
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
IP address of the MAE	10.60.60.60/24	UMTS/LTE/NR: peer IP address for the OMCH
OMCH IP address of the gNodeB	10.36.36.188/24	NR: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the gNodeB	10.37.37.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.

Item	Instance	Remarks
OMCH IP address of the NodeB	10.31.31.188/24	UMTS: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the NodeB	10.33.33.188/24	UMTS: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of the GBTS	10.35.35.188/24	GSM: communication IP address of the GBTS. This is a logical IP address configured on the GBTS main control board.
OM/Signaling/Service IP address of the eGBTS	10.35.35.188/24	The same IP address is used for OM, signaling, and services for the eGBTS.
ESN of the GBTS	abcdefghijklmn	For the GBTS, the ESN needs to be planned. For the eGBTS, the ESN does not need to be planned.

NOTE

In this scenario, the GTMU, UMPT_L, and UMPT_U do not need to be configured with Ethernet port attributes because the GTMU and UMPT_L communicate with the UCIU through the backplane, the UCIU communicates with the UMPT_N through the M interface, and the UMPT_U communicates with the UMPT_N through the backplane.

5.4.3.14.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

MAE Side

On the MAE side, add two routes to the DHCP relay of the NodeB and eNodeB with the destination IP address of one route set to the OMCH IP address of the gNodeB and that of the other route set to the lower-level transmission port IP address of the gNodeB.

GSM Side (GBTS)

```
//Adding a tunnel from the GTMU to the UMPT_N
ADD BTSTUNNEL: IDTYPE=BYID, BTSID=10, SRCCN=0, SRCRN=0, SRCN=6, TN=0, DSTCN=0, DSTSRN=1,
```

```
DSTSN=7;
//Adding the IP address of the GTMU
ADD BTSDEVIP: IDTYPE=BYID, BTSID=10, PT=LOOPINTERFACE, PN=0, CN=0, SRN=0, SN=6,
IP="10.35.35.188", MASK="255.255.255.255";
//Setting the communication IP address of the GBTS. Unlike non-co-transmission scenarios, this scenario
requires that the GBTS use the logical IP address, that is, BTS Communication Type must be set to LOGICIP.
In non-co-transmission scenarios, the IP address of the GBTS can be set as required.
SET BTSIP: IDTYPE=BYID, BTSID=10, BTSCOMTYPE=LOGICIP, BTSIP="10.35.35.188", BSCIP="10.10.10.10",
BTSMUTIP=NO;
//Adding a route from the GBTS to the BSC. Unlike non-co-transmission scenarios, this scenario requires
that Route Type be set to OUTIF, Interface Type be set to TUNNEL, and Outgoing Interface No. be set to the
number of the added tunnel.
ADD BTSIPRT: IDTYPE=BYID, BTSID=10, RTIDX=1, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=OUTIF, CN=0, SRN=0, SN=6, ITFTYPE=TUNNEL, IFNO=0;
//Adding the ESN of the GBTS
ADD BTSESN: IDTYPE=BYID, BTSID=10, MAINDEVTAB="abcdefghijklmn", OMBEARBOARD=BACKBOARD;
//Optional Adding two routes from the BSC to the GBTS's DHCP relay with the destination IP address of
one route set to the OMCH IP address of the eNodeB and that of the other route set to the lower-level
transmission port IP address of the eNodeB
ADD IPRT: SRN=0, SN=16, DSTIP="10.20.20.188", DSTMASK="255.255.255.255", NEXTHOP="10.21.21.254",
PRIORITY=HIGH, REMARK="relay";
```

GSM Side (eGBTS)

If **GTRANSPARA.TRANSFCFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the GTMU to the UMPT_L
ADD TUNNEL: SCN=0, SSRN=0, SSN=6, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the eGBTS can be set as required.
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.35.35.188",
MASK="255.255.255.255", VRFID=0;
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to
TUNNEL.
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.21.21.1", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANSFCFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the GTMU to the UMPT_L
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=0, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFID=0;
//Setting an IP address for the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.35.35.188", MASK="255.255.255.255";
//Adding routes from the eGBTS to the BSC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPROUTE4: RTIDX=0, DSTIP="10.21.21.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.35.35.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

LTE Side

If **GTRANSPARA.TRANSFCFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:


```
//Adding a tunnel from the UMPT_L to the UMPT_N
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Adding the IP address of the eNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
eNodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the eNodeB can be
set as required.
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.32.32.1",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the eNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set
to TUNNEL.
ADD IPRT: RTIDX=0, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=10, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=1, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=0, SN=7, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=4000, RXBW=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, SN=7, SBT=BACK_BOARD, PT=TUNNEL, JNRSCGRP=DISABLE,
LOCALIP="10.32.32.1", PEERIP="10.50.50.50", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_L to the UMPT_N
ADD BACKTUNNEL: PORTID=0, SSN=7, DCN=0, DSRN=1, DSN=7;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Adding the IP address of the eNodeB. Unlike non-co-transmission scenarios, this scenario requires that the
eNodeB use the logical IP address. In non-co-transmission scenarios, the IP address of the eNodeB can be
set as required.
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
//Adding routes from the eNodeB to the MME/S-GW/MAE. Unlike non-co-transmission scenarios, this
scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTIDX=0, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=10, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=4000,
RXBW=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, TRANS CFGMODE=NEW, JNIPRSCGRP=DISABLE, BPT=TUNNEL, PORTID=0,
LOCALIP="10.32.32.1", PEERIP="10.50.50.50", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

UMTS Side

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_U to the UMPT_N
ADD TUNNEL: SCN=0, SSRN=1, SSN=6, TUNNELID=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the
NodeB use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the NodeB can be set as required.
ADD DEVIP: CN=0, SRN=1, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.33.33.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set to
TUNNEL.
```

```
ADD IPRT: RTIDX=0, SRN=1, SN=6, SBT=BACK_BOARD, DSTIP="10.15.15.15", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
ADD IPRT: RTIDX=2, SRN=1, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=6;
//Adding a transmission resource group
ADD RSCGRP: CN=0, SRN=1, SN=6, BEAR=IP, SBT=BACK_BOARD, PT=TUNNEL, RSCGRPID=0, RU=KBPS,
TXBW=3000, RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000,
TXPBS=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, SRN=1, SN=6, SBT=BACK_BOARD, PT=TUNNEL, PN=0, JNRSCGRP=DISABLE,
LOCALIP="10.33.33.188", PEERIP="10.11.11.11", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_U to the UMPT_N
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=6, DCN=0, DSRN=1, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=1, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the NodeB. Unlike non-co-transmission scenarios, this scenario requires that the
NodeB use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.33.33.188", MASK="255.255.255.255";
//Adding routes from the NodeB to the RNC/MAE. Unlike non-co-transmission scenarios, this scenario
requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPRROUTE4: RTIDX=0, DSTIP="10.15.15.15", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=2, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=0, PT=TUNNEL, PORTID=0, RSCGRPNO=0, RU=KBPS, TXBW=3000,
RXBW=3000, TXCBS=4000, TXEBS=4000, TXCIR=4000, RXCIR=4000, TXPIR=4000, RXPIR=4000, TXPBS=4000;
//Adding an IP path
ADD IPPATH: PATHID=0, TRANS CFGMODE=NEW, JNIPRSCGRP=DISABLE, BPT=TUNNEL, PORTID=0,
LOCALIP="10.33.33.188", PEERIP="10.11.11.11", PATHTYPE=ANY;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.31.31.188", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

NR Side

NOTE

In this scenario, the gNodeB uses the CI port for communication. Therefore, Ethernet ports are not required.

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding tunnels from the UMPT_N to the UMPT_L, GTMU, and UMPT_U
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=1, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=SHAPING;
ADD TUNNEL: SCN=0, SSRN=1, SSN=7, TUNNELID=6, DCN=0, DSRN=1, DSN=6, TUNNELTYPE=SHAPING;
//Setting IP addresses for the gNodeB
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.37.37.188",
MASK="255.255.255.255", VRFIDX=0;
ADD DEVIP: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.20.20.188",
MASK="255.255.255.0", VRFIDX=0;
//Adding uplink routes from the NodeB to the RNC/MAE through the gNodeB, an uplink route from the
GBTS/eGBTS to the BSC through the gNodeB, and uplink routes from the eNodeB to the MME/S-GW
through the gNodeB
ADD IPRT: RTIDX=0, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.15.15.15",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=1, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.60.60.60",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=2, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
```

```

ADD IPRT: RTIDX=10, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
ADD IPRT: RTIDX=11, CN=0, SRN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1", PREF=60;
//Adding downlink routes from the MAE/RNC to the NodeB through the gNodeB, from the BSC to the GBTS/
eGBTS through the gNodeB, and from the MAE/MME/S-GW to the eNodeB through the gNodeB. When IP-
based co-transmission for the GBTS/eGBTS, NodeB, eNodeB, and gNodeB is implemented through
backplane tunnels, the downlink routes from the RNC/MAE to the NodeB through the gNodeB, from the
MME/S-GW/MAE to the eNodeB through the gNodeB, and from the BSC to the GBTS/eGBTS through the
gNodeB must be configured on the gNodeB. In addition, Subboard Type must be set to BACK_BOARD and
Port Type must be set to TUNNEL.
ADD IPRT: RTIDX=3, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.31.31.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=1;
ADD IPRT: RTIDX=4, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.33.33.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=2;
ADD IPRT: RTIDX=5, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.35.35.188",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=3;
ADD IPRT: RTIDX=10, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.32.32.1",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=10;
ADD IPRT: RTIDX=11, CN=0, SRN=1, SN=7, SBT=BACK_BOARD, DSTIP="10.30.30.1",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=11;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, eNodeB, NodeB,
and gNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS, NodeB, and eNodeB are to be
deployed using DHCP. Therefore, DHCP relay needs to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/
eGBTS, eNodeB, NodeB, and gNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS, NodeB,
and eNodeB are to be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be
added on the gNodeB. For the NodeB, the DHCP server is the RNC or MAE. For the GBTS, the DHCP server
is the BSC. For the eGBTS and eNodeB, the DHCP server is the MAE.
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
ADD DHCPVRIP: DHCPVRIP="10.10.10.10";
//((Optional) Configuring VLANs. There are two methods of configuring different VLANs for GSM, UMTS,
LTE, and NR data for isolation:
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink route from the GBTS/eGBTS to the BSC through the gNodeB, the uplink route from
the NodeB to the RNC through the gNodeB, and the uplink route from the eNodeB to the MME/S-GW
through the gNodeB must be different from the next-hop address of the uplink route for the gNodeB. For
example, you can set the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through
the gNodeB to 10.20.20.101, the next-hop address of the uplink route from the NodeB to the RNC through
the gNodeB to 10.20.20.201, and the next-hop address of the uplink route from the eNodeB to the MME/S-
GW through the gNodeB to 10.20.20.202, which are different from the next-hop address (10.20.20.1) of the
uplink route for the gNodeB.
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=32, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.202", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=42, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, LTE, and NR
data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIOR=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIOR=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIOR=34, VLANID=42;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIOR=0, VLANID=22;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP,
VLANGROUPNO=1;

```

If **GTRANSPARA.TRANSFCFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```

//Adding tunnels from the UMPT_N to the UMPT_L, GTMU, and UMPT_U
ADD BACKTUNNEL: PORTID=0, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=7;
ADD BACKTUNNEL: PORTID=1, SSRN=1, SSN=7, DCN=0, DSRN=0, DSN=6;
ADD BACKTUNNEL: PORTID=10, SSRN=1, SSN=7, DCN=0, DSRN=1, DSN=6;
//Adding an interface to Ethernet port 1 on the UMPT_N (in slot 7)
SET ETHPORT: CN=0, SRN=1, SN=7, PN=1, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO,
DUPLEX=AUTO;
ADD INTERFACE: ITFIDX=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFIDX=0;

```

```
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=1, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting IP addresses for the gNodeB
ADD IPADDR4: ITFID=1, IP="10.37.37.188", MASK="255.255.255.255";
ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
//Adding uplink routes from the NodeB to the RNC/MAE through the gNodeB, an uplink route from the
GBTS/eGBTS to the BSC through the gNodeB, and uplink routes from the eNodeB to the MME/S-GW
through the gNodeB
ADD IPRROUTE4: RTIDX=0, DSTIP="10.15.15.15", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTIDX=1, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTIDX=2, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1", PREF=60;
ADD IPRROUTE4: RTIDX=10, DSTIP="10.40.40.40", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1", PREF=60;
ADD IPRROUTE4: RTIDX=11, DSTIP="10.50.50.50", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1", PREF=60;
//Adding downlink routes from the MAE/RNC to the NodeB through the gNodeB, from the BSC to the GBTS/
eGBTS through the gNodeB, and from the MAE/MME/S-GW to the eNodeB through the gNodeB. When IP-
based co-transmission for the GBTS/eGBTS, NodeB, eNodeB, and gNodeB is implemented through
backplane tunnels, the downlink routes from the RNC/MAE to the NodeB through the gNodeB, from the
MME/S-GW/MAE to the eNodeB through the gNodeB, and from the BSC to the GBTS/eGBTS through the
gNodeB must be configured on the gNodeB. In addition, Port Type must be set to TUNNEL.
ADD IPRROUTE4: RTIDX=3, DSTIP="10.31.31.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=4, DSTIP="10.33.33.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=5, DSTIP="10.35.35.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=1;
ADD IPRROUTE4: RTIDX=10, DSTIP="10.32.32.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=10;
ADD IPRROUTE4: RTIDX=11, DSTIP="10.30.30.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=11;
//((Optional) Enabling DHCP relay. When co-transmission is enabled for the GBTS/eGBTS, eNodeB, NodeB,
and gNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS, NodeB, and eNodeB are to be
deployed using DHCP. Therefore, DHCP relay needs to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. When co-transmission is enabled for the GBTS/
eGBTS, eNodeB, NodeB, and gNodeB, the gNodeB needs to work as the relay if the GBTS/eGBTS, NodeB,
and eNodeB are to be deployed using DHCP. Therefore, the IP address of the DHCP server needs to be
added on the gNodeB. For the NodeB, the DHCP server is the RNC or MAE. For the GBTS, the DHCP server
is the BSC. For the eGBTS and eNodeB, the DHCP server is the MAE.
//Adding the IP address of the DHCP server for the eGBTS/NodeB/eNodeB
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//Adding the IP address of the DHCP server for the GBTS
ADD DHCPVRIP: DHCPVRIP="10.10.10.10";
//((Optional) Configuring VLANs
//There are two methods of configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation
when the VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
addresses of the uplink route from the GBTS/eGBTS to the BSC through the gNodeB, the uplink route from
the NodeB to the RNC through the gNodeB, and the uplink route from the eNodeB to the MME/S-GW
through the gNodeB must be different from the next-hop address of the uplink route for the gNodeB. For
example, you can set the next-hop address of the uplink route from the GBTS/eGBTS to the BSC through
the gNodeB to 10.20.20.101, the next-hop address of the uplink route from the NodeB to the RNC through
the gNodeB to 10.20.20.201, and the next-hop address of the uplink route from the eNodeB to the MME/S-
GW through the gNodeB to 10.20.20.202, which are different from the next-hop address (10.20.20.1) of the
uplink route for the gNodeB.
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=32, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.202", MASK="255.255.255.0", VLANMODE=SINGLEVLAN,
VLANID=42, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, LTE, and NR
```

```
data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=26, VLANID=32;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=34, VLANID=42;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
ADD VLANMAP: VRFID=0, NEXTHOP="10.20.20.1", MASK="255.255.255.0", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
//Configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation when the INTERFACE MO is
used
//Configuring the VLAN of the interface for GSM data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for UMTS data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;
//Configuring the VLAN of the interface for NR data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;
```

5.4.3.14.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.14.4 Activation Verification

After co-transmission is enabled on the base station, check whether it is enabled by verifying the status of the IP links between the base station and the peer devices.

NR Side

On the gNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the gNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=1, SN=7, SRCIP="10.20.20.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
PING: CN=0, SRN=1, SN=7, SRCIP="10.20.20.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

LTE Side

On the eNodeB, ping the IP addresses of the MME/S-GW. If the IP addresses can be pinged, the transmission links between the eNodeB and the MME/S-GW are normal.

```
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
```

UMTS Side

After the configuration file is delivered to the NodeB and activated, perform the following step on the NodeB to verify that the transmission link between the NodeB and the RNC is normal:

On the NodeB, ping the IP address of the RNC. If the ping operations succeed, the transmission links are normal.

```
PING: CN=0, SRN=1, SN=6, SRCIP="10.33.33.188", DSTIP="10.15.15.15", CONTPING=DISABLE, APPTIF=NO;
```

GSM Side

After the configuration file is delivered to the GBTS/eGBTS and activated, perform the following step on the BSC to verify that the transmission link between the BSC and the GBTS/eGBTS is normal:

On the BSC, ping the IP address of the GTMUc. If the ping operations succeed, the transmission links are normal.

PING IP: SIPADDR="10.10.10.10", DESTIP="10.35.35.188", NEXTHOP="10.21.21.254", CONTPING=NO;

5.4.3.15 Co-transmission on the Co-MPT GULTM MBTS Side in IP over FE/GE Mode

5.4.3.15.1 Data Preparation

Figure 5-25 shows an example of co-transmission on the co-MPT GUL MBTS side. In this scenario, an FE/GE port on the UMPT_GUL serves as the co-transmission port of the GUL MBTS and connects to the MBSC/MME/S-GW. Table 5-15 describes the IP address plan.

Figure 5-25 IP co-transmission on the co-MPT GUL MBTS side

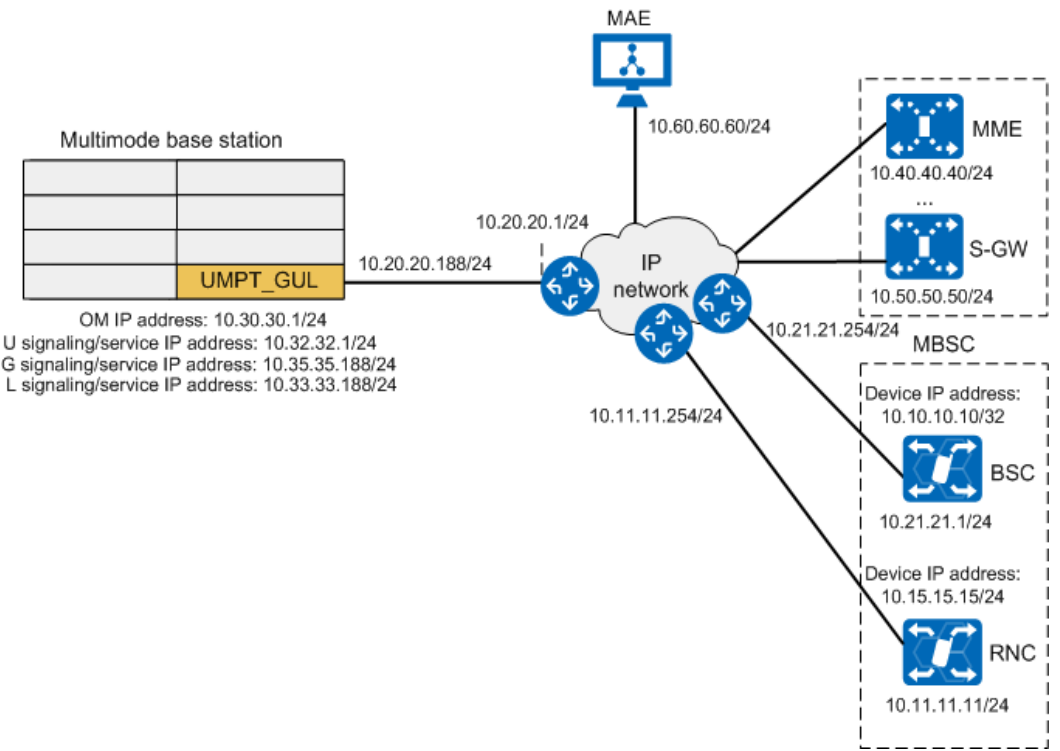


Table 5-15 Data plan

Item	Instance	Remarks
Device IP address of the BSC	10.10.10.10/32	None

Item	Instance	Remarks
Port IP address of the BSC	10.21.21.1/24	None
IP address of the port on the router that connects to the BSC	10.21.21.254/24	None
IP address of the port on the router that connects to the RNC	10.11.11.254/24	None
IP address of the port on the router that connects to the UMPT_GUL	10.20.20.1/24	None
IP address of the port on the UMPT_GUL	10.20.20.188/24	Device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment. This IP address is configured on the co-transmission port.
OMCH IP address	10.30.30.1/24	Management-plane IP address of the MBTS. This is a logical IP address configured on the UMPT_GUL. NOTE In this scenario, only one OMCH IP address is configured for the MBTS because only one OMCH is required.
Signaling/Service IP address of UMTS	10.32.32.1/24	UMTS: control-plane and user-plane IP address. This is a logical IP address configured on the UMPT_GUL.
Device IP address of the RNC	10.15.15.15/24	None
Port IP address of the RNC	10.11.11.11/24	None
IP address of the MAE	10.60.60.60/24	Peer IP address for the OMCH of the MBTS
Signaling/Service IP address of GSM	10.35.35.188/24	GSM: control-plane and user-plane IP address. This is a logical IP address configured on the UMPT_GUL.
IP address of the MME	10.40.40.40/24	LTE: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	LTE: peer IP address for the IP path

Item	Instance	Remarks
Signaling/Service IP address of LTE	10.33.33.188/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the UMPT_GUL.
ESN of the MBTS	abcdefghijklmn	None

5.4.3.15.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTE

In this scenario, the MBTS is configured with only one UMPT_GUL. Transport data is configured for the MBTS as common data instead of for the GSM, UMTS, and LTE sides separately.

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a port IP address
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH, PN=0, IP="10.20.20.188",
MASK="255.255.255.0", VRFIDX=0;
//Adding the IP address of the NodeB
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=1, IP="10.32.32.1",
MASK="255.255.255.255", VRFIDX=0;
//Adding the IP address of the eGBTS
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=2, IP="10.35.35.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding the IP address of the eNodeB
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=3, IP="10.33.33.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding an uplink route from the MBTS to the RNC with Route Type set to NEXTHOP and Next Hop IP set
to the IP address of the directly connected router
ADD IPRT: RTIDX=0, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.15.15.15", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
//Adding an uplink route from the MBTS to the BSC
ADD IPRT: RTIDX=1, CN=1, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
//Adding an uplink route from the MBTS to the MME/S-GW
ADD IPRT: RTIDX=2, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=3, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
//(Optional) Configuring VLANs. There are two methods of configuring different VLANs for GSM, UMTS,
and LTE data for isolation:
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
address of the uplink route from the MBTS to the BSC, the next-hop address of the uplink route from the
MBTS to the RNC, and the next-hop address of the uplink route from the MBTS to the MME/S-GW must be
different from each other. For example, you can set the next-hop address of the uplink route from the
MBTS to the BSC to 10.20.20.101, which is different from the next-hop address (10.20.20.1) of the uplink
```



```
route from the MBTS to the RNC and the next-hop IP address (10.20.20.10) of the uplink route from the
MBTS to the MME/S-GW.
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.10", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=20, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, and LTE data
packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=34, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
ADD VLANMAP: NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding an interface to the co-transmission port
ADD ETHPORT: CN=0, SRN=1, SN=7, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO,
DUPLEX=AUTO;
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71;
//Adding the INTERFACE setting to the UMTS loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Adding the INTERFACE setting to the GSM loopback interface
ADD LOOPBACK: PORTID=800, SN=7;
ADD INTERFACE: ITFID=2, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=800, VRFIDX=0;
//Adding the INTERFACE setting to the LTE loopback interface
ADD LOOPBACK: PORTID=900, SN=7;
ADD INTERFACE: ITFID=3, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=900, VRFIDX=0;
//Adding a port IP address
ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
//Adding the IP address of the NodeB
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
//Adding the IP address of the eGBTS
ADD IPADDR4: ITFID=2, IP="10.35.35.188", MASK="255.255.255.255";
//Adding the IP address of the eNodeB
ADD IPADDR4: ITFID=3, IP="10.33.33.188", MASK="255.255.255.255";
//Adding an uplink route from the MBTS to the RNC with Route Type set to NEXTHOP and Next Hop IP set
to the IP address of the directly connected router
ADD IPRROUTE4: RTIDX=0, DSTIP="10.15.15.15", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
//Adding an uplink route from the MBTS to the BSC
ADD IPRROUTE4: RTIDX=1, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
//Adding an uplink route from the MBTS to the MME/S-GW
ADD IPRROUTE4: RTIDX=2, DSTIP="10.40.40.0", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTIDX=3, DSTIP="10.50.50.0", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
//(Optional) Configuring VLANs
//There are two methods of configuring different VLANs for GSM, UMTS, and LTE data for isolation when
the VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
address of the uplink route from the MBTS to the BSC, the next-hop address of the uplink route from the
MBTS to the RNC, and the next-hop address of the uplink route from the MBTS to the MME/S-GW must be
different from each other. For example, you can set the next-hop address of the uplink route from the
MBTS to the BSC to 10.20.20.101, which is different from the next-hop address (10.20.20.1) of the uplink
route from the MBTS to the RNC and the next-hop IP address (10.20.20.10) of the uplink route from the
MBTS to the MME/S-GW.
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.10", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=2, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=20, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, and LTE data
packets and configuring the mapping between a VLAN group and the next hop
```

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=34, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
ADD VLANMAP: NEXTHOPIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
//Configuring different VLANs for GSM, UMTS, and LTE data for isolation when the INTERFACE MO is used
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFIDX=0;
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFIDX=0;
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFIDX=0;
```

5.4.3.15.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.15.4 Activation Verification

After the configuration file is delivered to the MBTS and activated, perform the following steps on the MBTS side to check whether the transmission links between the MBTS and the peer ends are normal. If the IP addresses can be pinged, the transmission links are normal.

- On the MBTS, ping the IP address of the RNC.
PING: CN=0, SRN=0, SN=7, SRCIP="10.32.32.1", DSTIP="10.15.15.15", CONTPING=DISABLE, APPTIF=NO;
- On the MBTS, ping the IP address of the BSC.
PING: CN=0, SRN=0, SN=7, SRCIP="10.35.35.1", DSTIP="10.10.10.10", CONTPING=DISABLE, APPTIF=NO;
- On the MBTS, ping the IP address of the MME.
PING: CN=0, SRN=0, SN=7, SRCIP="10.33.33.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
- On the MBTS, ping the IP address of the S-GW.
PING: CN=0, SRN=0, SN=7, SRCIP="10.33.33.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;

5.4.3.16 Co-transmission on the Co-MPT MBTS Side in IPv4/IPv6 Dual-Stack over FE/GE/10GE/25GE/50GE Mode

5.4.3.16.1 Data Preparation

Scenario 1: LN MBTS

[Figure 5-26](#) shows an example of co-transmission on the co-MPT LN MBTS side in NSA networking. In this scenario, an Ethernet port on the UMPT_LN serves as the co-transmission port of the LN MBTS and connects to the MME/S-GW. [Table 5-16](#) describes the IP address plan.

Figure 5-26 IP dual-stack co-transmission on the co-MPT LN MBTS side in NSA networking

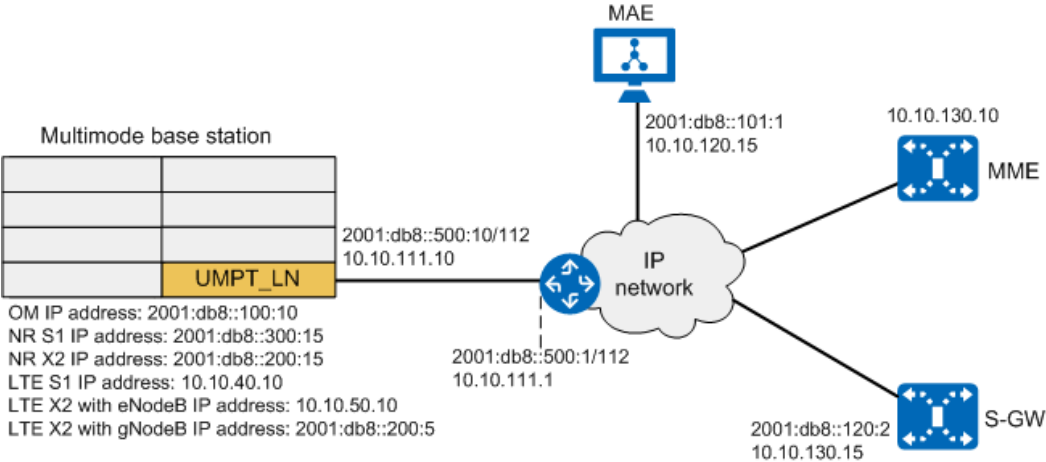


Table 5-16 Data plan of scenario 1

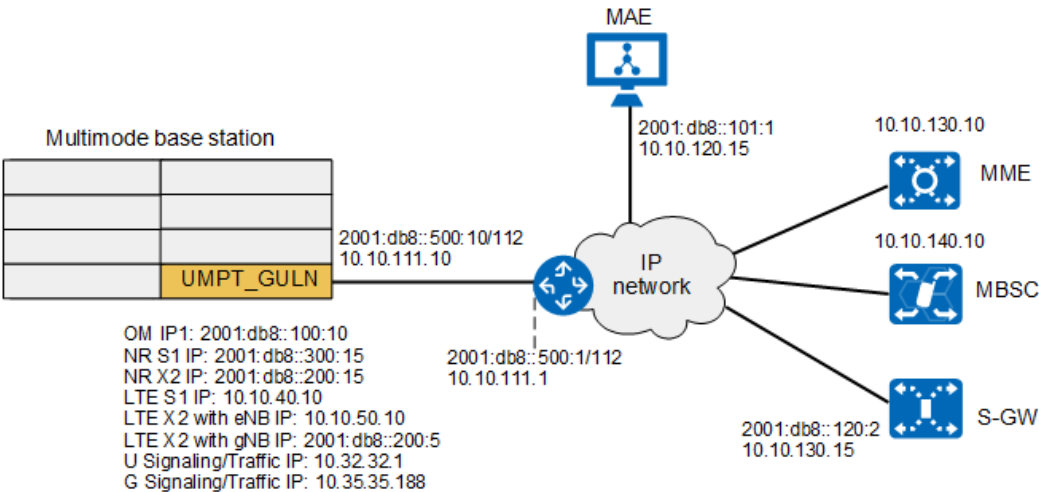
Item	Instance	Remarks
IP address of the Ethernet port on the router that connects to the UMPT_LN	2001:db8::500:1/112 10.10.111.1	Dual-stack IP addresses must be planned for Ethernet ports in dual-stack transmission.
IP addresses of the Ethernet port on the UMPT_LN	2001:db8::500:10/112 10.10.111.10/28	Dual-stack IP addresses must be planned for Ethernet ports in dual-stack transmission.
OMCH IP address	2001:db8::100:10/128	Management-plane IPv6 address of the MBTS. This is a logical IP address. NOTE In this scenario, only one OMCH IP address is configured for the MBTS because only one OMCH is required. Either IPv6 or IPv4 transmission can be used.
User-plane IP address of the NR S1 interface	2001:db8::300:15/128	In NSA networking, the S1 interface has only the user-plane IP address.
Control-plane/user-plane IP address of the NR X2 interface	2001:db8::200:15/128	NR: IPv6 transmission is used over the X2 interface.
Control-plane/user-plane IP address of the LTE S1 interface	10.10.40.10/32	LTE: IPv4 transmission is used over the S1 interface.
IP address of the MAE	10.60.60.60/24	Peer IP address for the OMCH of the MBTS

Item	Instance	Remarks
Control-plane/user-plane IP address of the LTE X2 interface	10.10.50.10/32 2001:db8::200:5/128	Dual-stack IP addresses for the LTE X2 interface: The X2 interface between eNodeBs uses IPv4 transmission. The X2 interface between the eNodeB and gNodeB uses IPv6 transmission.
IP address of the MME	10.10.130.10/24	LTE: IPv4 transmission is used over the S1 interface.
IPv4 address of the S-GW	10.10.130.15/24	LTE: IPv4 transmission is used over the S1 user plane, and the S-GW must be configured with an IPv4 address.
IPv6 address of the S-GW	2001:db8::120:2/128	NR: IPv6 transmission is used over the S1 user plane, and the S-GW must be configured with an IPv6 address.
ESN of the MBTS	abcdefghijklmn	None

Scenario 2: GULN MBTS

Figure 5-27 shows an example of co-transmission on the co-MPT GULN MBTS side. In this scenario, an Ethernet port on the UMPT_GULN serves as the co-transmission port of the MBTS and connects to the MBSC/MME/S-GW. Table 5-16 describes the IP address plan.

Figure 5-27 IP dual-stack co-transmission on the co-MPT GULN MBTS side



5.4.3.16.2 Using MML Commands

Before using MML commands, refer to 5.3 Requirements and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTE

In this scenario, only one UMPT_LN or UMPT_GULN needs to be configured for the MBTS, and transport data needs to be configured for the MBTS as common data. IPv6 or dual-stack transmission supports only the new transmission model, that is, **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** must be set to **NEW**.

Scenario 1

```
//Configuring VLAN interfaces
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=1, VLANID=100, DSCP2PCMAPID=0,
MTU4=1500;
ADD INTERFACE: ITFID=1, ITFTYPE=VLAN, PT=ETH, PORTID=1, VLANID=200, DSCP2PCMAPID=0,
MTU6=1500, IPV6SW=ENABLE;
//Configuring an IPv4 loopback interface
ADD LOOPBACK: PORTID=1, CN=0, SRN=0, SN=7;
ADD INTERFACE: ITFID=3, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=8, VRFID=0;
//Configuring an IPv6 loopback interface
ADD LOOPBACK: PORTID=2, CN=0, SRN=0, SN=7;
ADD INTERFACE: ITFID=4, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=8, VRFID=0, IPV6SW=ENABLE;
//Configuring an IP address for an IPv4 Ethernet port
ADD IPADDR4: ITFID=0, IP="10.10.111.10", MASK="255.255.255.0";
//Configuring an IP address for an IPv6 Ethernet port
ADD IPADDR6: IPADDR6ID="1", ITFID=1, IPV6="2001:db8::500:10", PFXLEN=112;
//Configuring the OMCH IPv6 address
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8::100:10", PEERIP6="2001:db8::100:1",
PEERIP6PFXLEN=112;
//Configuring IPv6 addresses for gNodeB S1 and X2 interfaces
ADD IPADDR6: IPADDR6ID="2", ITFID=4, IPV6="2001:db8::300:15", PFXLEN=128;
ADD IPADDR6: IPADDR6ID="3", ITFID=4, IPV6="2001:db8::200:15", PFXLEN=128;
//Configuring IPv4 addresses for eNodeB S1 and X2 interfaces and an IPv6 address for eNodeB X2 interface
ADD IPADDR4: ITFID=3, IP="10.10.40.10", MASK="255.255.255.255";
ADD IPADDR4: ITFID=3, IP="10.10.50.10", MASK="255.255.255.255";
ADD IPADDR6: IPADDR6ID="4", ITFID=4, IPV6="2001:db8::200:5", PFXLEN=128;
//Adding an uplink IPv6 route from the gNodeB to the S-GW with Route Type set to NEXTHOP and Next
Hop IP set to the IP address of the directly connected router
ADD IPRROUTE6: RTIDX=0, DSTIP="2001:db8::120:0", PFXLEN=112, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8::500:1", PREF=60;
//Adding an uplink route from the MBTS to the MAE
ADD IPRROUTE6: RTIDX=1, DSTIP="2001:db8::101:0", PFXLEN=112, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8::500:1", PREF=60;
//Adding an IPv6 route to the peer end of the X2 interface
ADD IPRROUTE6: RTIDX=2, DSTIP="2001:db8::200:0", PFXLEN=112, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8::500:1", PREF=60;
//Adding an uplink IPv4 route from the eNodeB to the MME/S-GW
ADD IPRROUTE4: RTIDX=10, DSTIP="10.10.130.0", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.111.1";
//Adding an IPv4 route to the peer eNodeB of the X2 interface
ADD IPRROUTE4: RTIDX=11, DSTIP="10.10.50.0", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.111.1";
```

Scenario 2

In addition to the MML commands used in scenario 1, the following MML commands are required in scenario 2:

The user-plane and control-plane configurations for GSM and UMTS on the MBTS side are consistent with those described in [5.4.3.15.2 Using MML Commands](#). The following configurations are added on the base station controller side.

On the RNC side

```
//Querying the IP address of the NodeB
LST UNODEBIP: LstType=ByNodeBId, NodeBId=0, LstFormat=VERTICAL;
//Removing the IP address of the NodeB
RMV UNODEBIP: IDTYPE=BYID, NODEBID=0;
```

On the BSC side

```
//Querying the IP address of the eGBTS
LST BTSOAMIP: IDTYPE=BYID, BTSID=7, LSTFORMAT=VERTICAL;
//Removing the IP address of the eGBTS
RMV BTSOAMIP: IDTYPE=BYID, BTSID=7;
```

5.4.3.16.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.16.4 Activation Verification

After the configuration file is delivered to the MBTS and activated, perform the following steps on the MBTS side to verify that the transmission links between the MBTS and the peer ends are normal. If the IP addresses can be pinged, the transmission links are normal.

- On the MBTS, ping the IPv4 address of the MME.
PING: CN=0, SRN=0, SN=7, SRCIP="10.10.40.10", DSTIP="10.10.130.10", CONTPING=DISABLE, APPTIF=NO;
- On the MBTS, ping the IPv4 address of the S-GW.
PING: CN=0, SRN=0, SN=7, SRCIP="10.10.40.10", DSTIP="10.10.130.15", CONTPING=DISABLE, APPTIF=NO;
- On the MBTS, ping the IPv4 address of the peer eNodeB.
PING: CN=0, SRN=0, SN=7, SRCIP="10.10.50.10", DSTIP="10.10.50.15", CONTPING=DISABLE, APPTIF=NO;
- On the MBTS, ping the IPv6 address of the S-GW.
PING6: DSTIPTYPE=IPADDR6_TYPE_GLOBAL_UNICAST, SRCIP="2001:db8::200:15", DSTIP="2001:db8::120:2", CONTPING=DISABLE;
- On the MBTS, ping the IPv6 address of the peer end of the X2 interface.
PING6: DSTIPTYPE=IPADDR6_TYPE_GLOBAL_UNICAST, SRCIP="2001:db8::200:15", DSTIP="2001:db8::200:10", CONTPING=DISABLE;

5.4.3.17 Co-transmission on the Hybrid-MPT GULN/GUTN/GUMN MBTS Side in IP over FE/GE Mode

In this scenario, the GSM base station is an eGBTS. The main control board of the gNodeB must be a UMPTe/UMPTg/UMPTga/UMPTHa/UMPTHb.

5.4.3.17.1 Data Preparation

This section uses UMPT_GUL+UMPT_N, with UMPT_N providing a co-transmission port in NSA networking, as an example to describe co-transmission on the hybrid-

MPT GULN/GUTN/GUMN MBTS side in IP over FE/GE mode. In this scenario, a 10GE/25GE/50GE port on the UMPT_N serves as the co-transmission port of the GULN/GUTN/GUMN MBTS and connects to the MBSC/MME/S-GW. The UMPT_N and UMPT_GUL are interconnected through the backplane. **Figure 5-28** shows an example of the network topology, and **Table 5-17** describes the IP address plan.

Figure 5-28 Co-transmission on the hybrid-MPT GULN/GUTN/GUMN MBTS side

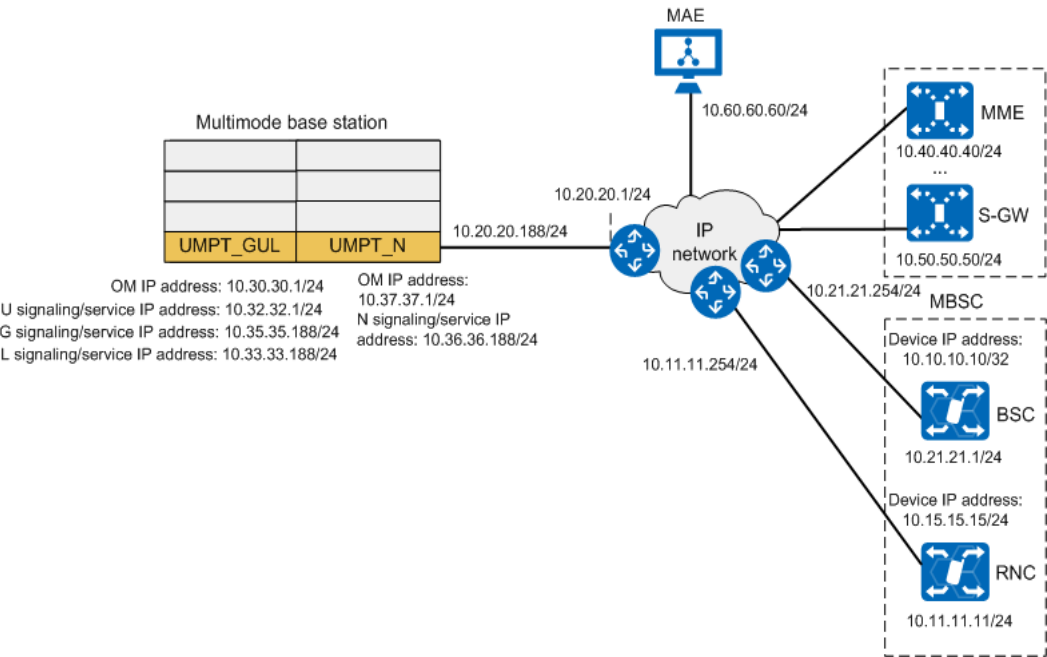


Table 5-17 Data plan

Item	Instance	Remarks
Device IP address of the BSC	10.10.10.10/32	None
Port IP address of the BSC	10.21.21.1/24	None
IP address of the port on the router that connects to the BSC	10.21.21.254/24	None
IP address of the port on the router that connects to the RNC	10.11.11.254/24	None
IP address of the port on the router that connects to the UMPT_N	10.20.20.1/24	None

Item	Instance	Remarks
IP address of the port on the UMPT_N	10.20.20.188/24	Device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment. This IP address is configured on the co-transmission port.
OMCH IP address of the GUL base station	10.30.30.1/24	GUL: management-plane IP address. This is a logical IP address configured on the UMPT_GUL.
Signaling/Service IP address of UMTS	10.32.32.1/24	UMTS: control-plane and user-plane IP address. This is a logical IP address configured on the UMPT_GUL.
Device IP address of the RNC	10.15.15.15/24	None
Port IP address of the RNC	10.11.11.11/24	None
IP address of the MAE	10.60.60.60/24	Peer IP address for the OMCH
Signaling/Service IP address of the eGBTS	10.35.35.188/24	The same IP address is used for signaling and services for the eGBTS.
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
Signaling/Service IP address of LTE	10.33.33.188/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the UMPT_UL.
OMCH IP address of NR	10.37.37.1/24	NR: management-plane IP address. This is a logical IP address configured on the UMPT_N.
Signaling/Service IP address of NR	10.36.36.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the UMPT_N.
ESN of the MBTS	abcdefghijklmn	None

5.4.3.17.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Co-MPT GUL MBTS Side

If **GTRANSPARA.TRANSFCFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_GUL to the UMPT_N
ADD TUNNEL: SCN=0, SSRN=0, SSN=6, TUNNELID=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Setting the IP address of the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address, that is, Port Type must be set to LOOPINT. In non-co-transmission
scenarios, the IP address of the eGBTS can be set as required.
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.35.35.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding the IP address of the NodeB
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.32.32.1",
MASK="255.255.255.255", VRFIDX=0;
//Adding the IP address of the eNodeB
ADD DEVIP: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=LOOPINT, PN=0, IP="10.33.33.188",
MASK="255.255.255.255", VRFIDX=0;
//Adding routes from the eGBTS/NodeB/eNodeB to the MAE. Unlike non-co-transmission scenarios, this
scenario requires that Subboard Type be set to BACK_BOARD, Route Type be set to IF, and Port Type be set
to TUNNEL.
ADD IPRT: RTIDX=0, SRN=0, SN=6, SBT=BACK_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=0;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

If **GTRANSPARA.TRANSFCFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```
//Adding a tunnel from the UMPT_GUL to the UMPT_N
ADD BACKTUNNEL: PORTID=0, SSN=6, DCN=0, DSRN=0, DSN=7, TUNNELTYPE=SHAPING;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SRN=0, SN=6;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Setting an IP address for the eGBTS. Unlike non-co-transmission scenarios, this scenario requires that the
eGBTS use the logical IP address.
ADD IPADDR4: ITFID=1, IP="10.35.35.188", MASK="255.255.255.255";
//Adding the IP address of the NodeB
ADD IPADDR4: ITFID=1, IP="10.32.32.1", MASK="255.255.255.255";
//Adding the IP address of the eNodeB
ADD IPADDR4: ITFID=0, IP="10.33.33.188", MASK="255.255.255.0";
//Adding routes from the eGBTS/NodeB/eNodeB to the MAE. Unlike non-co-transmission scenarios, this
scenario requires that Route Type be set to IF and Port Type be set to TUNNEL.
ADD IPROUTE4: RTIDX=0, DSTIP="10.60.60.60", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//Adding an OMCH
ADD OMCH: FLAG=MASTER, BEAR=IPV4, IP="10.30.30.1", MASK="255.255.255.0", PEERIP="10.60.60.60",
PEERMASK="255.255.255.0", CHECKTYPE=NONE, BRT=NO;
```

NR Side

If **GTRANSPARA.TRANSFCFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

```
//Adding a tunnel from the UMPT_N to the UMPT_GUL
ADD TUNNEL: SCN=0, SSRN=0, SSN=7, TUNNELID=6, DCN=0, DSRN=0, DSN=6,
TUNNELTYPE=NON_SHAPING;
//Adding IP addresses for the gNodeB
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=LOOPINT, PN=1, IP="10.36.36.188",
MASK="255.255.255.255", VRFIDX=0;
```

```

ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.20.20.188",
MASK="255.255.255.0", VRFIDX=0;
//Adding uplink routes from the co-MPT GUL MBTS to the RNC/BSC through the gNodeB with Route Type
set to NEXTHOP and Next Hop IP set to the IP address of the directly connected router
ADD IPRT: RTIDX=0, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.15.15.15", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=3, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
//Adding uplink routes from the co-MPT GUL MBTS to the MME/S-GW/MAE through the gNodeB
ADD IPRT: RTIDX=1, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.40.40.40", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=2, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.50.50.50", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
ADD IPRT: RTIDX=5, CN=0, SN=7, SBT=BASE_BOARD, DSTIP="10.60.60.60", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.20.20.1";
//Adding downlink routes from the BSC/RNC/S-GW/MME/MAE to the GUL MBTS through the gNodeB.
When IP-based co-transmission for the co-MPT GUL MBTS and the gNodeB is implemented through the
backplane tunnel, the downlink route from the BSC/RNC/S-GW/MME/MAE to the co-MPT GUL MBTS
through the gNodeB must be configured on the gNodeB. In addition, Subboard Type must be set to
BACK_BOARD and Port Type must be set to TUNNEL.
ADD IPRT: RTIDX=4, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.35.35.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=1;
ADD IPRT: RTIDX=6, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.32.32.1", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=2;
ADD IPRT: RTIDX=7, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.33.33.188", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=3;
ADD IPRT: RTIDX=8, CN=0, SN=7, SBT=BACK_BOARD, DSTIP="10.30.30.1", DSTMASK="255.255.255.255",
RTTYPE=IF, IFT=TUNNEL, IFNO=4;
//((Optional) Enabling DHCP relay. If the eGBTS/NodeB/eNodeB is to be deployed using DHCP, the gNodeB
needs to work as the relay. Therefore, DHCP relay needs to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. If the eGBTS/NodeB/eNodeB is to be deployed using
DHCP, the gNodeB needs to work as the relay. Therefore, the IP address of the DHCP server needs to be
added on the gNodeB. For the NodeB, the DHCP server is the RNC or MAE. For the eGBTS/eNodeB, the
DHCP server is the MAE.
//Adding the IP address of the DHCP server for the eGBTS/eNodeB
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//Adding the IP address of the DHCP server for the NodeB
ADD DHCPVRIP: DHCPVRIP="10.15.15.15";
//((Optional) Configuring VLANs. There are two methods of configuring different VLANs for GSM, UMTS,
LTE, and NR data for isolation:
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
address of the uplink route from the eGBTS to the BSC, the next-hop address of the uplink route from the
NodeB to the RNC, and the next-hop address of the uplink route from the eNodeB to the MME and S-GW
must be different from each other. For example, you can set the next-hop address of the uplink route from
the eGBTS to the BSC to 10.20.20.101, the next-hop address of the uplink route from the NodeB to the RNC
to 10.20.20.201, and the next-hop address of the uplink route from the eNodeB to the MME/S-GW to
10.20.20.202, which are different from the next-hop address (10.20.20.1) of the uplink route from the
gNodeB to the MME/S-GW.
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=32, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.202", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=42, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=20, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, LTE, and NR
data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=34, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=22;
ADD VLANMAP: NEXTHOIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=VLANGROUP,
VLANGROUPNO=1;

```

If **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

```

//Adding a tunnel from the UMPT_N to the UMPT_GUL
ADD BACKTUNNEL: PORTID=0, SSN=7, DCN=0, DSRN=0, DSN=6, TUNNELTYPE=NON_SHAPING;

```

```
//Adding an interface to Ethernet port 1 on the UMPT_L (in slot 6)
SET ETHPORT: CN=0, SRN=0, SN=7, PN=1, SBT=BASE_BOARD, PORTID=71, PA=COPPER, SPEED=AUTO,
DUPLEX=AUTO;
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=71, VRFIDX=0;
//Adding the INTERFACE setting to a loopback interface
ADD LOOPBACK: PORTID=700, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=700, VRFIDX=0;
//Adding IP addresses for the gNodeB
ADD IPADDR4: ITFID=1, IP="10.36.36.188", MASK="255.255.255.255";
ADD IPADDR4: ITFID=0, IP="10.20.20.188", MASK="255.255.255.0";
//Adding uplink routes from the co-MPT GUL MBTS to the RNC/BSC through the gNodeB with Route Type
set to NEXTHOP and Next Hop IP set to the IP address of the directly connected router
ADD IPRROUTE4: RTIDX=0, DSTIP="10.15.15.15", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTIDX=3, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
//Adding uplink routes from the co-MPT GUL MBTS to the MME/S-GW/MAE through the gNodeB
ADD IPRROUTE4: RTIDX=1, DSTIP="10.40.40.0", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTIDX=2, DSTIP="10.50.50.0", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
ADD IPRROUTE4: RTIDX=5, DSTIP="10.60.60.0", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.20.20.1";
//Adding downlink routes from the BSC/RNC/S-GW/MME/MAE to the GUL MBTS through the gNodeB.
When IP-based co-transmission for the co-MPT GUL MBTS and the gNodeB is implemented through the
backplane tunnel, the downlink route from the BSC/RNC/S-GW/MME/MAE to the co-MPT GUL MBTS
through the gNodeB must be configured on the gNodeB. In addition, Port Type must be set to TUNNEL.
ADD IPRROUTE4: RTIDX=4, DSTIP="10.35.35.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=6, DSTIP="10.32.32.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=7, DSTIP="10.33.33.188", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
ADD IPRROUTE4: RTIDX=8, DSTIP="10.30.30.1", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL,
PORTID=0;
//((Optional) Enabling DHCP relay. If the eGBTS/NodeB/eNodeB is to be deployed using DHCP, the gNodeB
needs to work as the relay. Therefore, DHCP relay needs to be enabled for the gNodeB.
SET DHCPRELAYSWITCH: ES=ENABLE;
//((Optional) Adding the IP address of the DHCP server. If the eGBTS/NodeB/eNodeB is to be deployed using
DHCP, the gNodeB needs to work as the relay. Therefore, the IP address of the DHCP server needs to be
added on the gNodeB. For the NodeB, the DHCP server is the RNC or MAE. For the eGBTS/eNodeB, the
DHCP server is the MAE.
//Adding the IP address of the DHCP server for the eGBTS/eNodeB
ADD DHCPVRIP: DHCPVRIP="10.60.60.60";
//Adding the IP address of the DHCP server for the NodeB
ADD DHCPVRIP: DHCPVRIP="10.15.15.15";
//((Optional) Configuring VLANs
//There are two methods of configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation
when the VLANMAP MO is used.
//Method 1 (recommended): Configuring differentiated next-hop addresses. Specifically, the next-hop
address of the uplink route from the eGBTS to the BSC, the next-hop address of the uplink route from the
NodeB to the RNC, and the next-hop address of the uplink route from the eNodeB to the MME and S-GW
must be different from each other. For example, you can set the next-hop address of the uplink route from
the eGBTS to the BSC to 10.20.20.101, the next-hop address of the uplink route from the NodeB to the RNC
to 10.20.20.201, and the next-hop address of the uplink route from the eNodeB to the MME/S-GW to
10.20.20.202, which are different from the next-hop address (10.20.20.1) of the uplink route from the
gNodeB to the MME/S-GW.
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.101", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=22, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.201", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=32, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.202", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=42, SETPRIO=DISABLE;
ADD VLANMAP: VRFIDX=0, NEXTHOIP="10.20.20.1", MASK="255.255.255.255",
VLANMODE=SINGLEVLAN, VLANID=20, SETPRIO=DISABLE;
//Method 2 (not recommended): Configuring differentiated DSCP values for the GSM, UMTS, LTE, and NR
data packets and configuring the mapping between a VLAN group and the next hop
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=40, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRIO=34, VLANID=22;
```

```
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=26, VLANID=22;
ADD VLANCLASS: VLANGROUPNO=1, TRAFFIC=USERDATA, SRVPRI=0, VLANID=22;
ADD VLANMAP: NEXTHOPIP="10.20.20.1", MASK="255.255.255.255", VLANMODE=VLANGROUP,
VLANGROUPNO=1;
//Configuring different VLANs for GSM, UMTS, LTE, and NR data for isolation when the INTERFACE MO is
used
//Configuring the VLAN of the interface for GSM data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for UMTS data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=33, VRFID=0;
//Configuring the VLAN of the interface for LTE data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;
//Configuring the VLAN of the interface for NR data
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=71, VLANID=44, VRFID=0;
```

5.4.3.17.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3.17.4 Activation Verification

After the configuration file is delivered to the MBTS and activated, perform the following steps on the MBTS side to verify that the transmission links between the MBTS and the peer ends are normal.

- On the BSC, ping the IP address of the eGBTS.
PING: SIPADDR="10.10.10.10", DESTIP="10.35.35.188", NEXTHOP="10.21.21.254", CONTPING=NO;
- On the co-MPT GUL MBTS, ping the IP address of the RNC. If the IP address can be pinged, the transmission link is normal.
PING: CN=0, SRN=0, SN=6, SRCIP="10.32.32.1", DSTIP="10.15.15.15", CONTPING=DISABLE, APPTIF=NO;
- On the co-MPT GUL MBTS, ping the IP address of the MME.
PING: CN=0, SRN=0, SN=6, SRCIP="10.33.33.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
- On the co-MPT GUL MBTS, ping the IP address of the S-GW.
PING: CN=0, SRN=0, SN=6, SRCIP="10.33.33.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;
- On the gNodeB, ping the IP address of the MME.
PING: CN=0, SRN=0, SN=7, SRCIP="10.36.36.188", DSTIP="10.40.40.40", CONTPING=DISABLE, APPTIF=NO;
- On the gNodeB, ping the IP address of the S-GW.
PING: CN=0, SRN=0, SN=7, SRCIP="10.36.36.188", DSTIP="10.50.50.50", CONTPING=DISABLE, APPTIF=NO;

5.4.3.18 Reconstruction from Transmission Through the Main Control Board Panel on an eNodeB to Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN/TN/MN MBTS Side in IP over FE/GE Mode

This section describes how to reconstruct the transmission mode from transmission through the main control board panel on an eNodeB to main-control-board-based IP co-transmission through backplane interconnection on the LN MBTS side.

5.4.3.18.1 Data Preparation

Figure 5-29 shows an example of reconstruction from transmission through the main control board panel on an eNodeB to main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LN MBTS side, and Table 5-18 describes the IP address plan. The main control board in this scenario must be a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga/UMPTga.

Figure 5-29 Reconstruction from transmission through the main control board panel on an eNodeB to main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LN MBTS side

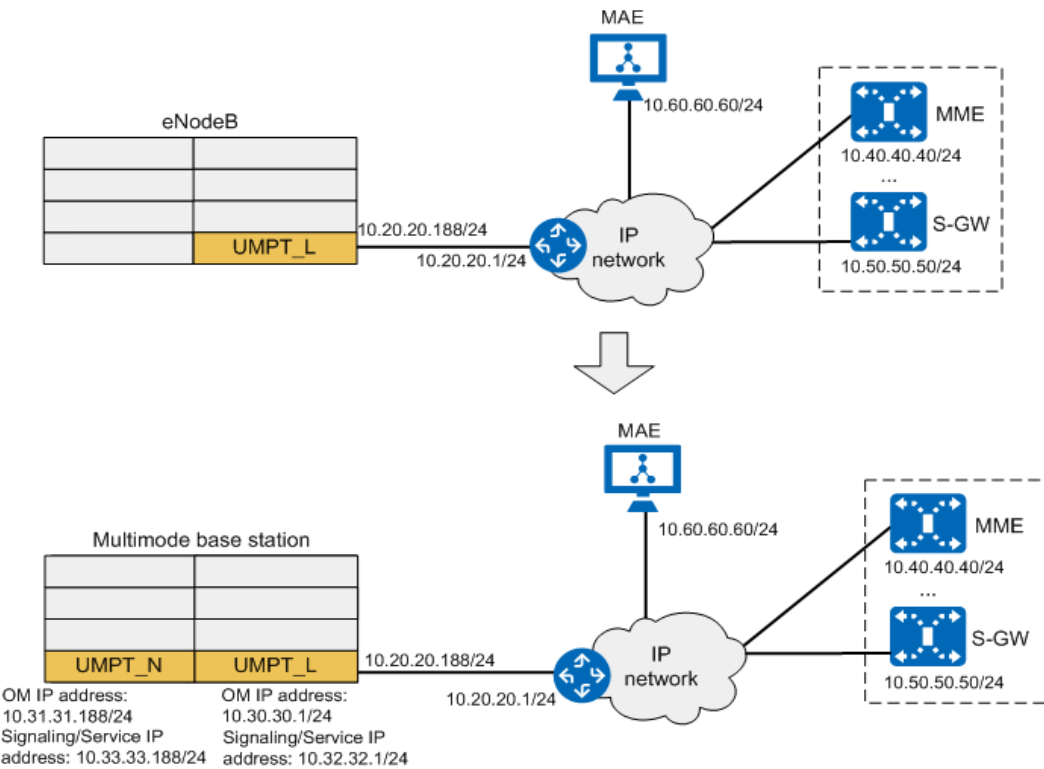


Table 5-18 Data plan

Item	Instance	Remarks
IP address of the port on the router that connects to the UMPT_L	10.20.20.1/24	None
IP address of FE port 1 on the UMPT_L of the eNodeB	10.20.20.188/24	None
OMCH IP address of the eNodeB	10.30.30.1/24	LTE: management-plane IP address. This is a logical IP address configured on the eNodeB main control board.

Item	Instance	Remarks
Signaling/Service IP address of the eNodeB	10.32.32.1/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the eNodeB main control board.
Signaling/Service IP address of the gNodeB	10.33.33.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the gNodeB main control board.
OMCH IP address of the gNodeB	10.31.31.188/24	NR: management-plane IP address. This is a logical IP address configured on the gNodeB main control board.
IP address of the MME	10.40.40.40/24	LTE/NR: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	LTE/NR: peer IP address for the IP path
IP address of the MAE	10.60.60.60/24	NR/LTE: peer IP address for the OMCH

5.4.3.18.2 Reconstruction Preparations and Procedure

Reconstruction Preparations

- Check the system.
 - Perform a health check on the system to check whether network KPIs are normal and whether major alarms were reported. If network KPIs are abnormal or major alarms were reported, record them for KPI or alarm comparison before and after reconstruction.
 - Run the **LST VER** command to check the software version of an eNodeB. MBTSs of V100R007C00 and later support IP-based co-transmission through backplane interconnection.
- Get boards ready and obtain software versions.
 - Get the UMPT_L board and all gNodeB boards ready.
 - Obtain the software versions of the MAE and MBTS.
- Back up data.
 - Back up the measurement results of original performance counters reported one week before reconstruction.
 - Run the **BKP CFGFILE** command to back up the eNodeB configuration file before reconstruction. The configuration file is in .xml format. Then, run the **ULD CFGFILE** command to upload the configuration file to the local PC. (On the MAE-Deployment, choose **LTE Application > Network Adjustment > Export Base Station Bulk Configuration Data.**)
- Make the reconfiguration scripts.

- For details about the reconfiguration on the LTE side, see [LTE Side](#).
 - For details about the reconfiguration on the NR side, see [NR Side](#).
5. Make the rollback scripts.
- For details about rollback scripts on the LTE side, see [Rollback on the LTE Side](#).
 - For details about rollback scripts on the gNodeB side, see [Rollback on the NR Side](#).
6. Prepare for automatic deployment.
- MAE-based deployment is applied for the LN MBTS and the following preparations are required before the deployment.
- Upload the new configuration file for the LN MBTS to the MAE.
 - Upload the eNodeB configuration file backed up before the reconstruction to the MAE.
 - Enable the DHCP relay function for the router directly connected to the eNodeB.

Reconstruction Procedure

Step 1 Reconstruct hardware. Install the gNodeB hardware.

Step 2 Modify data configurations for the eNodeB.

Perform eNodeB deployment. For details, see *3900 & 5900 Series Base Station Commissioning Guide* in *3900 & 5900 Series Base Station Product Documentation*.

Step 3 Perform gNodeB deployment. For details, see *3900 & 5900 Series Base Station Commissioning Guide* in *3900 & 5900 Series Base Station Product Documentation*.

Step 4 Verify services after reconstruction. For details, see [5.4.3.6.4 Activation Verification](#) in [5.4.3.6 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN/TN/MN MBTS Side in IP over FE/GE Mode](#).

Step 5 (Optional) Perform reconstruction rollback. Perform the rollback operation if reconstruction fails.

- Restore the hardware configuration and network topology to those before reconstruction.
- Execute the rollback scripts. For details, see [5.4.3.18.4 Rollback](#).

----End

5.4.3.18.3 Reconfiguration

LTE Side

For details, see [LTE Side](#) in [5.4.3.6 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN/TN/MN MBTS Side in IP over FE/GE Mode](#).

NR Side

For details, see [NR Side](#) in [5.4.3.6 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN/TN/MN MBTS Side in IP over FE/GE Mode](#).

5.4.3.18.4 Rollback

Rollback on the LTE Side

If the transmission link between the eNodeB and the MAE is normal, download the backup of the original configuration file from the MAE to the eNodeB.

If the transmission link between the eNodeB and the MAE is disconnected, the eNodeB performs automatic deployment by using the backup of the original configuration file.

Rollback on the NR Side

Remove the gNodeB on the MAE-Deployment. (On the MAE-Deployment, choose **Base Station** > **NR**. Select a site and right-click **Delete**.)

5.4.3.19 Reconstruction from Main-Control-Board-based IP Co-transmission Through Panel Interconnection on the Separate-MPT LG/TG/MG MBTS Side to Main-Control-Board-based IP Co-transmission Through Backplane Interconnection on the Separate-MPT LG+N/TG+N/MG+N MBTS Side in IP over FE/GE Mode

This section describes how to reconstruct the transmission mode from main-control-board-based IP co-transmission through panel interconnection on the separate-MPT LG/TG/MG MBTS side to main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LG+N/TG+N/MG+N MBTS side. The GSM base stations in this scenario can only be GBTSs.

5.4.3.19.1 Data Preparation

[Figure 5-30](#) shows an example of reconstruction from main-control-board-based IP co-transmission through panel interconnection on the separate-MPT LG/TG/MG MBTS side to main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LG+N/TG+N/MG+N MBTS side. [Table 5-19](#) describes the IP address plan.

Figure 5-30 Reconstruction from main-control-board-based IP co-transmission through panel interconnection on the separate-MPT LG/TG/MG MBTS side to main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LG+N/TG+N/MG+N MBTS side

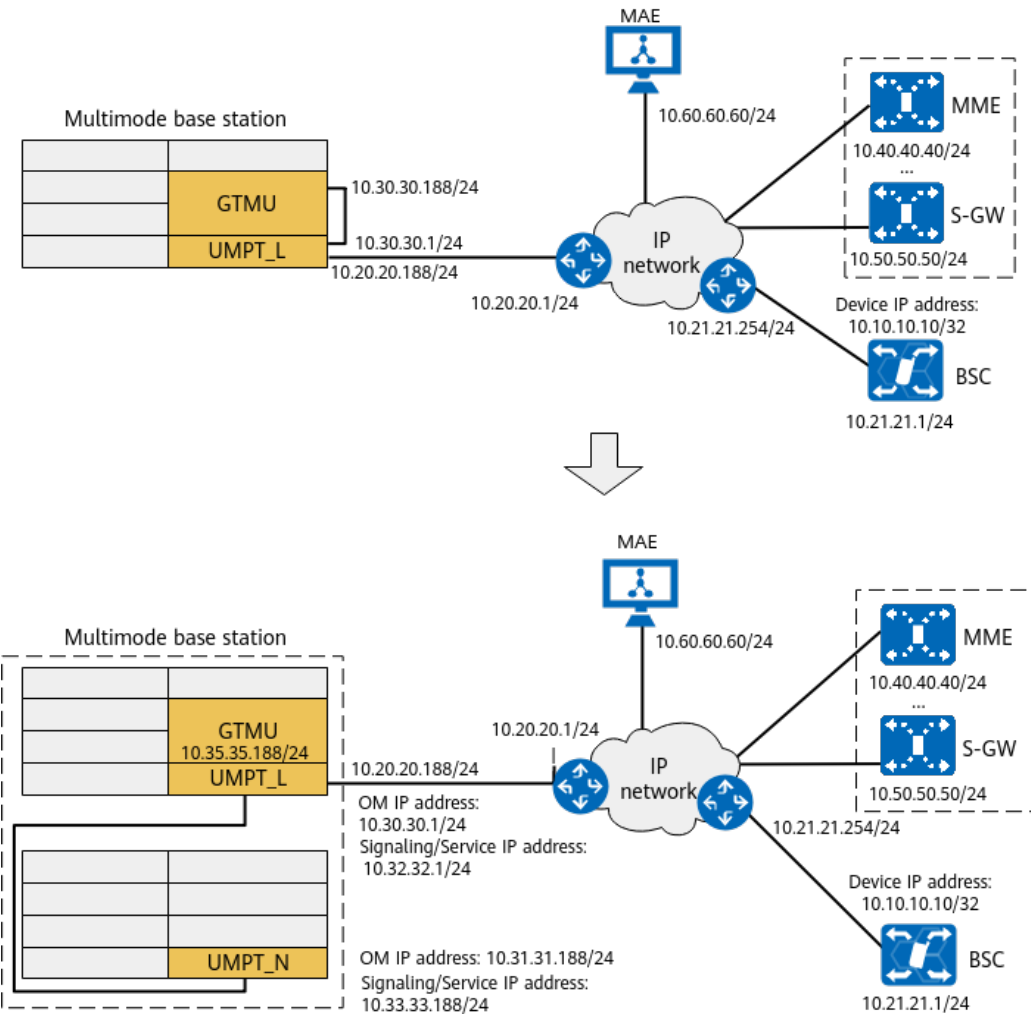


Table 5-19 Data plan

Item	Instance	Remarks
Device IP address of the BSC	10.10.10.10/32	None
Port IP address of the BSC	10.21.21.1/24	None
Port IP address of the MAE	10.60.60.60/24	NR/LTE: peer IP address for the OMCH
IP address of the port on the router that connects to the BSC	10.21.21.254/24	None

Item	Instance	Remarks
IP address of the port on the router that connects to the eNodeB	10.20.20.1/24	None
IP address of 10GE port 1 on the eNodeB UMPT_L	10.20.20.188/24	LTE: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment This IP address is configured on the co-transmission port.
OMCH IP address of the eNodeB	10.30.30.1/24	LTE: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the eNodeB	10.32.32.1/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of the GBTS	10.35.35.188/24	GSM: communication IP address of the GBTS. This is a logical IP address configured on the GBTS main control board.
OMCH IP address of the gNodeB	10.31.31.188/24	NR: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the gNodeB	10.33.33.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the eNodeB main control board.
IP address of the MME	10.40.40.40/24	NR/LTE: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	NR/LTE: peer IP address for the IP path
ESN of the GBTS	abcdefghijklmn	None

5.4.3.19.2 Reconstruction Preparations and Procedure

Reconstruction Preparations

1. Check the system.
 - Perform a health check on the system to check whether network KPIs are normal and whether major alarms were reported. If network KPIs are abnormal or major alarms were reported, record them for KPI or alarm comparison before and after reconstruction.

- Check software versions. Run the **DSP BTSVER** command on the BSC to check the software version of the GBTS or the **LST VER** command on the eNodeB to check the software version of the eNodeB. MBTSs of V100R007C00 and later support IP-based co-transmission through backplane interconnection.
- 2. Get boards ready and obtain software versions.
 - Get the eNodeB and gNodeB hardware ready. The main control board for the eNodeB is a UMPT_L, and the main control board for the gNodeB is a UMPT_N. Optical modules on the UMPT_N and UMPT_L must meet certain specifications. For details about the specifications of optical modules required by the UMPT, go to *3900 & 5900 Series Base Station Product Documentation* > **Product Description** > **Hardware Description** > **BBU Hardware Description**, select a BBU, choose its board section, choose **UMPT** > **Port**, and see the **Transmission Mode** column.
 - Obtain the software versions of the BSC, MAE, and MBTS.
- 3. Back up data.
 - Back up the measurement results of original performance counters reported one week before reconstruction.
 - Back up GBTS data configurations before reconstruction on the MAE-Deployment. (On the MAE-Deployment, choose **GSM Application** > **Network Deployment** > **Export GBTS Summary Data**.)
 - Run the **BKP CFGFILE** command to back up the eNodeB configuration file. The configuration file is in .xml format. Then, run the **ULD CFGFILE** command to upload the configuration file to the local PC. (On the MAE-Deployment, choose **LTE Application** > **Network Adjustment** > **Export Base Station Bulk Configuration Data**.)
- 4. Make the reconfiguration scripts.
 - Reconfiguration on the GSM side has the following scripts:
 - IP rehomeing scripts. For details, see [GSM Side](#).
 - Scripts for reconstruction. For details, see [GSM Side \(GBTS\)](#) in [5.4.3.7 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LG+N/TG+N/MG+N MBTS Side in IP over FE/GE Mode](#).
 - For details about the reconfiguration on the LTE side, see [LTE Side](#).
 - For details about the reconfiguration on the NR side, see [NR Side](#).
- 5. Make the rollback scripts.
 - For details about rollback scripts on the LTE side, see [Rollback on the LTE Side](#).
 - For details about rollback scripts on the GSM side, see [Rollback on the GSM Side](#).
 - For details about rollback scripts on the NR side, see [Rollback on the NR Side](#).
- 6. Prepare for automatic deployment.

MAE-based deployment is applied for the LG+N MBTS. Upload the following files to the MAE and enable the DHCP relay function for the router that is directly connected to the eNodeB: new configuration files for the eNodeB and

gNodeB, and the eNodeB configuration file backed up before the reconstruction.

Reconstruction Procedure

Step 1 Upgrade the software.

If the software version of the MBTS does not meet the reconstruction requirements, upgrade the software before reconstruction.

Step 2 Reconstruct hardware and modify data configurations.

1. Download the eNodeB configuration file in .xml format from the MAE to the eNodeB by choosing **Software > NE File Transfer > From OSS Client to NE** on the MAE-Access.
2. Execute the BSC IP rehome scripts.
3. Get the gNodeB configuration file and the gNodeB deployment list ready on the MAE.

NOTE

GBTS and eNodeB services are interrupted when the following steps are performed. Before performing these steps, power off the GTMU. Power on the GTMU after eNodeB deployment is complete. This is because the IP rehome scripts take effect only after the GBTS resets. In addition, the IP rehome scripts are effective within specified time.

4. Run the **SET CFGFILEENB** command to activate the eNodeB configuration file on the MAE.
`SET CFGFILEENB: FLAG=ENABLE, RSTMODE=IMMEDIATELY;`
5. Instruct site engineers to remove the connections between panels and remove the GTMU board.
6. Deactivate the GBTS on the BSC, and execute the scripts for GBTS reconstruction to reactivate the GBTS.
7. Check that the eNodeB maintenance link is normal on the MAE. After the eNodeB is automatically deployed, install and power on the GTMU of the GBTS to make the GBTS rehome scripts take effect.
8. Wait 10 minutes, and check that services processed on the GBTS and eNodeB are normal on the MAE.
9. Install the gNodeB and connect the cable between the UMPT_L and the UMPT_N on the gNodeB. Then, power on the BBU of the gNodeB to deploy the gNodeB.

Step 3 Verify services after reconstruction. For details, see [5.4.3.7.4 Activation Verification](#) in [5.4.3.7 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LG+N/TG+N/MG+N MBTS Side in IP over FE/GE Mode](#).

Step 4 (Optional) Perform reconstruction rollback. Perform the rollback operation if reconstruction fails.

- Execute the rollback scripts. For details about the rollback scripts, see [5.4.3.19.4 Rollback](#).

- Restore the hardware configuration and network topology to those before reconstruction.

----End

5.4.3.19.3 Reconfiguration

LTE Side

For details, see [LTE Side](#) in [5.4.3.7 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LG+N/TG+N/MG+N MBTS Side in IP over FE/GE Mode](#).

GSM Side

```
//Importing the communication IP address of the base station
IMP BTSIPPARA: IDTYPE=BYID, BTSID=10, CFGSW=ENABLE, IPPHYTRANSTYPE=IP_OVER_FE/GE,
BTSCOMTYPE=LOGICIP, BTSIP="10.35.35.188", BSCIP="10.10.10.10";
//Importing the device IP address of the base station
IMP BTSDEVIP: IDTYPE=BYID, BTSID=10, CFGSW=ENABLE, PT=LOOPINTERFACE, PN=0, CN=0, SRN=0, SN=6,
IPIDX=0, IP="10.35.35.188", MASK="255.255.255.255";
//Importing the configuration of a tunnel from the GTMU to the UMPT_L
IMP BTSTUNNEL: IDTYPE=BYID, BTSID=10, CFGSW=ENABLE, SRCN=0, SRCRN=0, SRCSN=6, TN=0,
DSTCN=0, DSTSRN=0, DSTSN=7;
//Importing the configuration of a route to the tunnel from the GBTS to the BSC
IMP BTSIPRT: IDTYPE=BYID, BTSID=10, CFGSW=ENABLE, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=OUTIF, CN=0, SRN=0, SN=6, ITFType=TUNNEL, IFNO=0;
//Activating the IP rehomeing data
ACT BTSIMPDATA: IDTYPE=BYID, BTSID=10;
```

NR Side

For details, see [NR Side](#) in [5.4.3.7 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LG+N/TG+N/MG+N MBTS Side in IP over FE/GE Mode](#).

5.4.3.19.4 Rollback

Rollback on the GSM Side

```
//Deactivating the base station
DEA BTS: IDTYPE=BYID, BTSID=10;
//Removing the IP path
//Removing the IP path between the GBTS and the BSC
RMV IPPATH: ANI=1017, PATHID=0;
//Removing an adjacent node
//Removing an Abis adjacent node
RMV ADJNODE: ANI=1017;
//Removing the base station
RMV BTS: IDTYPE=BYID, BTSID=10;
```

Execute the configuration scripts that are backed up before reconstruction. (On the MAE-Deployment, choose **Advanced** > **System** > **Script Executor**.)

Rollback on the LTE Side

If the transmission link between the eNodeB and the MAE is normal, download the backup of the original configuration file from the MAE to the eNodeB.

If the transmission link between the eNodeB and the MAE is disconnected, the eNodeB performs automatic deployment by using the backup of the original configuration file.

Rollback on the NR Side

Remove the gNodeB on the MAE-Deployment. (On the MAE-Deployment, choose **Base Station > NR**. Select a site and right-click **Delete**.)

5.4.3.20 Reconstruction from Main-Control-Board-based Co-transmission Through Panel Interconnection on the Separate-MPT LU/TU/MU MBTS Side to Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LU+N/TU+N/MU+N MBTS Side in IP over FE/GE Mode

This section describes how to reconstruct the transmission mode from main-control-board-based IP co-transmission through panel interconnection on the separate-MPT LU/TU/MU MBTS side to main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LU+N/TU+N/MU+N MBTS side.

5.4.3.20.1 Data Preparation

[Figure 5-31](#) shows an example of reconstruction from main-control-board-based IP co-transmission through panel interconnection on the separate-MPT LU/TU/MU MBTS side to main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LU+N/TU+N/MU+N MBTS side. [Table 5-20](#) describes the IP address plan.

Figure 5-31 Reconstruction from main-control-board-based IP co-transmission through panel interconnection on the separate-MPT LU/TU/MU MBTS side to main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LU+N/TU+N/MU+N MBTS side

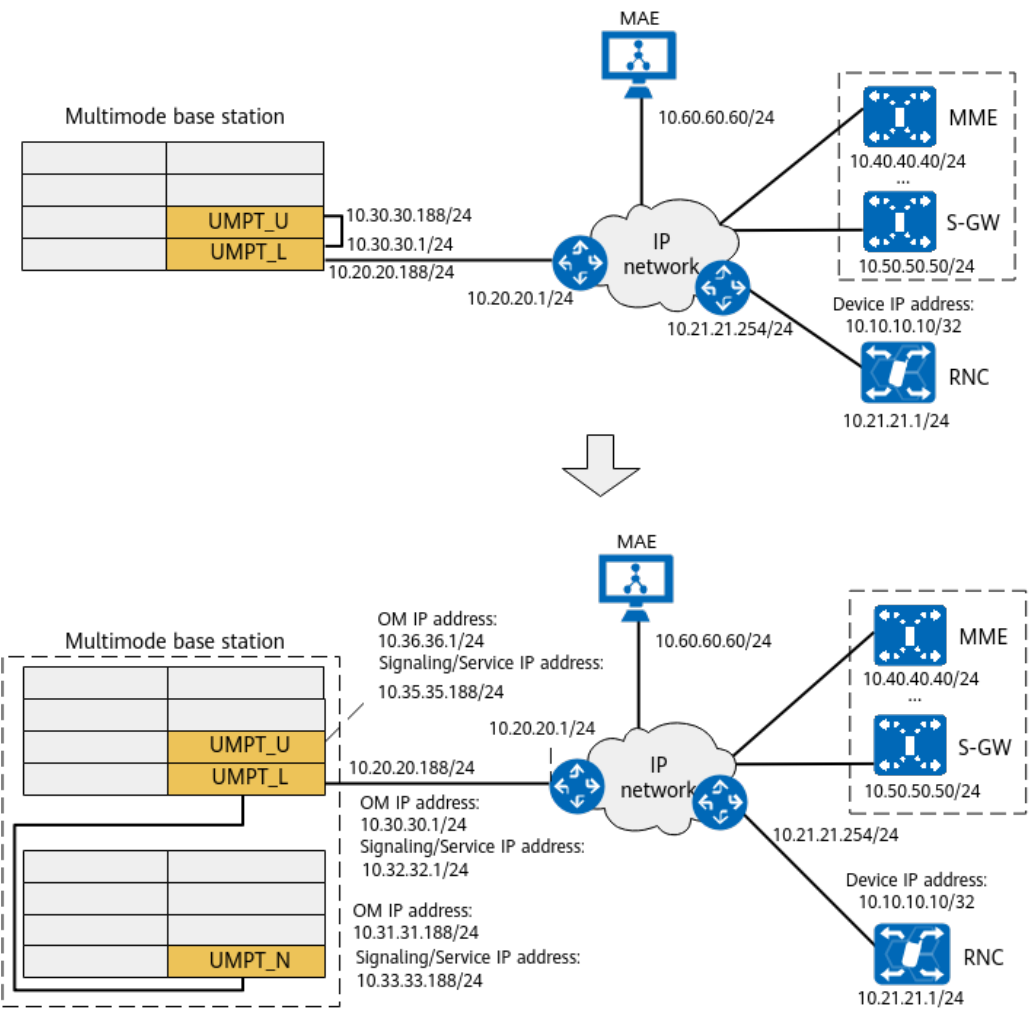


Table 5-20 Data plan

Item	Instance	Remarks
Device IP address of the RNC	10.10.10.10/32	None
Port IP address of the RNC	10.21.21.1/24	None
Port IP address of the MAE	10.60.60.60/24	NR/LTE/UMTS: peer IP address for the OMCH
IP address of the port on the router that connects to the RNC	10.21.21.254/24	None

Item	Instance	Remarks
IP address of the port on the router that connects to the eNodeB	10.20.20.1/24	None
IP address of 10GE port 1 on the eNodeB UMPT_L	10.20.20.188/24	LTE: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment This IP address is configured on the co-transmission port.
OMCH IP address of the eNodeB	10.30.30.1/24	LTE: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the eNodeB	10.32.32.1/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the NodeB	10.35.35.188/24	UMTS: communication IP address of the NodeB. This is a logical IP address configured on the NodeB main control board.
OMCH IP address of the NodeB	10.36.36.1/24	UMTS: management-plane IP address. This is a logical IP address configured on the main control board.
OMCH IP address of the gNodeB	10.31.31.188/24	NR: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the gNodeB	10.33.33.188/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the eNodeB main control board.
IP address of the MME	10.40.40.40/24	NR/LTE: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	NR/LTE: peer IP address for the IP path

5.4.3.20.2 Reconstruction Preparations and Procedure

Reconstruction Preparations

1. Check the system.
 - Perform a health check on the system to check whether network KPIs are normal and whether major alarms were reported. If network KPIs are

- abnormal or major alarms were reported, record them for KPI or alarm comparison before and after reconstruction.
- Run the **LST VER** command to check the software version of a NodeB and an eNodeB. MBTSs of V100R007C00 and later support IP-based co-transmission through backplane interconnection.
2. Get boards ready and obtain software versions.
 - Get the eNodeB and gNodeB hardware ready. The main control board for the eNodeB is a UMPT_L, and the main control board for the gNodeB is a UMPT_N. Optical modules on the UMPT_N and UMPT_L must meet certain specifications. For details about the specifications of optical modules required by the UMPT, go to *3900 & 5900 Series Base Station Product Documentation > Product Description > Hardware Description > BBU Hardware Description*, select a BBU, choose its board section, choose **UMPT > Port**, and see the **Transmission Mode** column.
 - Obtain the software versions of the RNC, MAE, and MBTS.
 3. Back up data.
 - Back up the measurement results of original performance counters reported one week before reconstruction.
 - Run the **BKP CFGFILE** command to back up the NodeB configuration file. The configuration file is in .xml format. Then, run the **ULD CFGFILE** command to upload the configuration file to the local PC. (On the MAE-Deployment, choose **UMTS Application > Network Adjustment > Export Base Station Bulk Configuration Data.**)
 - Run the **BKP CFGFILE** command to back up the eNodeB configuration file. The configuration file is in .xml format. Then, run the **ULD CFGFILE** command to upload the configuration file to the local PC. (On the MAE-Deployment, choose **LTE Application > Network Adjustment > Export Base Station Bulk Configuration Data.**)
 4. Make the reconfiguration scripts.
 - For details about the reconfiguration on the UMTS side, see [UMTS Side](#).
 - For details about the reconfiguration on the LTE side, see [LTE Side](#).
 - For details about the reconfiguration on the NR side, see [NR Side](#).
 5. Make the rollback scripts.
 - For details about rollback scripts on the LTE side, see [Rollback on the LTE Side](#).
 - For details about rollback scripts on the UMTS side, see [Rollback on the UMTS Side](#).
 - For details about rollback scripts on the NR side, see [Rollback on the NR Side](#).
 6. Prepare for automatic deployment.

MAE-based deployment is applied for the LU+N MBTS. Upload the following files to the MAE and enable the DHCP relay function for the router that is directly connected to the eNodeB: new configuration files for the eNodeB and gNodeB, and the eNodeB configuration file backed up before the reconstruction.

Reconstruction Procedure

Step 1 Upgrade the software.

If the software version of the MBTS does not meet the reconstruction requirements, upgrade the software before reconstruction.

Step 2 Reconstruct hardware and modify data configurations.

1. Download the eNodeB and NodeB configuration files in .xml format from the MAE to the eNodeB and NodeB by choosing **Software > NE File Transfer > From OSS Client to NE** on the MAE-Access.
2. Get the gNodeB configuration file and the gNodeB deployment list ready on the MAE.

NOTE

NodeB and eNodeB services are interrupted when the following steps are performed.

3. Run the **SET CFGFILEENB** command to activate the eNodeB and NodeB configuration files on the MAE.
`SET CFGFILEENB: FLAG=ENABLE, RSTMODE=IMMEDIATELY;`
4. Instruct site engineers to remove the connections between panels and remove the UMPT_U board.
5. Wait 10 minutes, and check that services processed on the NodeB and eNodeB are normal on the MAE.
6. Install the gNodeB and connect the cable between the UMPT_L and the UMPT_N on the gNodeB. Then, power on the BBU of the gNodeB to deploy the gNodeB.

Step 3 Verify services after reconstruction. For details, see [5.4.3.9.4 Activation Verification](#) in [5.4.3.9 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LU+N/TU+N/MU+N MBTS Side in IP over FE/GE Mode](#).

Step 4 (Optional) Perform reconstruction rollback. Perform the rollback operation if reconstruction fails.

- Execute the rollback scripts. For details about the rollback scripts, see [5.4.3.20.4 Rollback](#).
- Restore the hardware configuration and network topology to those before reconstruction.

-----End

5.4.3.20.3 Reconfiguration

LTE Side

For details, see [LTE Side](#) in [5.4.3.9 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LU+N/TU+N/MU+N MBTS Side in IP over FE/GE Mode](#).

UMTS Side

For details, see [UMTS Side](#) in [5.4.3.9 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LU+N/TU+N/MU+N MBTS Side in IP over FE/GE Mode](#).

NR Side

For details, see [NR Side](#) in [5.4.3.9 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LU+N/TU+N/MU+N MBTS Side in IP over FE/GE Mode](#).

5.4.3.20.4 Rollback

Rollback on the UMTS Side

If the transmission link between the NodeB and the MAE is normal, download the backup of the original configuration file from the MAE to the NodeB.

If the transmission link between the NodeB and the MAE is disconnected, the NodeB performs automatic deployment by using the backup of the original configuration file.

Rollback on the LTE Side

If the transmission link between the eNodeB and the MAE is normal, download the backup of the original configuration file from the MAE to the eNodeB.

If the transmission link between the eNodeB and the MAE is disconnected, the eNodeB performs automatic deployment by using the backup of the original configuration file.

Rollback on the NR Side

Remove the gNodeB on the MAE-Deployment. (On the MAE-Deployment, choose **Base Station > NR**. Select a site and right-click **Delete**.)

5.4.3.21 Reconstruction from Main-Control-Board-based Co-transmission Through Panel Interconnection on the Separate-MPT L+GU/T+GU/M+GU MBTS Side to Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN+GU/TN+GU/MN+GU MBTS Side in IP over FE/GE Mode

This section describes how to reconstruct the transmission mode from main-control-board-based IP co-transmission through panel interconnection on the separate-MPT L+GU/T+GU/M+GU MBTS side to main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LN+GU/TN+GU/MN+GU MBTS side. The GSM base stations in this scenario can only be GBTSS.

5.4.3.21.1 Data Preparation

[Figure 5-32](#) shows an example of reconstruction from main-control-board-based IP co-transmission through panel interconnection on the separate-MPT L+GU/T

+GU/M+GU MBTS side to main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LN+GU/TN+GU/MN+GU MBTS side. **Table 5-21** describes the IP address plan.

Figure 5-32 Reconstruction from main-control-board-based IP co-transmission through panel interconnection on the separate-MPT L+GU/T+GU/M+GU MBTS side to main-control-board-based IP co-transmission through backplane interconnection on the separate-MPT LN+GU/TN+GU/MN+GU MBTS side

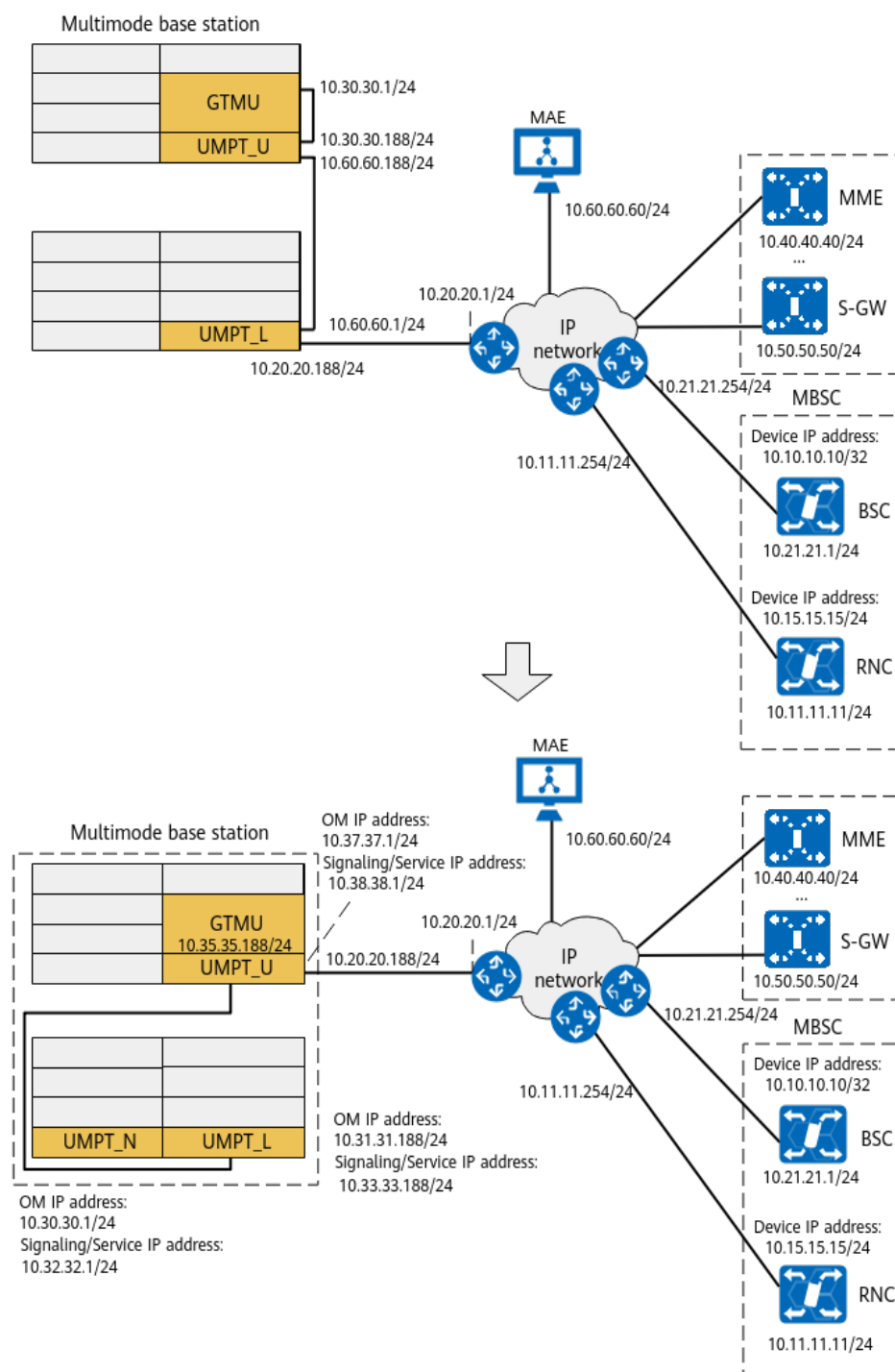


Table 5-21 Data plan

Item	Instance	Remarks
Device IP address of the BSC	10.10.10.10/32	None
Port IP address of the BSC	10.21.21.1/24	None
Device IP address of the RNC	10.15.15.15/32	None
Port IP address of the RNC	10.11.11.11/24	None
Port IP address of the MAE	10.60.60.60/24	NR/LTE/UMTS: peer IP address for the OMCH
IP address of the port on the router that connects to the BSC	10.21.21.254/24	None
IP address of the port on the router that connects to the RNC	10.23.23.254/24	None
IP address of the port on the router that connects to the eNodeB	10.20.20.1/24	None
IP address of 10GE port 1 on the eNodeB UMPT_L	10.20.20.188/24	LTE: device IP address (old model)/IPv4 address (new model) during configuration on the MAE-Deployment This IP address is configured on the co-transmission port.
OMCH IP address of the eNodeB	10.31.31.188/24	LTE: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the eNodeB	10.33.33.188/24	LTE: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of FE port 1 on the UMPT_U of the NodeB	10.20.20.188/24	UMTS: device IP address during configuration on the MAE-Deployment This IP address is configured on the co-transmission port.
OMCH IP address of the NodeB	10.37.37.1/24	UMTS: management-plane IP address. This is a logical IP address configured on the main control board.

Item	Instance	Remarks
Signaling/Service IP address of the NodeB	10.38.38.1/24	UMTS: control-plane and user-plane IP address. This is a logical IP address configured on the main control board.
IP address of the GBTS	10.35.35.188/24	GSM: communication IP address of the GBTS. This is a logical IP address configured on the GBTS main control board.
OMCH IP address of the gNodeB	10.30.30.1/24	NR: management-plane IP address. This is a logical IP address configured on the main control board.
Signaling/Service IP address of the gNodeB	10.32.32.1/24	NR: control-plane and user-plane IP address. This is a logical IP address configured on the eNodeB main control board.
IP address of the MME	10.40.40.40/24	NR/LTE: peer IP address for the SCTP link
IP address of the S-GW	10.50.50.50/24	NR/LTE: peer IP address for the IP path
ESN of the GBTS	abcdefghijklmn	None

5.4.3.21.2 Reconstruction Preparations and Procedure

Reconstruction Preparations

- Check the system.
 - Perform a health check on the system to check whether network KPIs are normal and whether major alarms were reported. If network KPIs are abnormal or major alarms were reported, record them for KPI or alarm comparison before and after reconstruction.
 - Check software versions. Run the **DSP BTSVER** command on the BSC to check the software version of the GBTS or the **LST VER** command on the eNodeB/NodeB to check the software version of the eNodeB/NodeB. MBTSs of V100R007C00 and later support IP-based co-transmission through backplane interconnection.
- Get boards ready and obtain software versions.
 - Get the eNodeB and gNodeB hardware ready. The main control boards for the NodeB, eNodeB, and gNodeB are UMPT_U, UMPT_L, and UMPT_N, respectively. Optical modules on the UMPT_N and UMPT_L must meet certain specifications. For details about the specifications of optical modules required by the UMPT, go to *3900 & 5900 Series Base Station Product Documentation* > **Product Description** > **Hardware Description** > **BBU Hardware Description**, select a BBU, choose its board section, choose **UMPT** > **Port**, and see the **Transmission Mode** column.

- Obtain the software versions of the BSC, RNC, MAE, and MBTS.
- 3. Back up data.
 - Back up the measurement results of original performance counters reported one week before reconstruction.
 - Back up GBTS data configurations before reconstruction on the MAE-Deployment. (On the MAE-Deployment, choose **GSM Application** > **Network Deployment** > **Export GBTS Summary Data**.)
 - Run the **BKP CFGFILE** command to back up the eNodeB configuration file. The configuration file is in .xml format. Then, run the **ULD CFGFILE** command to upload the configuration file to the local PC. (On the MAE-Deployment, choose **LTE Application** > **Network Adjustment** > **Export Base Station Bulk Configuration Data**.)
 - Run the **BKP CFGFILE** command to back up the NodeB configuration file. The configuration file is in .xml format. Then, run the **ULD CFGFILE** command to upload the configuration file to the local PC. (On the MAE-Deployment, choose **UMTS Application** > **Network Adjustment** > **Export Base Station Bulk Configuration Data**.)
- 4. Make the reconfiguration scripts.
 - Reconfiguration on the GSM side has the following scripts:
 - IP rehomings scripts. For details, see [GSM Side](#).
 - Scripts for reconstruction. For details, see [5.4.3.12 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN/TN/MN+GU MBTS Side in IP over FE/GE Mode in GSM Side \(GBTS\)](#).
 - For details about the reconfiguration on the LTE side, see [LTE Side](#).
 - For details about the reconfiguration on the UMTS side, see [UMTS Side](#).
 - For details about the reconfiguration on the NR side, see [NR Side](#).
- 5. Make the rollback scripts.
 - For details about rollback scripts on the LTE side, see [Rollback on the LTE Side](#).
 - For details about rollback scripts on the UMTS side, see [Rollback on the UMTS Side](#).
 - For details about rollback scripts on the GSM side, see [Rollback on the GSM Side](#).
 - For details about rollback scripts on the NR side, see [Rollback on the NR Side](#).
- 6. Prepare for automatic deployment.

MAE-based deployment is applied for the LN+GU/TN+GU/MN+GU MBTS. Upload the following files to the MAE and enable the DHCP relay function for the router that is directly connected to the eNodeB: new configuration files for the eNodeB, NodeB, and gNodeB, and the eNodeB and NodeB configuration files backed up before the reconstruction.

Reconstruction Procedure

Step 1 Upgrade the software.

If the software version of the MBTS does not meet the reconstruction requirements, upgrade the software before reconstruction.

Step 2 Reconstruct hardware and modify data configurations.

1. Download the eNodeB and NodeB configuration files in .xml format from the MAE to the eNodeB and NodeB by choosing **Software > NE File Transfer > From OSS Client to NE** on the MAE-Access.
2. Execute the BSC IP rehomeing scripts.
3. Get the gNodeB configuration file and the gNodeB deployment list ready on the MAE.

 NOTE

GBTS, eNodeB, and NodeB services are interrupted when the following steps are performed. Before performing these steps, power off the GTMU. Power on the GTMU after eNodeB and NodeB deployment is complete. This is because the IP rehomeing scripts take effect only after the GBTS resets. In addition, the IP rehomeing scripts are effective within specified time.

4. Run the **ACT CFGFILE** command to activate the eNodeB configuration file on the MAE.
`ACT CFGFILE: EFT=IMMEDIATELY;`
5. Run the **ACT CFGFILE** command to activate the NodeB configuration file on the MAE.
`ACT CFGFILE: EFT=IMMEDIATELY;`
6. Instruct site engineers to remove the connections between panels and remove the GTMU board.
7. Deactivate the GBTS on the BSC, and execute the scripts for GBTS reconstruction to reactivate the GBTS.
8. Check that the eNodeB and NodeB maintenance links are normal on the MAE. After the eNodeB and NodeB are automatically deployed, install and power on the GTMU of the GBTS to make the GBTS rehomeing scripts take effect.
9. Wait 10 minutes, and check that services processed on the GBTS, eNodeB, and NodeB are normal on the MAE.
10. Install the gNodeB and connect the cable between the UMPT_L and the UMPT_N on the gNodeB. Then, power on the BBU of the gNodeB to deploy the gNodeB.

Step 3 Verify services after reconstruction. For details, see [5.4.3.12 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN/TN/MN+GU MBTS Side in IP over FE/GE Mode](#) in [5.4.3.12.4 Activation Verification](#).

Step 4 (Optional) Perform reconstruction rollback. Perform the rollback operation if reconstruction fails.

- Execute the rollback scripts. For details about the rollback scripts, see [5.4.3.21.4 Rollback](#).
- Restore the hardware configuration and network topology to those before reconstruction.

----End

5.4.3.21.3 Reconfiguration

LTE Side

For details, see [LTE Side](#) in [5.4.3.12 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN/TN/MN+GU MBTS Side in IP over FE/GE Mode](#).

UMTS Side

For details, see [UMTS Side](#) in [5.4.3.12 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN/TN/MN+GU MBTS Side in IP over FE/GE Mode](#).

GSM Side

```
//Importing the communication IP address of the base station
IMP BTSIPPARA: IDTYPE=BYID, BTSID=10, CFGSW=ENABLE, IPPHYTRANSTYPE=IP_OVER_FE/GE,
BTSCOMTYPE=LOGICIP, BTSIP="10.35.35.188", BSCIP="10.10.10.10";
//Importing the device IP address of the base station
IMP BTSDEVIP: IDTYPE=BYID, BTSID=10, CFGSW=ENABLE, PT=LOOPINTERFACE, PN=0, CN=0, SRN=0, SN=6,
IPIDX=0, IP="10.35.35.188", MASK="255.255.255.255";
//Importing the configuration of a tunnel from the GTMU to the UMPT_L
IMP BTSTUNNEL: IDTYPE=BYID, BTSID=10, CFGSW=ENABLE, SRCN=0, SRCN=0, SRCN=6, TN=0,
DSTCN=0, DSTSRN=0, DSTSN=7;
//Importing the configuration of a route to the tunnel from the GBTS to the BSC
IMP BTSIPRT: IDTYPE=BYID, BTSID=10, CFGSW=ENABLE, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=OUTIF, CN=0, SRN=0, SN=6, ITFType=TUNNEL, IFNO=0;
//Activating the IP rehomeing data
ACT BTSIMPDATA: IDTYPE=BYID, BTSID=10;
```

NR Side

For details, see [NR Side](#) in [5.4.3.12 Main-Control-Board-based Co-transmission Through Backplane Interconnection on the Separate-MPT LN/TN/MN+GU MBTS Side in IP over FE/GE Mode](#).

5.4.3.21.4 Rollback

Rollback on the GSM Side

```
//Deactivating the base station
DEA BTS: IDTYPE=BYID, BTSID=10;
//Removing the IP path
//Removing the IP path between the GBTS and the BSC
RMV IPPATH: ANI=1017, PATHID=0;
//Removing an adjacent node
//Removing an Abis adjacent node
RMV ADJNODE: ANI=1017;
//Removing the base station
RMV BTS: IDTYPE=BYID, BTSID=10;
```

Execute the configuration scripts that are backed up before reconstruction. (On the MAE-Deployment, choose **Advanced** > **System** > **Script Executor**.)

Rollback on the LTE Side

If the transmission link between the eNodeB and the MAE is normal, download the backup of the original configuration file from the MAE to the eNodeB.

If the transmission link between the eNodeB and the MAE is disconnected, the eNodeB performs automatic deployment by using the backup of the original configuration file.

Rollback on the UMTS Side

If the transmission link between the NodeB and the MAE is normal, download the backup of the original configuration file from the MAE to the NodeB.

If the transmission link between the NodeB and the MAE is disconnected, the NodeB performs automatic deployment by using the backup of the original configuration file.

Rollback on the NR Side

Remove the gNodeB on the MAE-Deployment. (On the MAE-Deployment, choose **Base Station > NR**. Select a site and right-click **Delete**.)

5.4.4 Network Monitoring

None

6 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

- Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.
- End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see the *Glossary*.

9 Reference Documents

- *IPv4 Transmission*
- *IPv6 Transmission*
- *Bandwidth Sharing of Multimode Base Station Co-Transmission*
- *SRAN Networking and Evolution Overview*
- *BBU Interconnection*
- *VRF*
- *Equipment Security*
- *ATM Transport*
- *BSC6900 GU Initial Configuration Guide in BSC6900 GU Product Documentation*
- *BSC6910 GU Initial Configuration Guide in BSC6910 GU Product Documentation*
- *3900 & 5900 Series Base Station Initial Configuration Guide in 3900 & 5900 Series Base Station Product Documentation*
- *3900 & 5900 Series Base Station Commissioning Guide in 3900 & 5900 Series Base Station Product Documentation*